

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
CENTRAL VALLEY REGION**

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**ORDER R5-2014-0049  
NPDES NO. CA0079502**

**WASTE DISCHARGE REQUIREMENTS FOR THE  
CITY OF ROSEVILLE  
DRY CREEK WASTEWATER TREATMENT PLANT  
PLACER COUNTY**

The following Discharger is subject to waste discharge requirements (WDR's) set forth in this Order:

**Table 1. Discharger Information**

|                         |                                      |
|-------------------------|--------------------------------------|
| <b>Discharger</b>       | City of Roseville                    |
| <b>Name of Facility</b> | Dry Creek Wastewater Treatment Plant |
| <b>Facility Address</b> | 1800 Booth Road                      |
|                         | Roseville, CA 95747                  |
|                         | Placer County                        |

**Table 2. Discharge Location**

| <b>Discharge Point</b> | <b>Effluent Description</b>  | <b>Discharge Point Latitude (North)</b> | <b>Discharge Point Longitude (West)</b> | <b>Receiving Water</b> |
|------------------------|------------------------------|-----------------------------------------|-----------------------------------------|------------------------|
| 001                    | Treated Municipal Wastewater | 38° 43' 53"                             | 121° 18' 58"                            | Dry Creek              |

**Table 3. Administrative Information**

|                                                                                                                                                                                                                                                                                   |                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| This Order was adopted on:                                                                                                                                                                                                                                                        | <b>28 March 2014</b>                               |
| This Order shall become effective on:                                                                                                                                                                                                                                             | <b>1 May 2014</b>                                  |
| This Order shall expire on:                                                                                                                                                                                                                                                       | <b>30 April 2019</b>                               |
| The Discharger shall file a Report of Waste Discharge as an application for reissuance of WDR's in accordance with title 23, California Code of Regulations, and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than: | <b>180 days prior to the Order expiration date</b> |
| The U.S. Environmental Protection Agency (U.S. EPA) and the California Regional Water Quality Control Board, Central Valley Region have classified this discharge as follows:                                                                                                     | Major                                              |

I, **PAMELA CREEDON**, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on **the date indicated above**.

*Original Signed by*

\_\_\_\_\_  
**PAMELA C. CREEDON**, Executive Officer

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## I. FACILITY INFORMATION

Information describing the City of Roseville, Dry Creek Wastewater Treatment Plant (Facility) is summarized in Table 1 and in sections I and II of the Fact Sheet (Attachment F). Section I of the Fact Sheet also includes information regarding the Facility's permit application.

## II. FINDINGS

The California Regional Water Quality Control Board, Central Valley Region (hereinafter Central Valley Water Board), finds:

- A. Legal Authorities.** This Order serves as WDR's pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters.
- B. Background and Rationale for Requirements.** The Central Valley Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E and G through H are also incorporated into this Order.
- C. Provisions and Requirements Implementing State Law.** The provisions/requirements in subsections IV.B, IV.C, and V.B are included to implement State law only. These provisions/requirements are not required or authorized under the federal CWA; consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations.
- D. Monitoring and Reporting.** 40 CFR 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 authorize the Central Valley Water Board to require technical and monitoring reports. The Monitoring and Reporting Program establishes monitoring and reporting requirements to implement federal and State requirements. The Monitoring and Reporting Program is provided in Attachment E.

The technical and monitoring reports in this Order are required in accordance with Water Code section 13267, which states the following in subsection (b)(1), *"In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged or discharges, or is suspected of having discharged discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports."*

The Discharger owns and operates the Facility subject to this Order. The monitoring reports required by this Order are necessary to determine compliance with this Order. The need for the monitoring reports is discussed in the Fact Sheet.

- E. Notification of Interested Parties.** The Central Valley Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDR's for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet.
- F. Consideration of Public Comment.** The Central Valley Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED that Order R5-2008-0077-01 is rescinded upon the effective date of this Order except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order. This action in no way prevents the Central Valley Water Board from taking enforcement action for past violations of the previous Order.

### III. DISCHARGE PROHIBITIONS

- A.** Discharge of wastewater from the Facility, as the Facility is specifically described in the Fact Sheet in section II.B, in a manner different from that described in this Order is prohibited.
- B.** The by-pass or overflow of wastes to surface waters is prohibited, except as allowed by Federal Standard Provisions I.G. and I.H. (Attachment D).
- C.** Neither the discharge nor its treatment shall create a nuisance as defined in section 13050 of the Water Code.
- D.** The Discharger shall not allow pollutant-free wastewater to be discharged into the treatment or disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that are essentially free of pollutants.

### IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

#### A. Effluent Limitations – Discharge Point 001

##### 1. Final Effluent Limitations – Discharge Point 001

The Discharger shall maintain compliance with the following effluent limitations at Discharge Point 001, with compliance measured at Monitoring Location EFF-001 (unless otherwise noted) as described in the Monitoring and Reporting Program, Attachment E:

- a. The Discharger shall maintain compliance with the effluent limitations specified in Table 4:

**Table 4. Effluent Limitations**

| Parameter                                | Units                | Effluent Limitations |                |               |                       |                       |
|------------------------------------------|----------------------|----------------------|----------------|---------------|-----------------------|-----------------------|
|                                          |                      | Average Monthly      | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Conventional Pollutants                  |                      |                      |                |               |                       |                       |
| Biochemical Oxygen Demand (5-day @ 20°C) | mg/L                 | 10                   | 15             | 20            | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 1,500                | 2,250          | 3,000         | --                    | --                    |
| pH                                       | standard units       | --                   | --             | --            | 6.5                   | 8.5                   |
| Total Suspended Solids                   | mg/L                 | 10                   | 15             | 20            | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 1,500                | 2,250          | 3,000         | --                    | --                    |

| Parameter                      | Units                | Effluent Limitations |                |               |                       |                       |
|--------------------------------|----------------------|----------------------|----------------|---------------|-----------------------|-----------------------|
|                                |                      | Average Monthly      | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Non-Conventional Pollutants    |                      |                      |                |               |                       |                       |
| Ammonia Nitrogen, Total (as N) | mg/L                 | 0.8                  | --             | 2.2           | --                    | --                    |
|                                | lbs/day <sup>1</sup> | 120                  | --             | 330           | --                    | --                    |
| Nitrate Plus Nitrite (as N)    | mg/L                 | 10                   | --             | --            | --                    | --                    |

<sup>1</sup> Based on an average dry weather flow of 18 million gallons per day (MGD).

- b. **Percent Removal:** The average monthly percent removal of 5-day biochemical oxygen demand (BOD<sub>5</sub>) and total suspended solids (TSS) shall not be less than 85 percent.
- c. **Acute Whole Effluent Toxicity.** Survival of aquatic organisms in 96-hour bioassays of undiluted waste shall be no less than:
  - i. 70%, minimum for any one bioassay; and
  - ii. 90%, median for any three consecutive bioassays.
- d. **Total Coliform Organisms.** Effluent total coliform organisms at Monitoring Location UVS-002 shall not exceed:
  - i. 2.2 most probable number (MPN) per 100 mL, as a 7-day median;
  - ii. 23 MPN/100 mL, more than once in any 30-day period; and
  - iii. 240 MPN/100 mL, at any time.
- e. **Average Dry Weather Flow.** The average dry weather discharge flow shall not exceed 18 MGD.
- f. **Mercury, total.** The total annual mass discharge of total mercury shall not exceed 5.5 pounds/year.

## 2. Interim Effluent Limitations-Not Applicable

### B. Land Discharge Specifications – Not Applicable

### C. Recycling Specifications – Not Applicable

## V. RECEIVING WATER LIMITATIONS

### A. Surface Water Limitations

The discharge shall not cause the following in Dry Creek:

1. **Bacteria.** The fecal coliform concentration, based on a minimum of not less than five samples for any 30-day period, to exceed a geometric mean of 200 MPN/100 mL, nor more than 10 percent of the total number of fecal coliform samples taken during any 30-day period to exceed 400 MPN/100 mL.
2. **Biostimulatory Substances.** Water to contain biostimulatory substances which promote aquatic growths in concentrations that cause nuisance or adversely affect beneficial uses.
3. **Chemical Constituents.** Chemical constituents to be present in concentrations that adversely affect beneficial uses.
4. **Color.** Discoloration that causes nuisance or adversely affects beneficial uses.

5. **Dissolved Oxygen:**
  - a. The monthly median of the mean daily dissolved oxygen concentration to fall below 85 percent of saturation in the main water mass;
  - b. The 95 percentile dissolved oxygen concentration to fall below 75 percent of saturation; nor
  - c. The dissolved oxygen concentration to be reduced below 7.0 mg/L at any time.
6. **Floating Material.** Floating material to be present in amounts that cause nuisance or adversely affect beneficial uses.
7. **Oil and Grease.** Oils, greases, waxes, or other materials to be present in concentrations that cause nuisance, result in a visible film or coating on the surface of the water or on objects in the water, or otherwise adversely affect beneficial uses.
8. **pH.** The pH to be depressed below 6.5 nor raised above 8.5.
9. **Pesticides:**
  - a. Pesticides to be present, individually or in combination, in concentrations that adversely affect beneficial uses;
  - b. Pesticides to be present in bottom sediments or aquatic life in concentrations that adversely affect beneficial uses;
  - c. Total identifiable persistent chlorinated hydrocarbon pesticides to be present in the water column at concentrations detectable within the accuracy of analytical methods approved by USEPA or the Executive Officer;
  - d. Pesticide concentrations to exceed those allowable by applicable antidegradation policies (see State Water Board Resolution No. 68-16 and 40 CFR 131.12);
  - e. Pesticide concentrations to exceed the lowest levels technically and economically achievable;
  - f. Pesticides to be present in concentration in excess of the maximum contaminant levels (MCLs) set forth in CCR, Title 22, division 4, chapter 15; nor
  - g. Thiobencarb to be present in excess of 1.0 µg/L.
10. **Radioactivity:**
  - a. Radionuclides to be present in concentrations that are harmful to human, plant, animal, or aquatic life nor that result in the accumulation of radionuclides in the food web to an extent that presents a hazard to human, plant, animal, or aquatic life.
  - b. Radionuclides to be present in excess of the MCLs specified in Table 64442 of section 64442 and Table 64443 of section 64443 of Title 22 of the California Code of Regulations.
11. **Suspended Sediments.** The suspended sediment load and suspended sediment discharge rate of surface waters to be altered in such a manner as to cause nuisance or adversely affect beneficial uses.
12. **Settleable Substances.** Substances to be present in concentrations that result in the deposition of material that causes nuisance or adversely affects beneficial uses.
13. **Suspended Material.** Suspended material to be present in concentrations that cause nuisance or adversely affect beneficial uses.

14. **Taste and Odors.** Taste- or odor-producing substances to be present in concentrations that impart undesirable tastes or odors to fish flesh or other edible products of aquatic origin, or that cause nuisance, or otherwise adversely affect beneficial uses.
15. **Temperature:**
  - a. The natural temperature to be increased by more than 5°F. Compliance to be determined based on the difference in temperature at Monitoring Locations RSW-001 and RSW-002.
  - b. The natural temperature at Monitoring Location RSW-002 to exceed a daily average temperature of 15.6°C (60°F) and a daily maximum temperature of 16.7°C (62°F) between 1 November through 31 March.
16. **Toxicity.** Toxic substances to be present, individually or in combination, in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.
17. **Turbidity:**
  - a. Shall not exceed 2 Nephelometric Turbidity Units (NTU) where natural turbidity is less than 1 NTU;
  - b. Shall not increase more than 1 NTU where natural turbidity is between 1 and 5 NTUs;
  - c. Shall not increase more than 20 percent where natural turbidity is between 5 and 50 NTUs;
  - d. Shall not increase more than 10 NTU where natural turbidity is between 50 and 100 NTUs; nor
  - e. Shall not increase more than 10 percent where natural turbidity is greater than 100 NTUs.

**B. Groundwater Limitations – Not Applicable**

**VI. PROVISIONS**

**A. Standard Provisions**

1. The Discharger shall comply with all Standard Provisions included in Attachment D.
2. The Discharger shall comply with the following provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply:
  - a. If the Discharger's wastewater treatment plant is publicly owned or subject to regulation by California Public Utilities Commission, it shall be supervised and operated by persons possessing certificates of appropriate grade according to Title 23, CCR, division 3, chapter 26.
  - b. After notice and opportunity for a hearing, this Order may be terminated or modified for cause, including, but not limited to:
    - i. violation of any term or condition contained in this Order;
    - ii. obtaining this Order by misrepresentation or by failing to disclose fully all relevant facts;
    - iii. a change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge; and

iv. a material change in the character, location, or volume of discharge.

The causes for modification include:

- *New regulations.* New regulations have been promulgated under section 405(d) of the CWA, or the standards or regulations on which the permit was based have been changed by promulgation of amended standards or regulations or by judicial decision after the permit was issued.
- *Land application plans.* When required by a permit condition to incorporate a land application plan for beneficial reuse of sewage sludge, to revise an existing land application plan, or to add a land application plan.
- *Change in sludge use or disposal practice.* Under 40 CFR 122.62(a)(1), a change in the Discharger's sludge use or disposal practice is a cause for modification of the permit. It is cause for revocation and reissuance if the Discharger requests or agrees.

The Central Valley Water Board may review and revise this Order at any time upon application of any affected person or the Central Valley Water Board's own motion.

- c. If a toxic effluent standard or prohibition (including any scheduled compliance specified in such effluent standard or prohibition) is established under section 307(a) of the CWA, or amendments thereto, for a toxic pollutant that is present in the discharge authorized herein, and such standard or prohibition is more stringent than any limitation upon such pollutant in this Order, the Central Valley Water Board will revise or modify this Order in accordance with such toxic effluent standard or prohibition.

The Discharger shall comply with effluent standards and prohibitions within the time provided in the regulations that establish those standards or prohibitions, even if this Order has not yet been modified.

- d. This Order shall be modified, or alternately revoked and reissued, to comply with any applicable effluent standard or limitation issued or approved under sections 301(b)(2)(C) and (D), 304(b)(2), and 307(a)(2) of the CWA, if the effluent standard or limitation so issued or approved:
- i. Contains different conditions or is otherwise more stringent than any effluent limitation in the Order; or
  - ii. Controls any pollutant limited in the Order.

The Order, as modified or reissued under this paragraph, shall also contain any other requirements of the CWA then applicable.

- e. The provisions of this Order are severable. If any provision of this Order is found invalid, the remainder of this Order shall not be affected.
- f. The Discharger shall take all reasonable steps to minimize any adverse effects to waters of the State or users of those waters resulting from any discharge or sludge use or disposal in violation of this Order. Reasonable steps shall include such accelerated or additional monitoring as necessary to determine the nature and impact of the non-complying discharge or sludge use or disposal.
- g. The Discharger shall ensure compliance with any existing or future pretreatment standard promulgated by USEPA under section 307 of the CWA, or amendment thereto, for any discharge to the municipal system.



- h. A copy of this Order shall be maintained at the discharge facility and be available at all times to operating personnel. Key operating personnel shall be familiar with its content.
- i. Safeguard to electric power failure:
  - i. The Discharger shall provide safeguards to assure that, should there be reduction, loss, or failure of electric power, the discharge shall comply with the terms and conditions of this Order.
  - ii. Upon written request by the Central Valley Water Board, the Discharger shall submit a written description of safeguards. Such safeguards may include alternate power sources, standby generators, retention capacity, operating procedures, or other means. A description of the safeguards provided shall include an analysis of the frequency, duration, and impact of power failures experienced over the past 5 years on effluent quality and on the capability of the Discharger to comply with the terms and conditions of the Order. The adequacy of the safeguards is subject to the approval of the Central Valley Water Board.
  - iii. Should the treatment works not include safeguards against reduction, loss, or failure of electric power, or should the Central Valley Water Board not approve the existing safeguards, the Discharger shall, within 90 days of having been advised in writing by the Central Valley Water Board that the existing safeguards are inadequate, provide to the Central Valley Water Board and USEPA a schedule of compliance for providing safeguards such that in the event of reduction, loss, or failure of electric power, the Discharger shall comply with the terms and conditions of this Order. The schedule of compliance shall, upon approval of the Central Valley Water Board, become a condition of this Order.
- j. The Discharger, upon written request of the Central Valley Water Board, shall file with the Board a technical report on its preventive (failsafe) and contingency (cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. This report may be combined with that required under the Central Valley Water Board Standard Provision contained in section VI.A.2.i of this Order.

The technical report shall:

- i. Identify the possible sources of spills, leaks, untreated waste by-pass, and contaminated drainage. Loading and storage areas, power outage, waste treatment unit outage, and failure of process equipment, tanks and pipes should be considered.
- ii. Evaluate the effectiveness of present facilities and procedures and state when they became operational.
- iii. Predict the effectiveness of the proposed facilities and procedures and provide an implementation schedule containing interim and final dates when they will be constructed, implemented, or operational.

The Central Valley Water Board, after review of the technical report, may establish conditions which it deems necessary to control accidental discharges and to minimize the effects of such events. Such conditions shall be incorporated as part of this Order, upon notice to the Discharger.

- k. A publicly owned treatment works whose waste flow has been increasing, or is projected to increase, shall estimate when flows will reach hydraulic and treatment capacities of its treatment and disposal facilities. The projections shall be made in January, based on the last 3 years' average dry weather flows, peak wet weather flows and total annual flows, as appropriate. When any projection shows that capacity of any part of the facilities may be exceeded in 4 years, the Discharger shall notify the Central Valley Water Board by 31 January. A copy of the notification shall be sent to appropriate local elected officials, local permitting agencies and the press. Within 120 days of the notification, the Discharger shall submit a technical report showing how it will prevent flow volumes from exceeding capacity or how it will increase capacity to handle the larger flows. The Central Valley Water Board may extend the time for submitting the report.
- l. The Discharger shall submit technical reports as directed by the Executive Officer. All technical reports required herein that involve planning, investigation, evaluation, or design, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code, sections 6735, 7835, and 7835.1. To demonstrate compliance with Title 16, CCR, sections 415 and 3065, all technical reports must contain a statement of the qualifications of the responsible registered professional(s). As required by these laws, completed technical reports must bear the signature(s) and seal(s) of the registered professional(s) in a manner such that all work can be clearly attributed to the professional responsible for the work.
- m. The Central Valley Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13385, 13386, and 13387.
- n. For publicly owned treatment works, prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a permanent decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. (Water Code section 1211).
- o. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
- p. In the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, maximum daily effluent limitation, hourly average effluent limitation, or receiving water limitation of this Order, the Discharger shall notify the Central Valley Water Board by telephone (916) 464-3291 within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing within five days, unless the Central Valley Water Board waives confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and prevent recurrence including, where applicable, a schedule of implementation. Other noncompliance requires written notification as above at the time of the normal monitoring report.

**B. Monitoring and Reporting Program (MRP) Requirements**

The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E.

**C. Special Provisions**

**1. Reopener Provisions**

- a. Conditions that necessitate a major modification of a permit are described in 40 CFR 122.62, including, but not limited to:
  - i. If new or amended applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA, or amendments thereto, this permit may be reopened and modified in accordance with the new or amended standards.
  - ii. When new information, that was not available at the time of permit issuance, would have justified different permit conditions at the time of issuance.
- b. **Mercury.** If mercury is found to be causing toxicity based on acute or chronic toxicity test results, or if a TMDL program is adopted, this Order shall be reopened and the mass effluent limitation modified (higher or lower) or an effluent concentration limitation imposed. If the Central Valley Water Board determines that a mercury offset program is feasible for Dischargers subject to a NPDES permit, then this Order may be reopened to reevaluate the mercury mass loading limitation(s) and the need for a mercury offset program for the Discharger.
- c. **Whole Effluent Toxicity.** As a result of a Toxicity Reduction Evaluation (TRE), this Order may be reopened to include a numeric chronic toxicity limitation, a new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE. Additionally, if the State Water Board revises the SIP's toxicity control provisions that would require the establishment of numeric chronic toxicity effluent limitations, this Order may be reopened to include a numeric chronic toxicity effluent limitation based on the new provisions.
- d. **Water Effects Ratios (WER) and Metal Translators.** A default WER of 1.0 has been used in this Order for calculating criteria for applicable inorganic constituents. In addition, default dissolved-to-total metal translators have been used to convert water quality objectives from dissolved to total recoverable. If the Discharger performs studies to determine site-specific WERs and/or site-specific dissolved-to-total metal translators, this Order may be reopened to modify the effluent limitations for the applicable constituents.

**2. Special Studies, Technical Reports and Additional Monitoring Requirements**

- a. **Chronic Whole Effluent Toxicity.** For compliance with the Basin Plan's narrative toxicity objective, this Order requires the Discharger to conduct chronic whole effluent toxicity (WET) testing, as specified in MRP section V. Furthermore, this Provision requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity. If the discharge exceeds the numeric toxicity monitoring trigger during accelerated monitoring established in this Provision, the Discharger is required to initiate a TRE in accordance with an approved TRE Work Plan, and take actions to mitigate the impact of the discharge and prevent recurrence of toxicity. A TRE is a site-specific study conducted in a stepwise process to identify the source(s) of toxicity and the effective control measures for effluent toxicity. TREs are designed to identify the causative agents and sources of whole effluent toxicity, evaluate the effectiveness of the toxicity

control options, and confirm the reduction in effluent toxicity. This Provision includes procedures for accelerated chronic toxicity monitoring and TRE initiation.

- i. **Accelerated Monitoring and TRE Initiation.** When the numeric toxicity monitoring trigger is exceeded during regular chronic toxicity monitoring, and the testing meets all test acceptability criteria, the Discharger shall initiate accelerated monitoring as required in the Accelerated Monitoring Specifications. The Discharger shall initiate a TRE to address effluent toxicity if any WET testing results exceed the numeric toxicity monitoring trigger during accelerated monitoring.
- ii. **Numeric Toxicity Monitoring Trigger.** The numeric toxicity monitoring trigger to initiate a TRE is  $>1 \text{ TUc}$  (where  $\text{TUc} = 100/\text{NOEC}$ ). The monitoring trigger is not an effluent limitation; it is the toxicity threshold at which the Discharger is required to begin accelerated monitoring and initiate a TRE.
- iii. **Accelerated Monitoring Specifications.** If the numeric toxicity monitoring trigger is exceeded during regular chronic toxicity testing, the Discharger shall initiate accelerated monitoring within 14-days of notification by the laboratory of the exceedance. Accelerated monitoring shall consist of four chronic toxicity tests conducted once every two weeks using the species that exhibited toxicity. The following protocol shall be used for accelerated monitoring and TRE initiation:
  - (a) If the results of four consecutive accelerated monitoring tests do not exceed the monitoring trigger, the Discharger may cease accelerated monitoring and resume regular chronic toxicity monitoring. However, notwithstanding the accelerated monitoring results, if there is adequate evidence of a pattern of effluent toxicity, the Executive Officer may require that the Discharger initiate a TRE.
  - (b) If the source(s) of the toxicity is easily identified (e.g., temporary plant upset), the Discharger shall make necessary corrections to the facility and shall continue accelerated monitoring until four consecutive accelerated tests do not exceed the monitoring trigger. Upon confirmation that the effluent toxicity has been removed, the Discharger may cease accelerated monitoring and resume regular chronic toxicity monitoring.
  - (c) If the result of any accelerated toxicity test exceeds the monitoring trigger, the Discharger shall cease accelerated monitoring and begin a TRE to investigate the cause(s) of, and identify corrective actions to reduce or eliminate effluent toxicity. Within thirty (30) days of notification by the laboratory of any test result exceeding the monitoring trigger during accelerated monitoring, the Discharger shall submit a TRE Action Plan to the Central Valley Water Board including, at minimum:
    - (1) Specific actions the Discharger will take to investigate and identify the cause(s) of toxicity, including a TRE WET monitoring schedule;
    - (2) Specific actions the Discharger will take to mitigate the impact of the discharge and prevent the recurrence of toxicity; and
    - (3) A schedule for these actions.

Within sixty (60) days of notification by the laboratory of the test results, the Discharger shall submit to the Central Valley Water Board a TRE Workplan for approval by the Executive Officer. The TRE Workplan shall

outline the procedures for identifying the source(s) of, and reducing or eliminating effluent toxicity. The TRE Workplan must be developed in accordance with USEPA guidance<sup>1</sup>.

**3. Best Management Practices and Pollution Prevention**

- a. **Salinity Evaluation and Minimization Plan.** The Discharger shall continue to implement a salinity evaluation and minimization plan to identify and address sources of salinity from the Facility. The Discharger shall provide annual reports demonstrating reasonable progress in the reduction of salinity in its discharge to Dry Creek in accordance with section X.D.1 of the Monitoring and Reporting Program (Attachment E).
- b. **Pollution Prevention Plan (PPP) for Mercury.** The Discharger shall continue to implement a PPP for mercury. Progress reports shall be submitted in accordance with section X.D.1 of the Monitoring and Reporting Program (Attachment E).

**4. Construction, Operation and Maintenance Specifications**

- a. **Filtration System Operating Specifications.** To ensure the filtration system is operating properly to facilitate adequate disinfection of the wastewater, the turbidity of the filter effluent measured at Monitoring Location FIL-001 shall not exceed:
  - i. 2 NTU as a daily average;
  - ii. 5 NTU more than 5 percent of the time within a 24-hour period; and
  - iii. 10 NTU, at any time.
- b. **Ultraviolet Light (UV) Disinfection System Operating Specifications.** The UV disinfection system must be operated in accordance with an operations and maintenance program that assures adequate disinfection, and shall meet the following minimum specifications to provide virus inactivation equivalent to Title 22 Disinfected Tertiary Recycled Water:
  - i. **UV Dose.** The minimum hourly average UV dose in the UV reactor shall be 100 millijoules per square centimeter (mJ/cm<sup>2</sup>).
  - ii. **UV Transmittance.** The minimum hourly average UV transmittance (at 254 nanometers) in the wastewater measured at Monitoring Location UVS-001 shall not fall below 55 percent.
  - iii. The lamp sleeves and cleaning system components must be visually inspected per the manufacturer's operations manual for physical wear (scoring, solarization, seal leaks, cleaning fluid levels, etc.) and to check the efficacy of the cleaning system.
  - iv. The lamp sleeves must be cleaned periodically as necessary to meet the UV dose requirements.
  - v. Lamps must be replaced per the manufacturer's operations manual, or sooner, if there are indications the lamps are failing to provide adequate disinfection. Lamp age and lamp replacement records must be maintained.
- c. **Emergency Storage Basin Operating Requirements**
  - i. The emergency storage basins shall be used only to prevent overwhelming of the treatment process (up to 100-year flood protection), to store partially

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<sup>1</sup> See the Fact Sheet (Attachment F, section VI.B.2.a.) for a list of USEPA guidance documents that must be considered in development of the TRE Workplan.

treated wastewater, or to prevent plant upsets by diverting influent that would be harmful to the treatment process.

- ii. Objectionable odors originating from the emergency storage basins shall not be perceivable beyond the limits of the wastewater treatment and disposal areas (or property owned by the Discharger).
- iii. As a means of discerning compliance with item ii above, the dissolved oxygen content in the upper zone (1 foot) of wastewater in the emergency storage basins shall not be less than 1.0 mg/L.
- iv. The emergency storage basins containing water for more than 7 consecutive days shall not have a pH less than 6.5 or greater than 8.5.
- v. The emergency storage basins shall be managed to prevent breeding of mosquitoes. In particular:
  - (a) An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface;
  - (b) Weeds shall be minimized; and
  - (c) Vegetation, debris, and dead algae shall not accumulate on the water surface.
- vi. Public contact with wastewater shall be precluded through means as fences, signs, and other acceptable alternatives.
- vii. Freeboard shall never be less than 2 feet (measured vertically to the lowest point of overflow). In no case shall wind/wave action cause overtopping of levees (freeboard of more than 2 feet may be necessary).
- viii. Wastewater contained in the emergency storage basins must be returned to the treatment facility to receive treatment in accordance with the Effluent Limitations and Discharge Specifications in section IV.A of this Order prior to discharge to Dry Creek.
- ix. The Discharger shall keep a log related to the use of the emergency storage basins. In particular, the Discharger shall record the following when any type of wastewater is directed to any basin:
  - (a) The date(s) when the wastewater is directed to each basin;
  - (b) The type(s) of wastewater (e.g., untreated due to plant upset, tertiary treated) directed to each basin;
  - (c) The total volume of wastewater directed to each basin;
  - (d) The duration of time wastewater is collected in each basin; prior to redirection back to the wastewater treatment plant; and
  - (e) The date when all wastewater in each basin has been redirected to the wastewater treatment plant.

The emergency storage basin log shall be submitted with the monthly self-monitoring reports required in Section X.B of the Monitoring and Reporting Program (Attachment E).

## 5. Special Provisions for Municipal Facilities (POTWs Only)

### a. Pretreatment Requirements

- i. The Discharger shall be responsible and liable for the performance of all Control Authority pretreatment requirements contained in 40 CFR Part 403, including any subsequent regulatory revisions to 40 CFR Part 403. For violations of pretreatment requirements, the Discharger shall be subject to enforcement actions, penalties, fines, and other remedies by USEPA or other appropriate parties, as provided in the CWA. USEPA may initiate enforcement action against a nondomestic user for noncompliance with applicable standards and requirements as provided in the CWA.
- ii. The Discharger shall enforce the requirements promulgated under sections 307(b), 307(c), 307(d), and 402(b) of the CWA with timely, appropriate and effective enforcement actions. The Discharger shall cause all nondomestic users subject to federal categorical standards to achieve compliance no later than the date specified in those requirements or, in the case of a new nondomestic user, upon commencement of the discharge.
- iii. The Discharger shall perform the pretreatment functions as required in 40 CFR Part 403 including, but not limited to:
  - (a) Implement the necessary legal authorities as provided in 40 CFR 403.8(f)(1);
  - (b) Enforce the pretreatment requirements under 40 CFR 403.5 and 403.6;
  - (c) Implement the programmatic functions as provided in 40 CFR 403.8(f)(2); and
  - (d) Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR 403.8(f)(3).

- b. **Sludge/Biosolids Treatment or Discharge Specifications.** Sludge in this document means the solid, semisolid, and liquid residues removed during primary, secondary, or advanced wastewater treatment processes. Solid waste refers to grit and screening material generated during preliminary treatment. Residual sludge means sludge that will not be subject to further treatment at the wastewater treatment plant. Biosolids refer to sludge that has been treated and tested and shown to be capable of being beneficially and legally used pursuant to federal and state regulations as a soil amendment for agricultural, silvicultural, horticultural, and land reclamation activities as specified under 40 CFR Part 503.

- i. Collected screenings, residual sludge, biosolids, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste, as set forth in Title 27, CCR, division 2, subdivision 1, section 20005, et seq. Removal for further treatment, storage, disposal, or reuse at sites (e.g., landfill, composting sites, soil amendment sites) that are operated in accordance with valid waste discharge requirements issued by a Regional Water Board will satisfy these specifications.

Sludge and solid waste shall be removed from screens, sumps, ponds, clarifiers, etc. as needed to ensure optimal plant performance.

The treatment of sludge generated at the Facility shall be confined to the Facility property and conducted in a manner that precludes infiltration of waste

constituents into soils in a mass or concentration that will violate groundwater limitations in section V.B. of this Order. In addition, the storage of residual sludge, solid waste, and biosolids on Facility property shall be temporary and controlled, and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or concentration that will violate groundwater limitations included in section V.B. of this Order.

- ii. The use, disposal, storage, and transportation of biosolids shall comply with existing federal and state laws and regulations, including permitting requirements and technical standards included in 40 CFR Part 503. If the State Water Board and the Central Valley Water Board are given the authority to implement regulations contained in 40 CFR Part 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR Part 503 whether or not they have been incorporated into this Order.
  - iii. The Discharger shall comply with Section IX.A. Biosolids of the Monitoring and Reporting Program, Attachment E.
  - iv. Any proposed change in biosolids use or disposal practice from a previously approved practice shall be reported to the Executive Officer and USEPA Regional Administrator at least **90 days** in advance of the change.
  - v. **By 180 days of the permit effective date**, the Discharger shall submit a biosolids use or disposal plan to the Central Valley Water Board. The plan shall describe at a minimum:
    - (a) Sources and amounts of biosolids generated annually.
    - (b) Location(s) of on-site storage and description of the containment area.
    - (c) Plans for ultimate disposal. For landfill disposal, include the present classification of the landfill and the name and location of the landfill.
  - vi. The Discharger is encouraged to comply with the "Manual of Good Practice for Agricultural Land Application of Biosolids" developed by the California Water Environment Association.
  - vii. Use of biosolids as a soil amendment shall comply with valid WDRs issued by the State or Regional Water Boards. In most cases, this means the WDRs contained in State Water Board Water Quality Order No. 2004-0012-DWQ, *General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities* (Biosolids General Order). For a biosolids use project to be covered by the Biosolids General Order, the Discharger must file a complete Notice of Intent and receive a Notice of Applicability for each project, or be under contract with a biosolids disposal contractor regulated by valid WDRs.
- c. **Collection System.** On 2 May 2006, the State Water Board adopted State Water Board Order 2006-0003-DWQ, Statewide General WDRs for Sanitary Sewer Systems. The Discharger shall be subject to the requirements of Order 2006-0003-DWQ and any future revisions thereto. Order 2006-0003-DWQ requires that all public agencies that currently own or operate sanitary sewer systems apply for coverage under the general WDRs. The Discharger has applied for and has been approved for coverage under Order 2006-0003-DWQ for operation of its wastewater collection system.



## 6. Other Special Provisions

- a. Wastewater shall be oxidized, coagulated (as needed), filtered, and adequately disinfected pursuant to the Department of Public Health (DPH; formerly the Department of Health Services) reclamation criteria, CCR, Title 22, division 4, chapter 3, (Title 22), or equivalent.
- b. By **1 October 2014**, the Discharger shall submit a *Groundwater Monitoring Well Destruction Workplan* for the wells that are not used at the Facility. The workplan shall describe the proposed abandonment procedures, which shall comply with *California Well Standards Bulletin 74-90* (June 1991); *State of California Bulletin 94-81* (December 1981); and any more stringent standards adopted by the state or county pursuant to Water Code section 13801. At a minimum, the workplan shall specifically address the monitoring wells listed in Table E-1 of the Monitoring and Reporting Program, Attachment E. For each well that will not be destroyed, provide an explanation of the need to maintain the well for monitoring purposes.
- c. By **1 October 2015**, a *Well Destruction Report* shall be submitted to the Central Valley Water Board that describes in detail the methods used to abandon each well and includes copies of the well abandonment permits issued by the City of Roseville Environmental Utilities Department.

## 7. Compliance Schedules – Not Applicable

## VII. COMPLIANCE DETERMINATION

- A. **BOD<sub>5</sub> and TSS Effluent Limitations (Sections IV.A.1.a and IV.A.1.b).** Compliance with the final effluent limitations for BOD<sub>5</sub> and TSS required in Limitations and Discharge Requirements section IV.A.1.a shall be ascertained by 24-hour composite samples. Compliance with effluent limitations required in Limitations and Discharge Requirements section IV.A.1.b for percent removal shall be calculated using the arithmetic mean of BOD<sub>5</sub> and TSS in effluent samples collected over a monthly period as a percentage of the arithmetic mean of the values for influent samples collected at approximately the same times during the same period.
- B. **Total Mercury Mass Loading Effluent Limitations (Section IV.A.1.f).** The procedures for calculating mass loadings are as follows:
  1. The total pollutant mass load for each individual calendar month shall be determined using an average of all concentration data collected that month and the corresponding total monthly flow. All Facility effluent monitoring data collected under the monitoring and reporting program, pretreatment program, and any special studies shall be used for these calculations. The total annual mass loading shall be the sum of the individual calendar months.
  2. In calculating compliance, the Discharger shall count all non-detect measures at one-half of the detection level. If compliance with the effluent limitation is not attained due to the non-detect contribution, the Discharger shall improve and implement available analytical capabilities and compliance shall be evaluated with consideration of the detection limits.
- C. **Average Dry Weather Flow Effluent Limitations (Section IV.A.1.e).** The average dry weather discharge flow represents the daily average flow when groundwater is at or near normal and runoff is not occurring. Compliance with the average dry weather flow effluent limitations will be determined annually based on the average daily flow over three consecutive dry weather months (e.g., July, August, and September).

**D. Total Coliform Organisms Effluent Limitations (Section IV.A.1.d).** For each day that an effluent sample is collected and analyzed for total coliform organisms, the 7-day median shall be determined by calculating the median concentration of total coliform bacteria in the effluent utilizing the bacteriological results of the last 7 days. For example, if a sample is collected on a Wednesday, the result from that sampling event and all results from the previous 6 days (i.e., Tuesday, Monday, Sunday, Saturday, Friday, and Thursday) are used to calculate the 7-day median. If the 7-day median of total coliform organisms exceeds a most probable number (MPN) of 2.2 per 100 milliliters, the Discharger will be considered out of compliance. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are <2, in which case the median value shall be the lower of the two data points.

**E. Mass Effluent Limitations.** The mass effluent limitations contained in the Final Effluent Limitations IV.A.1.a are based on the permitted average dry weather flow and calculated as follows:.

$$\text{Mass (lbs/day)} = \text{Flow (MGD)} \times \text{Concentration (mg/L)} \times 8.34 \text{ (conversion factor)}$$

If the effluent flow exceeds the permitted average dry weather flow during wet-weather seasons, the effluent mass limitations contained in Final Effluent Limitations IV.A.1.a shall not apply. If the effluent flow is below the permitted average dry weather flow during wet-weather seasons, the effluent mass limitations do apply.

**F. Priority Pollutant Effluent Limitations.** Compliance with effluent limitations for priority pollutants shall be determined in accordance with Section 2.4.5 of the SIP, as follows:

1. Dischargers shall be deemed out of compliance with an effluent limitation, if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).
2. Dischargers shall be required to conduct a Pollutant Minimization Program (PMP) in accordance with section 2.4.5.1 of the SIP when there is evidence that the priority pollutant is present in the effluent above an effluent limitation and either:
  - a. A sample result is reported as detected, but not quantified (DNQ) and the effluent limitation is less than the RL; or
  - b. A sample result is reported as non-detect (ND) and the effluent limitation is less than the method detection limit (MDL).
3. When determining compliance with an average monthly effluent limitation (AMEL) and more than one sample result is available in a month, the discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of DNQ or ND. In those cases, the discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
  - a. The data set shall be ranked from low to high, reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
  - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

4. If a sample result, or the arithmetic mean or median of multiple sample results, is below the RL, and there is evidence that the priority pollutant is present in the effluent above an effluent limitation and the discharger conducts a PMP (as described in section 2.4.5.1), the discharger shall not be deemed out of compliance.

## ATTACHMENT A – DEFINITIONS

### **Arithmetic Mean (m)**

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

Arithmetic mean =  $m = Sx / n$       where: Sx is the sum of the measured ambient water concentrations, and n is the number of samples.

### **Average Monthly Effluent Limitation (AMEL)**

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

### **Average Weekly Effluent Limitation (AWEL)**

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

### **Bioaccumulative**

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

### **Carcinogenic**

Pollutants are substances that are known to cause cancer in living organisms.

### **Coefficient of Variation (CV)**

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

### **Daily Discharge**

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

### **Detected, but Not Quantified (DNQ)**

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

### **Dilution Credit**

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

### **Effluent Concentration Allowance (ECA)**

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in U.S. EPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

### **Enclosed Bays**

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

### **Estimated Chemical Concentration**

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

### **Estuaries**

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

### **Inland Surface Waters**

All surface waters of the state that do not include the ocean, enclosed bays, or estuaries.

### **Instantaneous Maximum Effluent Limitation**

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

### **Instantaneous Minimum Effluent Limitation**

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

### **Maximum Daily Effluent Limitation (MDEL)**

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

### **Median**

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of

measurements (n) is odd, then the median =  $X_{(n+1)/2}$ . If n is even, then the median =  $(X_{n/2} + X_{(n/2)+1})/2$  (i.e., the midpoint between the  $n/2$  and  $n/2+1$ ).

#### **Method Detection Limit (MDL)**

MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 C.F.R. part 136, Attachment B, revised as of July 3, 1999.

#### **Minimum Level (ML)**

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

#### **Mixing Zone**

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

#### **Not Detected (ND)**

Sample results which are less than the laboratory's MDL.

#### **Ocean Waters**

The territorial marine waters of the State as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. Discharges to ocean waters are regulated in accordance with the State Water Board's California Ocean Plan.

#### **Persistent Pollutants**

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

#### **Pollutant Minimization Program (PMP)**

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Central Valley Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

#### **Pollution Prevention**

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board) or Central Valley Water Board.

**Satellite Collection System**

The portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

**Source of Drinking Water**

Any water designated as municipal or domestic supply (MUN) in a Central Valley Water Board Basin Plan.

**Standard Deviation (s)**

Standard Deviation is a measure of variability that is calculated as follows:

$$s = \left( \frac{\sum (x - m)^2}{(n - 1)} \right)^{0.5}$$

where:

x is the observed value;

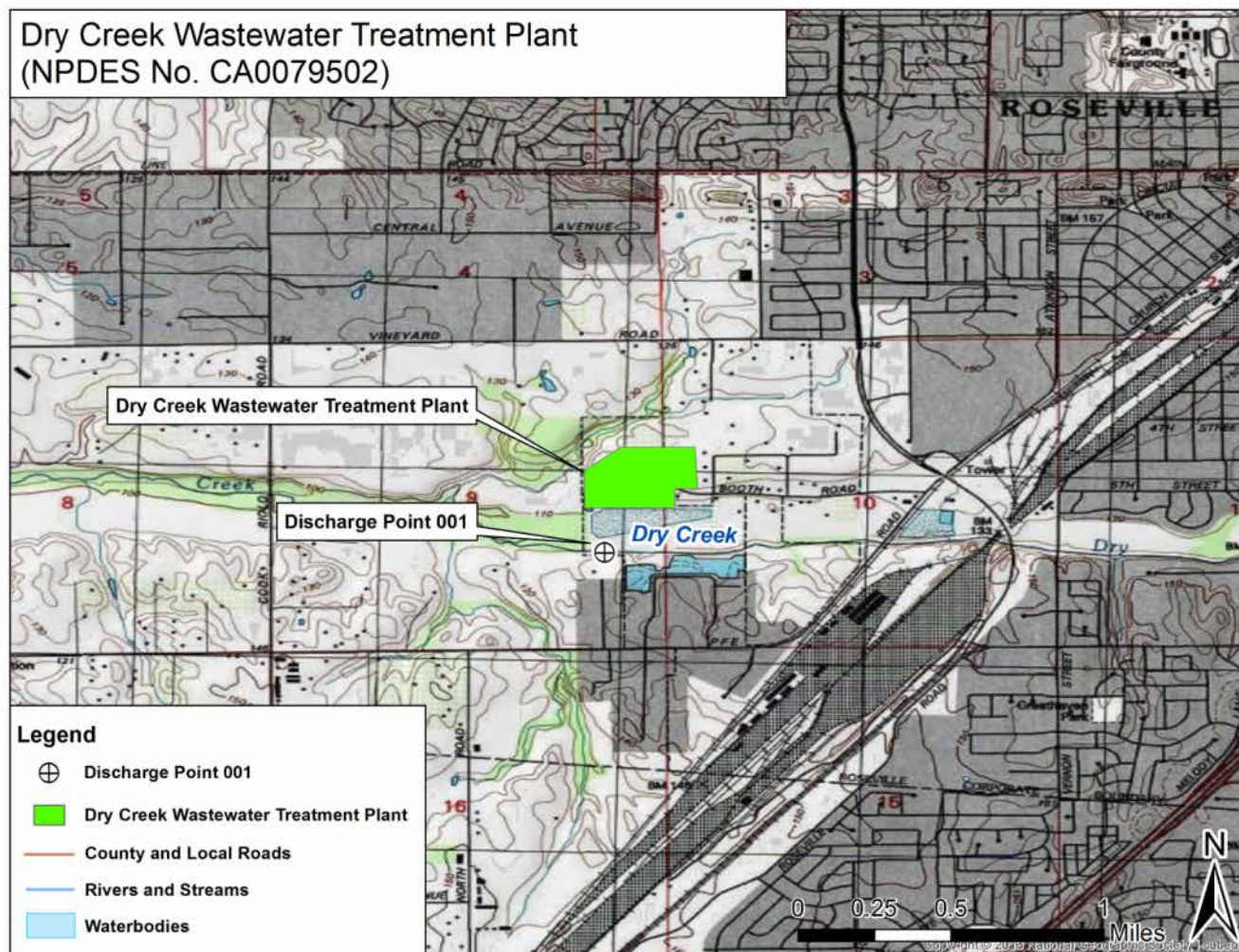
m is the arithmetic mean of the observed values; and

n is the number of samples.

**Toxicity Reduction Evaluation (TRE)**

TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

ATTACHMENT B – MAP





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## **ATTACHMENT D – STANDARD PROVISIONS**

### **I. STANDARD PROVISIONS – PERMIT COMPLIANCE**

#### **A. Duty to Comply**

1. The Discharger must comply with all of the conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification; or denial of a permit renewal application. (40 C.F.R. § 122.41(a).)
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 C.F.R. § 122.41(a)(1).)

#### **B. Need to Halt or Reduce Activity Not a Defense**

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 C.F.R. § 122.41(c).)

#### **C. Duty to Mitigate**

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 C.F.R. § 122.41(d).)

#### **D. Proper Operation and Maintenance**

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 C.F.R. § 122.41(e).)

#### **E. Property Rights**

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 C.F.R. § 122.41(g).)
2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 C.F.R. § 122.5(c).)

#### **F. Inspection and Entry**

The Discharger shall allow the Central Valley Water Board, State Water Board, U.S. EPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (40 C.F.R. § 122.41(i); Wat. Code, § 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (40 C.F.R. § 122.41(i)(1));

2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (40 C.F.R. § 122.41(i)(2));
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (40 C.F.R. § 122.41(i)(3)); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (40 C.F.R. § 122.41(i)(4).)

#### **G. Bypass**

1. Definitions
  - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 C.F.R. § 122.41(m)(1)(i).)
  - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 C.F.R. § 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 C.F.R. § 122.41(m)(2).)
3. Prohibition of bypass. Bypass is prohibited, and the Central Valley Water Board may take enforcement action against a Discharger for bypass, unless (40 C.F.R. § 122.41(m)(4)(i)):
  - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 C.F.R. § 122.41(m)(4)(i)(A));
  - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 C.F.R. § 122.41(m)(4)(i)(B)); and
  - c. The Discharger submitted notice to the Central Valley Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 C.F.R. § 122.41(m)(4)(i)(C).)
4. The Central Valley Water Board may approve an anticipated bypass, after considering its adverse effects, if the Central Valley Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 C.F.R. § 122.41(m)(4)(ii).)
5. Notice
  - a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. (40 C.F.R. § 122.41(m)(3)(i).)

- b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). (40 C.F.R. § 122.41(m)(3)(ii).)

#### **H. Upset**

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 C.F.R. § 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 C.F.R. § 122.41(n)(2).)
2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 C.F.R. § 122.41(n)(3)):
  - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 C.F.R. § 122.41(n)(3)(i));
  - b. The permitted facility was, at the time, being properly operated (40 C.F.R. § 122.41(n)(3)(ii));
  - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 C.F.R. § 122.41(n)(3)(iii)); and
  - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 C.F.R. § 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 C.F.R. § 122.41(n)(4).)

## **II. STANDARD PROVISIONS – PERMIT ACTION**

### **A. General**

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 C.F.R. § 122.41(f).)

### **B. Duty to Reapply**

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 C.F.R. § 122.41(b).)

### **C. Transfers**

This Order is not transferable to any person except after notice to the Central Valley Water Board. The Central Valley Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other

requirements as may be necessary under the CWA and the Water Code. (40 C.F.R. § 122.41(l)(3); § 122.61.)

### **III. STANDARD PROVISIONS – MONITORING**

- A.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 C.F.R. § 122.41(j)(1).)
- B.** Monitoring results must be conducted according to test procedures under 40 C.F.R. part 136 or, in the case of sludge use or disposal, approved under 40 C.F.R. part 136 unless otherwise specified in 40 C.F.R. part 503 unless other test procedures have been specified in this Order. (40 C.F.R. § 122.41(j)(4); § 122.44(i)(1)(iv).)

### **IV. STANDARD PROVISIONS – RECORDS**

- A.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 C.F.R. part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Central Valley Water Board Executive Officer at any time. (40 C.F.R. § 122.41(j)(2).)
- B.** Records of monitoring information shall include:
  - 1. The date, exact place, and time of sampling or measurements (40 C.F.R. § 122.41(j)(3)(i));
  - 2. The individual(s) who performed the sampling or measurements (40 C.F.R. § 122.41(j)(3)(ii));
  - 3. The date(s) analyses were performed (40 C.F.R. § 122.41(j)(3)(iii));
  - 4. The individual(s) who performed the analyses (40 C.F.R. § 122.41(j)(3)(iv));
  - 5. The analytical techniques or methods used (40 C.F.R. § 122.41(j)(3)(v)); and
  - 6. The results of such analyses. (40 C.F.R. § 122.41(j)(3)(vi).)
- C.** Claims of confidentiality for the following information will be denied (40 C.F.R. § 122.7(b)):
  - 1. The name and address of any permit applicant or Discharger (40 C.F.R. § 122.7(b)(1)); and
  - 2. Permit applications and attachments, permits and effluent data. (40 C.F.R. § 122.7(b)(2).)

### **V. STANDARD PROVISIONS – REPORTING**

#### **A. Duty to Provide Information**

The Discharger shall furnish to the Central Valley Water Board, State Water Board, or U.S. EPA within a reasonable time, any information which the Central Valley Water Board, State Water Board, or U.S. EPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Central Valley Water Board, State Water Board, or U.S. EPA copies of records required to be kept by this Order. (40 C.F.R. § 122.41(h); Wat. Code, § 13267.)

## **B. Signatory and Certification Requirements**

1. All applications, reports, or information submitted to the Central Valley Water Board, State Water Board, and/or U.S. EPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 C.F.R. § 122.41(k).)
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of U.S. EPA). (40 C.F.R. § 122.22(a)(3).)
3. All reports required by this Order and other information requested by the Central Valley Water Board, State Water Board, or U.S. EPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
  - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 C.F.R. § 122.22(b)(1));
  - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 C.F.R. § 122.22(b)(2)); and
  - c. The written authorization is submitted to the Central Valley Water Board and State Water Board. (40 C.F.R. § 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Central Valley Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 C.F.R. § 122.22(c).)
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 C.F.R. § 122.22(d).)

## **C. Monitoring Reports**

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 C.F.R. § 122.41(l)(4).)

2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Central Valley Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. (40 C.F.R. § 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 C.F.R. part 136, or another method required for an industry-specific waste stream under 40 C.F.R. subchapters N or O, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Central Valley Water Board. (40 C.F.R. § 122.41(l)(4)(ii).)
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 C.F.R. § 122.41(l)(4)(iii).)

**D. Compliance Schedules**

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 C.F.R. § 122.41(l)(5).)

**E. Twenty-Four Hour Reporting**

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. (40 C.F.R. § 122.41(l)(6)(i).)
2. The following shall be included as information that must be reported within 24 hours under this paragraph (40 C.F.R. § 122.41(l)(6)(ii)):
  - a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(A).)
  - b. Any upset that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(B).)
3. The Central Valley Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 C.F.R. § 122.41(l)(6)(iii).)

**F. Planned Changes**

The Discharger shall give notice to the Central Valley Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 C.F.R. § 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 C.F.R. § 122.41(l)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 C.F.R. § 122.41(l)(1)(ii).)

3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan. (40 C.F.R. § 122.41(l)(1)(iii).)

**G. Anticipated Noncompliance**

The Discharger shall give advance notice to the Central Valley Water Board or State Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 C.F.R. § 122.41(l)(2).)

**H. Other Noncompliance**

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. (40 C.F.R. § 122.41(l)(7).)

**I. Other Information**

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Central Valley Water Board, State Water Board, or U.S. EPA, the Discharger shall promptly submit such facts or information. (40 C.F.R. § 122.41(l)(8).)

**VI. STANDARD PROVISIONS – ENFORCEMENT**

- A. The Central Valley Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13385, 13386, and 13387.

**VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS**

**A. Publicly-Owned Treatment Works (POTWs)**

All POTWs shall provide adequate notice to the Central Valley Water Board of the following (40 C.F.R. § 122.42(b)):

1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 C.F.R. § 122.42(b)(1)); and
2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 C.F.R. § 122.42(b)(2).)
3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 C.F.R. § 122.42(b)(3).)



## ATTACHMENT E – MONITORING AND REPORTING PROGRAM

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## **ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)**

The Code of Federal Regulations (40 C.F.R. § 122.48) requires that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Central Valley Water Board to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements that implement federal and California regulations.

### **I. GENERAL MONITORING PROVISIONS**

- A.** Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge. All samples shall be taken at the monitoring locations specified below and, unless otherwise specified, before the monitored flow joins or is diluted by any other waste stream, body of water, or substance. Monitoring locations shall not be changed without notification to and the approval of the Central Valley Water Board.
- B.** Effluent samples shall be taken downstream of the last addition of wastes to the treatment or discharge works where a representative sample may be obtained prior to mixing with the receiving waters. Samples shall be collected at such a point and in such a manner to ensure a representative sample of the discharge.
- C.** Chemical, bacteriological, and bioassay analyses of any material required by this Order shall be conducted by a laboratory certified for such analyses by the Department of Public Health (DPH). Laboratories that perform sample analyses must be identified in all monitoring reports submitted to the Central Valley Water Board. In the event a certified laboratory is not available to the Discharger for any onsite field measurements such as pH, dissolved oxygen (DO), turbidity, temperature, and residual chlorine, such analyses performed by a noncertified laboratory will be accepted provided a Quality Assurance-Quality Control Program is instituted by the laboratory. A manual containing the steps followed in this program for any onsite field measurements such as pH, DO, turbidity, temperature, and residual chlorine must be kept onsite in the treatment facility laboratory and shall be available for inspection by Central Valley Water Board staff. The Discharger must demonstrate sufficient capability (qualified and trained employees, properly calibrated and maintained field instruments, etc.) to adequately perform these field measurements. The Quality Assurance-Quality Control Program must conform to USEPA guidelines or to procedures approved by the Central Valley Water Board.
- D.** Appropriate flow measurement devices and methods consistent with accepted scientific practices shall be selected and used to ensure the accuracy and reliability of measurements of the volume of monitored discharges. All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary, at least yearly, to ensure their continued accuracy. All flow measurement devices shall be calibrated at least once per year to ensure continued accuracy of the devices.
- E.** Monitoring results, including noncompliance, shall be reported at intervals and in a manner specified in this Monitoring and Reporting Program.
- F.** Laboratories analyzing monitoring samples shall be certified by DPH, in accordance with the provision of Water Code section 13176, and the laboratory must include quality assurance/quality control data with their reports.
- G.** U.S. EPA's requires major and selected minor permittees under the NPDES program to participate in the annual Discharge Monitoring Report–Quality Assurance (DMR-QA) Study program. The DMR-QA Study program evaluates the analytical ability of laboratories that routinely perform self-monitoring analyses required by NPDES permits. Therefore, the

Discharger shall submit annually the results of the DMR-QA Study test to the State Water Resources Control Board at the following address:

DMR-QA Coordinator  
 Office of Information Management and Analysis  
 State Water Resources Control Board  
 1001 I Street  
 Sacramento, CA 95814

- H. The Discharger shall file with the Central Valley Water Board technical reports on self-monitoring performed according to the detailed specifications contained in this Monitoring and Reporting Program.
- I. The results of all monitoring required by this Order shall be reported to the Central Valley Water Board, and shall be submitted in such a format as to allow direct comparison with the limitations and requirements of this Order. Unless otherwise specified, discharge flows shall be reported in terms of the monthly average and the daily maximum discharge flows.

## II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

**Table E-1. Monitoring Station Locations**

| Discharge Point Name | Monitoring Location Name | Monitoring Location Description                                                                                                                                                                                              |
|----------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --                   | INF-001                  | A location where a representative sample of the influent into the Facility can be collected.                                                                                                                                 |
| 001                  | EFF-001                  | A location(s) where a representative sample of the effluent can be collected downstream from the last connection through which wastes can be admitted into the outfall.<br>Latitude: 38° 43' 53" N Longitude: 121° 18' 58" W |
| --                   | RSW-001                  | In Dry Creek, approximately 200 feet upstream of Discharge Point 001.                                                                                                                                                        |
| --                   | RSW-002                  | In Dry Creek, approximately 200 feet downstream of Discharge Point 001.                                                                                                                                                      |
| --                   | LND-001                  | Monitoring within Emergency Storage Basin (East Pond).                                                                                                                                                                       |
| --                   | LND-002                  | Monitoring within Emergency Storage Basin (West Pond).                                                                                                                                                                       |
| --                   | GW-001                   | Groundwater monitoring well (identified as MW-01).                                                                                                                                                                           |
| --                   | GW-002                   | Groundwater monitoring well (identified as MW-02).                                                                                                                                                                           |
| --                   | GW-003                   | Groundwater monitoring well (identified as MW-03).                                                                                                                                                                           |
| --                   | BIO-001                  | A location where a representative sample of the biosolids can be obtained.                                                                                                                                                   |
| --                   | FIL-001                  | Monitoring of the filter effluent to be measured downstream of the filters and ultraviolet light (UV) disinfection system and upstream of the point of diversion to the emergency storage basins.                            |
| --                   | UVS-001                  | A location where a representative sample of wastewater can be collected immediately upstream of the UV disinfection system.                                                                                                  |
| --                   | UVS-002                  | A location where a representative sample of wastewater can be collected immediately downstream of the UV disinfection system.                                                                                                |

The North latitude and West longitude information in Table E-1 are approximate for administrative purposes.

### III. INFLUENT MONITORING REQUIREMENTS

#### A. Monitoring Location INF-001

1. The Discharger shall monitor influent to the facility at Monitoring Location INF-001 as follows:

**Table E-2. Influent Monitoring**

| Parameter                                | Units | Sample Type                  | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------------------------------|-------|------------------------------|----------------------------|---------------------------------|
| Flow                                     | MGD   | Meter                        | Continuous                 | --                              |
| <b>Conventional Pollutants</b>           |       |                              |                            |                                 |
| Biochemical Oxygen Demand (5-day @ 20°C) | mg/L  | 24-hr Composite <sup>1</sup> | 3/Week                     | 2                               |
| Total Suspended Solids                   | mg/L  | 24-hr Composite <sup>1</sup> | 3/Week                     | 2                               |

<sup>1</sup> 24-hour flow proportional composite.

<sup>2</sup> Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136; or by methods approved by the Central Valley Water Board or the State Water Board.

### IV. EFFLUENT MONITORING REQUIREMENTS

#### A. Monitoring Location EFF-001

1. The Discharger shall monitor treated wastewater at Monitoring Location EFF-001 as follows. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level:

**Table E-3. Effluent Monitoring**

| Parameter                                             | Units            | Sample Type                  | Minimum Sampling Frequency | Required Analytical Test Method |
|-------------------------------------------------------|------------------|------------------------------|----------------------------|---------------------------------|
| Flow                                                  | MGD              | Meter                        | Continuous                 | --                              |
| <b>Conventional Pollutants</b>                        |                  |                              |                            |                                 |
| Biochemical Oxygen Demand (5-day @ 20°C)              | mg/L             | 24-hr Composite <sup>1</sup> | 3/Week                     | 2                               |
|                                                       | lbs/day          | Calculate                    | 3/Week                     | --                              |
| pH                                                    | standard units   | Meter                        | Continuous <sup>3</sup>    | 2                               |
| Total Suspended Solids                                | mg/L             | 24-hr Composite <sup>1</sup> | 3/Week                     | 2                               |
|                                                       | lbs/day          | Calculate                    | 3/Week                     | --                              |
| <b>Priority Pollutants</b>                            |                  |                              |                            |                                 |
| Mercury, Total Recoverable                            | µg/L             | Grab                         | 1/Month                    | 2,4                             |
| Priority Pollutants and Other Constituents of Concern | See Section IX.C | See Section IX.C             | See Section IX.C           | 2                               |
| <b>Non-Conventional Pollutants</b>                    |                  |                              |                            |                                 |
| Ammonia Nitrogen, Total (as N)                        | mg/L             | Grab                         | 1/Week <sup>3,5</sup>      | 2                               |
| Electrical Conductivity @ 25°C                        | µmhos/cm         | Grab                         | 1/Week                     | 2                               |
| Hardness, Total (as CaCO <sub>3</sub> )               | mg/L             | Grab                         | 1/Month                    | 2                               |
| Nitrate Nitrogen, Total (as N)                        | mg/L             | Grab                         | 1/Month <sup>6</sup>       | 2                               |

| Parameter                      | Units  | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|--------------------------------|--------|-------------|----------------------------|---------------------------------|
| Nitrite Nitrogen, Total (as N) | mg/L   | Grab        | 1/Month <sup>6</sup>       | <sup>2</sup>                    |
| Temperature                    | °F(°C) | Grab        | 1/Day <sup>3</sup>         | <sup>2</sup>                    |
| Total Dissolved Solids         | mg/L   | Grab        | 1/Month                    | <sup>2</sup>                    |

<sup>1</sup> 24-hour flow proportional composite.

<sup>2</sup> Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136 or by methods approved by the Central Valley Water Board or the State Water Board.

<sup>3</sup> pH and temperature shall be recorded at the time of ammonia sample collection.

<sup>4</sup> Total mercury samples shall be grab samples taken using clean hands/dirty hands procedures, as described in U.S. EPA method 1669: Sampling Ambient Water for Trace Metals at EPA Water Quality Criteria Levels, for collection of equipment blanks (section 9.4.4.2), and shall be analyzed by U.S. EPA method 1631 (Revision E) with a reporting level of 0.5 ng/L.

<sup>5</sup> Concurrent with whole effluent toxicity monitoring.

<sup>6</sup> Monitoring for nitrite and nitrate shall be conducted concurrently.

## V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

**A. Acute Toxicity Testing.** The Discharger shall conduct acute toxicity testing to determine whether the effluent is contributing acute toxicity to the receiving water. The Discharger shall meet the following acute toxicity testing requirements:

1. Monitoring Frequency – The Discharger shall perform semi-annual acute toxicity testing, concurrent with effluent ammonia sampling.
2. Sample Types – The Discharger may use flow-through or static renewal testing. For static renewal testing, the samples shall be flow proportional 24-hour composite samples and shall be representative of the volume and quality of the discharge. The effluent samples shall be taken at Monitoring Location EFF-001.
3. Test Species – Test species shall be rainbow trout (*Oncorhynchus mykiss*).
4. Methods – The acute toxicity testing samples shall be analyzed using EPA-821-R-02-012, Fifth Edition. Temperature and pH shall be recorded at the time of sample collection. No pH adjustment may be made unless approved by the Executive Officer.
5. Test Failure – If an acute toxicity test does not meet all test acceptability criteria, as specified in the test method, the Discharger must re-sample and re-test as soon as possible, not to exceed 7 days following notification of test failure.

**B. Chronic Toxicity Testing.** The Discharger shall conduct three species chronic toxicity testing to determine whether the effluent is contributing chronic toxicity to the receiving water. The Discharger shall meet the following chronic toxicity testing requirements:

1. Monitoring Frequency – The Discharger shall perform quarterly three species chronic toxicity testing.
2. Sample Types – Effluent samples shall be flow proportional 24-hour composites and shall be representative of the volume and quality of the discharge. The effluent samples shall be taken at Monitoring Location EFF-001. The receiving water control shall be a grab sample obtained from Monitoring Location RSW-001, as identified in this Monitoring and Reporting Program.
3. Sample Volumes – Adequate sample volumes shall be collected to provide renewal water to complete the test in the event that the discharge is intermittent.

4. **Test Species** – Chronic toxicity testing measures sublethal (e.g., reduced growth, reproduction) and/or lethal effects to test organisms exposed to an effluent compared to that of the control organisms. The Discharger shall conduct chronic toxicity tests with:
  - The cladoceran, water flea, *Ceriodaphnia dubia* (survival and reproduction test);
  - The fathead minnow, *Pimephales promelas* (larval survival and growth test); and
  - The green alga, *Selenastrum capricornutum* (growth test).
5. **Methods** – The presence of chronic toxicity shall be estimated as specified in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition*, EPA/821-R-02-013, October 2002.
6. **Reference Toxicant** – As required by the SIP, all chronic toxicity tests shall be conducted with concurrent testing with a reference toxicant and shall be reported with the chronic toxicity test results.
7. **Dilutions** – For routine and accelerated chronic toxicity monitoring, it is not necessary to perform the test using a dilution series. The test may be performed using 100% effluent and one control. For TRE monitoring, the chronic toxicity testing shall be performed using the dilution series identified in Table E-4, below, unless an alternative dilution series is detailed in the submitted TRE Action Plan. A receiving water control or laboratory water control may be used as the diluent.

**Table E-4. TRE Monitoring Chronic Toxicity Testing Dilution Series**

| Sample          | Dilutions <sup>a</sup> (%) |    |    |    |      | Control |
|-----------------|----------------------------|----|----|----|------|---------|
|                 | 100                        | 75 | 50 | 25 | 12.5 |         |
| % Effluent      | 100                        | 75 | 50 | 25 | 12.5 | 0       |
| % Control Water | 0                          | 25 | 50 | 75 | 87.5 | 100     |

<sup>a</sup> Receiving water control or laboratory water control may be used as the diluent.

8. **Test Failure** – The Discharger must re-sample and re-test as soon as possible, but no later than fourteen (14) days after receiving notification of a test failure. A test failure is defined as follows:
    - a. The reference toxicant test or the effluent test does not meet all test acceptability criteria as specified in the *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition*, EPA/821-R-02-013, October 2002 (Method Manual), and its subsequent amendments or revisions; or
    - b. The percent minimum significant difference (PMSD) measured for the test exceeds the upper PMSD bound variability criterion in Table 6 on page 52 of the Method Manual. (A retest is only required in this case if the test results do not exceed the monitoring trigger specified in the Special Provision at section VI. 2.a.ii. of the Order.)
- C. WET Testing Notification Requirements.** The Discharger shall notify the Central Valley Water Board within 24-hours after the receipt of test results exceeding the monitoring trigger during regular or accelerated monitoring, or an exceedance of the acute toxicity effluent limitation.
- D. WET Testing Reporting Requirements.** All toxicity test reports shall include the contracting laboratory's complete report provided to the Discharger and shall be in accordance with the

appropriate "Report Preparation and Test Review" sections of the method manuals. At a minimum, whole effluent toxicity monitoring shall be reported as follows:

1. **Chronic WET Reporting.** Regular chronic toxicity monitoring results shall be reported to the Central Valley Water Board with the quarterly self-monitoring reports, as described in Table E-9, following completion of the test, and shall contain, at minimum:
  - a. The results expressed in TUC, measured as 100/NOEC, and also measured as 100/LC50, 100/EC25, 100/IC25, and 100/IC50, as appropriate.
  - b. The statistical methods used to calculate endpoints;
  - c. The statistical output page, which includes the calculation of the percent minimum significant difference (PMSD);
  - d. The dates of sample collection and initiation of each toxicity test; and
  - e. The results compared to the numeric toxicity monitoring trigger.

Additionally, the monthly discharger self-monitoring reports shall contain an updated chronology of chronic toxicity test results expressed in TUC, and organized by test species, type of test (survival, growth or reproduction), and monitoring frequency, i.e., either quarterly, monthly, accelerated, or Toxicity Reduction Evaluation (TRE).
2. **Acute WET Reporting.** Acute toxicity test results shall be submitted with the monthly discharger self-monitoring reports and reported as percent survival.
3. **TRE Reporting.** Reports for TREs shall be submitted in accordance with the schedule contained in the Discharger's approved TRE Workplan, or as amended by the Discharger's TRE Action Plan.
4. **Quality Assurance (QA).** The Discharger must provide the following information for QA purposes:
  - a. Results of the applicable reference toxicant data with the statistical output page giving the species, NOEC, LOEC, type of toxicant, dilution water used, concentrations used, PMSD, and dates tested.
  - b. The reference toxicant control charts for each endpoint, which include summaries of reference toxicant tests performed by the contracting laboratory.
  - c. Any information on deviations or problems encountered and how they were dealt with.

## VI. LAND DISCHARGE MONITORING REQUIREMENTS

### A. Monitoring Locations LND-001 and LND-002

1. When emergency storage basins contain water for more than 7 consecutive days, the Discharger shall monitor the emergency storage basins at Monitoring Locations LND-001 and LND-002 as follows:

**Table E-5. Land Discharge Monitoring Requirements**

| Parameter                      | Units               | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|--------------------------------|---------------------|-------------|----------------------------|---------------------------------|
| Freeboard                      | Feet <sup>1,2</sup> | --          | 1/Day                      | --                              |
| pH                             | standard units      | Grab        | 1/Week                     | <sup>3</sup>                    |
| Electrical Conductivity @ 25°C | µmhos/cm            | Grab        | 1/Week                     | <sup>3</sup>                    |
| Odors                          | Observation         | --          | 1/Week                     | --                              |

| Parameter        | Units       | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------|-------------|-------------|----------------------------|---------------------------------|
| Levee Condition  | Observation | --          | 1/Week                     | --                              |
| Dissolved Oxygen | mg/L        | Grab        | 1/Month                    | <sup>3</sup>                    |

<sup>1</sup> To be measured vertically to the lowest point of overflow.

<sup>2</sup> Include estimation of volume of wastewater in each pond.

<sup>3</sup> Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136 or by methods approved by the Central Valley Water Board or the State Water Board.

## VII. RECYCLING MONITORING REQUIREMENTS – NOT APPLICABLE

## VIII. RECEIVING WATER MONITORING REQUIREMENTS

### A. Monitoring Locations RSW-001 and RSW-002

- The Discharger shall monitor Dry Creek at Monitoring Locations RSW-001 and RSW-002 as follows:

**Table E-6. Receiving Water Monitoring Requirements**

| Parameter                                             | Units            | Sample Type       | Minimum Sampling Frequency    | Required Analytical Test Method |
|-------------------------------------------------------|------------------|-------------------|-------------------------------|---------------------------------|
| <b>Conventional Pollutants</b>                        |                  |                   |                               |                                 |
| pH                                                    | standard units   | Grab <sup>1</sup> | 1/Week                        | <sup>2</sup>                    |
| <b>Priority Pollutants</b>                            |                  |                   |                               |                                 |
| Priority Pollutants and Other Constituents of Concern | See Section IX.C | See Section IX.C  | See Section IX.C <sup>3</sup> | <sup>2,4</sup>                  |
| <b>Non-Conventional Pollutants</b>                    |                  |                   |                               |                                 |
| Dissolved Oxygen                                      | mg/L             | Grab <sup>1</sup> | 1/Week                        | <sup>2</sup>                    |
| Electrical Conductivity @ 25°C                        | µmhos/cm         | Grab <sup>1</sup> | 1/Week                        | <sup>2</sup>                    |
| Hardness, Total (as CaCO <sub>3</sub> )               | mg/L             | Grab              | 1/Month                       | <sup>2</sup>                    |
| Temperature                                           | °F(°C)           | Grab <sup>1</sup> | 1/Week                        | <sup>2</sup>                    |

<sup>1</sup> A hand-held field meter may be used, provided the meter utilizes a USEPA-approved algorithm/method and is calibrated and maintained in accordance with the manufacturer's instructions. A calibration and maintenance log for each meter used for monitoring required by this Monitoring and Reporting Program shall be maintained at the Facility.

<sup>2</sup> Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136 or by methods approved by the Central Valley Water Board or the State Water Board.

<sup>3</sup> Monitoring required at Monitoring Location RSW-001 only.

<sup>4</sup> For priority pollutant constituents the reporting level shall be consistent with Sections 2.4.2 and 2.4.3 of the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (See Attachment E, section IX.C).

- In conducting the receiving water sampling, a log shall be kept of the receiving water conditions throughout the reach bounded by Monitoring Locations RSW-001 and RSW-002. Attention shall be given to the presence or absence of:
  - Floating or suspended matter,
  - Discoloration,
  - Bottom deposits,
  - Aquatic life,
  - Visible films, sheens, or coatings,
  - Fungi, slimes, or objectionable growths,



- g. Potential nuisance conditions.

Notes on receiving water conditions shall be summarized in the monitoring report.

## IX. OTHER MONITORING REQUIREMENTS

### A. Biosolids

#### 1. Monitoring Location BIO-001

- A composite sample of sludge shall be collected quarterly at Monitoring Location BIO-001 in accordance with EPA's *POTW Sludge Sampling and Analysis Guidance Document, August 1989*, and tested for priority pollutants listed in 40 CFR Part 122, Appendix D, Tables II and III (excluding total phenols).
- Biosolids monitoring shall be conducted using the methods in *Test Methods for Evaluating Solid Waste, Physical/Chemical Methods* (EPA publication SW-846), as required in 40 CFR 503.8(b)(4). All results must be reported on a 100% dry weight basis. Records of all analyses must state on each page of the laboratory report whether the results are expressed in "100% dry weight" or "as is."
- Sampling records shall be retained for a minimum of 5 years. A log shall be maintained of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log must be complete enough to serve as a basis for part of the annual report.

### B. Ultraviolet Light (UV) Disinfection System

#### 1. Monitoring Locations UVS-001, UVS-002, and FIL-001

The Discharger shall monitor the UV disinfection system at Monitoring Locations UVS-001, UVS-002, and FIL-001 as follows:

**Table E-7. Ultraviolet Light Disinfection System Monitoring Requirements**

| Parameter                       | Units              | Sample Type | Monitoring Location | Minimum Sampling Frequency |
|---------------------------------|--------------------|-------------|---------------------|----------------------------|
| Flow                            | MGD                | Meter       | UVS-001             | Continuous <sup>1</sup>    |
| Turbidity                       | NTU                | Meter       | FIL-001             | Continuous <sup>1,2</sup>  |
| Number of UV banks in operation | Number             | Observation | N/A                 | Continuous <sup>1</sup>    |
| UV Transmittance                | Percent (%)        | Meter       | UVS-001             | Continuous <sup>1</sup>    |
| UV Dose <sup>3</sup>            | mJ/cm <sup>2</sup> | Calculated  | UVS-001             | Continuous <sup>1</sup>    |
| Total Coliform Organisms        | MPN/100 mL         | Grab        | UVS-002             | 5/Week                     |

<sup>1</sup> For continuous analyzers, the Discharger shall maintain a log documenting routine meter maintenance activities including date, time of day, and duration, in which the analyzer(s) is not in operation. If analyzer(s) fail to provide continuous monitoring for more than two hours and influent and/or effluent from the disinfection process is not diverted for retreatment, the Discharger shall obtain and report hourly manual and/or grab sample results. The Discharger shall not decrease power settings or reduce the number of UV lamp banks in operation while the continuous analyzers are out of service and water is being disinfected.

<sup>2</sup> Report daily average and maximum turbidity.

<sup>3</sup> Report daily minimum hourly UV dose and daily average UV dose. The minimum hourly average dose shall consist of lowest hourly average dose provided in any channel that had at least one bank of lamps operating during the hour interval. For channels that did not operate for the entire hour interval, the dose will be averaged based on the actual operation time.

### C. Effluent and Receiving Water Characterization

1. **Third Year, Quarterly Monitoring.** Quarterly priority pollutant samples shall be collected from the effluent and upstream receiving water (Monitoring Locations EFF-001 and RSW-001) and analyzed for the constituents listed in Table E-8, below. Quarterly monitoring shall be conducted during the third year of the permit term (four consecutive samples, evenly distributed throughout the year) and the results of such monitoring be submitted to the Central Valley Water Board with the quarterly SMR's described in Table E-9. Each individual monitoring event shall provide representative sample results for the effluent and upstream receiving water.
2. **Concurrent Sampling.** Effluent and receiving water sampling shall be performed at approximately the same time, on the same date.
3. **Sample type.** All receiving water samples shall be taken as grab samples. Effluent samples shall be taken as described in Table E-8, below.

**Table E-8. Effluent and Receiving Water Characterization Monitoring**

| Parameter                          | Units | Effluent Sample Type | Maximum Reporting Level <sup>1</sup> |
|------------------------------------|-------|----------------------|--------------------------------------|
| 1,1-Dichloroethane                 | µg/L  | Grab                 | 1                                    |
| 1,1-Dichloroethene                 | µg/L  | Grab                 | 0.5                                  |
| 1,1,1-Trichloroethane              | µg/L  | Grab                 | 2                                    |
| 1,1,2-Trichloroethane              | µg/L  | Grab                 | 0.5                                  |
| 1,1,2,2-Tetrachloroethane          | µg/L  | Grab                 | 0.5                                  |
| 1,2-Dichlorobenzene                | µg/L  | Grab                 | 2                                    |
| 1,2-Dichloroethane                 | µg/L  | Grab                 | 0.5                                  |
| cis-1,2-Dichloroethene             | µg/L  | Grab                 |                                      |
| 1,2-Dichloropropane                | µg/L  | Grab                 | 0.5                                  |
| 1,2,4-Trichlorobenzene             | µg/L  | Grab                 | 1                                    |
| 1,3-Dichlorobenzene                | µg/L  | Grab                 | 2                                    |
| 1,3-Dichloropropene                | µg/L  | Grab                 | 0.5                                  |
| 1,4-Dichlorobenzene                | µg/L  | Grab                 | 2                                    |
| Acrolein                           | µg/L  | Grab                 | 2                                    |
| Acrylonitrile                      | µg/L  | Grab                 | 2                                    |
| Benzene                            | µg/L  | Grab                 | 0.5                                  |
| Bromoform                          | µg/L  | Grab                 | 2                                    |
| Bromomethane                       | µg/L  | Grab                 | 2                                    |
| Carbon tetrachloride               | µg/L  | Grab                 | 0.5                                  |
| Chlorobenzene (mono chlorobenzene) | µg/L  | Grab                 | 2                                    |
| Chloroethane                       | µg/L  | Grab                 | 2                                    |
| 2- Chloroethyl vinyl ether         | µg/L  | Grab                 | 1                                    |
| Chloroform                         | µg/L  | Grab                 | 2                                    |
| Chloromethane                      | µg/L  | Grab                 | 2                                    |
| Dibromochloromethane               | µg/L  | Grab                 | 0.5                                  |
| Dichlorobromomethane               | µg/L  | Grab                 | 0.5                                  |
| Dichloromethane                    | µg/L  | Grab                 | 2                                    |
| Ethylbenzene                       | µg/L  | Grab                 | 2                                    |
| Hexachlorobenzene                  | µg/L  | Grab                 | 1                                    |
| Hexachlorobutadiene                | µg/L  | Grab                 | 1                                    |
| Hexachloroethane                   | µg/L  | Grab                 | 1                                    |
| Naphthalene                        | µg/L  | Grab                 | 10                                   |
| Tetrachloroethene                  | µg/L  | Grab                 | 0.5                                  |

| Parameter                             | Units | Effluent Sample Type         | Maximum Reporting Level <sup>1</sup> |
|---------------------------------------|-------|------------------------------|--------------------------------------|
| Toluene                               | µg/L  | Grab                         | 2                                    |
| trans-1,2-Dichloroethylene            | µg/L  | Grab                         | 1                                    |
| Trichloroethene                       | µg/L  | Grab                         | 2                                    |
| Vinyl chloride                        | µg/L  | Grab                         | 0.5                                  |
| Methyl-tert-butyl ether (MTBE)        | µg/L  | Grab                         |                                      |
| Trichlorofluoromethane                | µg/L  | Grab                         |                                      |
| 1,1,2-Trichloro-1,2,2-Trifluoroethane | µg/L  | Grab                         |                                      |
| Styrene                               | µg/L  | Grab                         |                                      |
| Xylenes                               | µg/L  | Grab                         |                                      |
| 1,2-Benzanthracene                    | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 1,2-Diphenylhydrazine                 | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |
| 2-Chlorophenol                        | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 2,4-Dichlorophenol                    | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 2,4-Dimethylphenol                    | µg/L  | 24-hr Composite <sup>2</sup> | 2                                    |
| 2,4-Dinitrophenol                     | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 2,4-Dinitrotoluene                    | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 2,4,6-Trichlorophenol                 | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| 2,6-Dinitrotoluene                    | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 2-Nitrophenol                         | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| 2-Chloronaphthalene                   | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| 3,3'-Dichlorobenzidine                | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 3,4-Benzofluoranthene                 | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| 4-Chloro-3-methylphenol               | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| 4,6-Dinitro-2-methylphenol            | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| 4-Nitrophenol                         | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| 4-Bromophenyl phenyl ether            | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| 4-Chlorophenyl phenyl ether           | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Acenaphthene                          | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |
| Acenaphthylene                        | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Anthracene                            | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Benzidine                             | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Benzo(a)pyrene (3,4-Benzopyrene)      | µg/L  | 24-hr Composite <sup>2</sup> | 2                                    |
| Benzo(g,h,i)perylene                  | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Benzo(k)fluoranthene                  | µg/L  | 24-hr Composite <sup>2</sup> | 2                                    |
| Bis(2-chloroethoxy) methane           | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Bis(2-chloroethyl) ether              | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |
| Bis(2-chloroisopropyl) ether          | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Bis(2-ethylhexyl) phthalate           | µg/L  | Grab                         | 5 <sup>3</sup>                       |
| Butyl benzyl phthalate                | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Chrysene                              | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Di-n-butylphthalate                   | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Di-n-octylphthalate                   | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Dibenzo(a,h)-anthracene               | µg/L  | 24-hr Composite <sup>2</sup> | 0.1                                  |
| Diethyl phthalate                     | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Dimethyl phthalate                    | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Fluoranthene                          | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Fluorene                              | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Hexachlorocyclopentadiene             | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Indeno(1,2,3-c,d)pyrene               | µg/L  | 24-hr Composite <sup>2</sup> | 0.05                                 |
| Isophorone                            | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |

| Parameter                             | Units | Effluent Sample Type         | Maximum Reporting Level <sup>1</sup> |
|---------------------------------------|-------|------------------------------|--------------------------------------|
| N-Nitrosodiphenylamine                | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |
| N-Nitrosodimethylamine                | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| N-Nitrosodi-n-propylamine             | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Nitrobenzene                          | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Pentachlorophenol                     | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |
| Phenanthrene                          | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Phenol                                | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |
| Pyrene                                | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Aluminum                              | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Antimony                              | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Arsenic                               | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Asbestos                              | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Barium                                | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Beryllium                             | µg/L  | 24-hr Composite <sup>2</sup> | 2                                    |
| Cadmium                               | µg/L  | 24-hr Composite <sup>2</sup> | 0.5                                  |
| Chromium (III)                        | µg/L  | 24-hr Composite <sup>2</sup> | 50                                   |
| Chromium (VI)                         | µg/L  | 24-hr Composite <sup>2</sup> | 10                                   |
| Copper                                | µg/L  | 24-hr Composite <sup>2</sup> | 2                                    |
| Cyanide                               | µg/L  | Grab                         | 5                                    |
| Fluoride                              | µg/L  | Grab                         | --                                   |
| Iron                                  | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Lead                                  | µg/L  | 24-hr Composite <sup>2</sup> | 0.5                                  |
| Mercury <sup>4</sup>                  | µg/L  | Grab                         | 0.5                                  |
| Manganese                             | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Molybdenum                            | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Nickel                                | µg/L  | 24-hr Composite <sup>2</sup> | 20                                   |
| Selenium                              | µg/L  | 24-hr Composite <sup>2</sup> | 5                                    |
| Silver                                | µg/L  | 24-hr Composite <sup>2</sup> | 0.25                                 |
| Thallium                              | µg/L  | 24-hr Composite <sup>2</sup> | 1                                    |
| Tributyltin                           | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Zinc                                  | µg/L  | 24-hr Composite <sup>2</sup> | 20                                   |
| 4,4'-DDD                              | µg/L  | 24-hr Composite <sup>2</sup> | 0.05                                 |
| 4,4'-DDE                              | µg/L  | 24-hr Composite <sup>2</sup> | 0.05                                 |
| 4,4'-DDT                              | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| alpha-Endosulfan                      | µg/L  | 24-hr Composite <sup>2</sup> | 0.02                                 |
| alpha-Hexachlorocyclohexane (BHC)     | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| Alachlor                              | µg/L  | 24-hr Composite <sup>2</sup> | --                                   |
| Aldrin                                | µg/L  | 24-hr Composite <sup>2</sup> | 0.005                                |
| beta-Endosulfan                       | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| beta-Hexachlorocyclohexane            | µg/L  | 24-hr Composite <sup>2</sup> | 0.005                                |
| Chlordane                             | µg/L  | 24-hr Composite <sup>2</sup> | 0.1                                  |
| delta-Hexachlorocyclohexane           | µg/L  | 24-hr Composite <sup>2</sup> | 0.005                                |
| Dieldrin                              | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| Endosulfan sulfate                    | µg/L  | 24-hr Composite <sup>2</sup> | 0.05                                 |
| Endrin                                | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| Endrin Aldehyde                       | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| Heptachlor                            | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| Heptachlor Epoxide                    | µg/L  | 24-hr Composite <sup>2</sup> | 0.01                                 |
| Lindane (gamma-Hexachlorocyclohexane) | µg/L  | 24-hr Composite <sup>2</sup> | 0.02                                 |

| Parameter                                     | Units          | Effluent Sample Type         | Maximum Reporting Level <sup>1</sup> |
|-----------------------------------------------|----------------|------------------------------|--------------------------------------|
| PCB-1016                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| PCB-1221                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| PCB-1232                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| PCB-1242                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| PCB-1248                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| PCB-1254                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| PCB-1260                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| Toxaphene                                     | µg/L           | 24-hr Composite <sup>2</sup> | 0.5                                  |
| Atrazine                                      | µg/L           | Grab                         | --                                   |
| Bentazon                                      | µg/L           | Grab                         | --                                   |
| Carbofuran                                    | µg/L           | Grab                         | --                                   |
| 2,4-D                                         | µg/L           | Grab                         | --                                   |
| Dalapon                                       | µg/L           | Grab                         | --                                   |
| 1,2-Dibromo-3-chloropropane (DBCP)            | µg/L           | Grab                         | --                                   |
| Di(2-ethylhexyl)adipate                       | µg/L           | Grab                         | --                                   |
| Dinoseb                                       | µg/L           | Grab                         | --                                   |
| Diquat                                        | µg/L           | Grab                         | --                                   |
| Endothal                                      | µg/L           | Grab                         | --                                   |
| Ethylene Dibromide                            | µg/L           | Grab                         | --                                   |
| Glyphosate                                    | µg/L           | Grab                         | --                                   |
| Methoxychlor                                  | µg/L           | Grab                         | --                                   |
| Molinate (Ordram)                             | µg/L           | Grab                         | --                                   |
| Oxamyl                                        | µg/L           | Grab                         | --                                   |
| Picloram                                      | µg/L           | Grab                         | --                                   |
| Simazine (Princep)                            | µg/L           | Grab                         | --                                   |
| Thiobencarb                                   | µg/L           | 24-hr Composite <sup>2</sup> | --                                   |
| 2,3,7,8-TCDD (Dioxin)                         | µg/L           | 24-hr Composite <sup>2</sup> | --                                   |
| 2,4,5-TP (Silvex)                             | µg/L           | Grab                         | --                                   |
| Diazinon                                      | µg/L           | 24-hr Composite <sup>2</sup> | 0.015                                |
| Chlorpyrifos                                  | µg/L           | 24-hr Composite <sup>2</sup> | 0.014                                |
| Boron                                         | µg/L           | 24-hr Composite <sup>2</sup> | --                                   |
| Chloride                                      | mg/L           | 24-hr Composite <sup>2</sup> | --                                   |
| Flow                                          | MGD            | --                           | --                                   |
| Hardness (as CaCO <sub>3</sub> ) <sup>4</sup> | mg/L           | Grab                         | --                                   |
| Foaming Agents (MBAS)                         | µg/L           | Grab                         | --                                   |
| Mercury, Methyl                               | ng/L           | Grab                         | 0.05                                 |
| Nitrate (as N) <sup>4</sup>                   | mg/L           | Grab                         | 2.0                                  |
| Nitrite (as N) <sup>4</sup>                   | mg/L           | Grab                         | 0.4                                  |
| pH <sup>4</sup>                               | standard units | Grab                         | 0.1                                  |
| Phosphorus, Total (as P)                      | mg/L           | 24-hr Composite <sup>2</sup> | --                                   |
| Specific conductance (EC) <sup>4</sup>        | µmhos/cm       | Grab                         | --                                   |
| Sulfate                                       | mg/L           | 24-hr Composite <sup>2</sup> | 500                                  |
| Sulfide (as S)                                | mg/L           | Grab                         | --                                   |
| Sulfite (as SO <sub>3</sub> )                 | mg/L           | 24-hr Composite <sup>2</sup> | --                                   |
| Temperature <sup>4</sup>                      | °F(°C)         | Grab                         | --                                   |
| Total Dissolved Solids (TDS) <sup>4</sup>     | mg/L           | Grab                         | --                                   |

| Parameter | Units | Effluent Sample Type | Maximum Reporting Level <sup>1</sup> |
|-----------|-------|----------------------|--------------------------------------|
|-----------|-------|----------------------|--------------------------------------|

- <sup>1</sup> The reporting levels required in this table for priority pollutant constituents are established based on Section 2.4.2 and Appendix 4 of the SIP.
- <sup>2</sup> 24-hour flow proportional composite.
- <sup>3</sup> In order to verify if bis (2-ethylhexyl) phthalate is truly present in the effluent and receiving water, the Discharger shall take steps to assure that sample containers, sampling apparatus, and analytical equipment are not sources of the detected contaminant.
- <sup>4</sup> The Discharger is not required to conduct effluent monitoring for constituents that have already been sampled in a given quarter, as required in Table E-3, except for hardness, pH, and temperature, which shall be conducted concurrently with the effluent and receiving water sampling.

## X. REPORTING REQUIREMENTS

### A. General Monitoring and Reporting Requirements

1. The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
2. Upon written request of the Central Valley Water Board, the Discharger shall submit a summary monitoring report. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year(s).
3. **Compliance Time Schedules.** For compliance time schedules included in the Order, the Discharger shall submit to the Central Valley Water Board, on or before each compliance due date, the specified document or a written report detailing compliance or noncompliance with the specific date and task. If noncompliance is reported, the Discharger shall state the reasons for noncompliance and include an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Central Valley Water Board by letter when it returns to compliance with the compliance time schedule.
4. The Discharger shall report to the Central Valley Water Board any toxic chemical release data it reports to the State Emergency Response Commission within 15 days of reporting the data to the Commission pursuant to section 313 of the "*Emergency Planning and Community Right to Know Act*" of 1986.

### B. Self-Monitoring Reports (SMR's)

1. The Discharger shall electronically submit SMR's using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). The CIWQS Web site will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal.
2. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. The Discharger shall submit monthly SMR's including the results of all required monitoring using U.S. EPA-approved test methods or other test methods specified in this Order. SMR's are to include all new monitoring results obtained since the last SMR was submitted. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

**Table E-9. Monitoring Periods and Reporting Schedule**

| <b>Sampling Frequency</b> | <b>Monitoring Period Begins On...</b> | <b>Monitoring Period</b>                                                                                              | <b>SMR Due Date</b>                                                 |
|---------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Continuous                | Permit effective date                 | All                                                                                                                   | Submit with monthly SMR                                             |
| 1/Day                     | Permit effective date                 | (Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling. | Submit with monthly SMR                                             |
| 1/Week                    | Permit effective date                 | Sunday through Saturday                                                                                               | Submit with monthly SMR                                             |
| 3/Week                    | Permit effective date                 | Sunday through Saturday                                                                                               | Submit with monthly SMR                                             |
| 5/Week                    | Permit effective date                 | Sunday through Saturday                                                                                               | Submit with monthly SMR                                             |
| 1/Month                   | Permit effective date                 | 1 <sup>st</sup> day of calendar month through last day of calendar month                                              | First day of second month following month of sampling               |
| 1/Quarter                 | Permit effective date                 | 1 January through 31 March<br>1 April through 30 June<br>1 July through 30 September<br>1 October through 31 December | 1 May<br>1 August<br>1 November<br>1 February of the following year |
| 2/Year                    | Permit effective date                 | 1 January through 30 June<br>1 July through 31 December                                                               | 1 August<br>1 February of the following year                        |
| 1/Year                    | Permit effective date                 | 1 January through 31 December                                                                                         | 1 February of the following year                                    |

4. **Reporting Protocols.** The Discharger shall report with each sample result the applicable Reporting Level (RL) and the current laboratory's Method Detection Limit (MDL), as determined by the procedure in 40 C.F.R. part 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ( $\pm$  a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.

- d. Dischargers are to instruct laboratories to establish calibration standards so that the Minimum Level (ML) value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.
5. **Multiple Sample Data.** When determining compliance with an AMEL, AWEL, or MDEL for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
  - a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
  - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
6. The Discharger shall submit SMR's in accordance with the following requirements:
  - a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.
  - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDR's; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
7. The Discharger shall submit in the SMR's calculations and reports in accordance with the following requirements:
  - a. **Average Dry Weather Flow.** The Discharger shall calculate and report the average dry weather flow for the effluent. The average dry weather flow shall be calculated as specified in Section VII.C of the Limitations and Discharge Requirements and reported in the December SMR.
  - b. **Mass Loading Limitations.** For BOD<sub>5</sub>, TSS, and ammonia, the Discharger shall calculate and report the mass loading (lbs/day) in the SMRs. The mass loading shall be calculated as follows:

$$\text{Mass Loading (lbs/day)} = \text{Flow (MGD)} \times \text{Concentration (mg/L)} \times 8.34$$

When calculating daily mass loading, the daily average flow and constituent concentration shall be used. For weekly average mass loading, the weekly average



flow and constituent concentration shall be used. For monthly average mass loading, the monthly average flow and constituent concentration shall be used.

- c. **Removal Efficiency (BOD<sub>5</sub> and TSS).** The Discharger shall calculate and report the percent removal of BOD<sub>5</sub> and TSS in the SMRs. The percent removal shall be calculated as specified in Section VII.A. of the Limitations and Discharge Requirements.
- d. **Total Coliform Organisms Effluent Limitations.** The Discharger shall calculate and report the 7-day median of total coliform organisms for the effluent. The 7-day median of total coliform organisms shall be calculated as specified in Section VII.D. of the Limitations and Discharge Requirements.
- e. **Dissolved Oxygen Receiving Water Limitations.** The Discharger shall measure or calculate, where appropriate, and report monthly in the self-monitoring report: i) the dissolved oxygen concentration, ii) the percent of saturation in the main water mass, and iii) the 95th percentile dissolved oxygen concentration.
- f. **Turbidity Receiving Water Limitations.** The Discharger shall calculate and report the turbidity increase in the receiving water applicable to the natural turbidity condition specified in Section V.A.17.a-e. of the Limitations and Discharge Requirements.
- g. **Temperature Receiving Water Limitations.** The Discharger shall calculate and report the temperature increase in the receiving water based on the difference in temperature at Monitoring Locations RSW-001 and RSW-002.
- h. **Total Calendar Annual Mass Loading Mercury Effluent Limitations.** The Discharger shall calculate and report the total calendar annual mercury mass loading for the effluent in the December SMR. The total calendar year annual mass loading shall be calculated as specified in Section VII.B. of the Limitations and Discharge Requirements.

**C. Discharge Monitoring Reports (DMR's)**

1. At any time during the term of this permit, the State Water Board or Central Valley Water Board may notify the Discharger to electronically submit DMR's. Until such notification is given specifically for the submittal of DMR's, the Discharger shall submit DMR's in accordance with the requirements described below.
2. DMR's must be signed and certified as required by the standard provisions (Attachment D). The Discharger shall submit the original DMR to the address listed below:

| STANDARD MAIL                                                                                                                            | FEDEX/UPS/<br>OTHER PRIVATE CARRIERS                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| State Water Resources Control Board<br>Division of Water Quality<br>c/o DMR Processing Center<br>PO Box 100<br>Sacramento, CA 95812-1000 | State Water Resources Control Board<br>Division of Water Quality<br>c/o DMR Processing Center<br>1001 I Street, 15 <sup>th</sup> Floor<br>Sacramento, CA 95814 |

3. All discharge monitoring results must be reported on the official U.S. EPA pre-printed DMR forms (EPA Form 3320-1) or on self-generated forms that follow the exact same format of EPA Form 3320-1.

#### D. Other Reports

1. **Special Study Reports and Progress Reports.** As specified in the Special Provisions contained in section VI of the Order, special study and progress reports shall be submitted in accordance with the following reporting requirements.

**Table E-10. Reporting Requirements for Special Provisions Reports**

| Special Provision                                                                      | Reporting Requirements         |
|----------------------------------------------------------------------------------------|--------------------------------|
| Salinity Evaluation and Minimization Plan, Annual Reports (Special Provision VI.C.3.a) | <b>1 February,</b><br>annually |
| Pollution Prevention Plan for Mercury, Progress Reports (Special Provision VI.C.3.b)   | <b>1 February,</b><br>annually |

2. The Discharger shall report the results of any special studies, acute and chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special Provisions VI.C. The Discharger shall submit reports with the first monthly SMR scheduled to be submitted on or immediately following the report due date.
3. Within 60 days of permit effective date, the Discharger shall submit a report outlining reporting levels (RLs), method detection limits, and analytical methods for approval. The Discharger shall comply with the monitoring and reporting requirements for CTR constituents as outlined in section 2.3 and 2.4 of the SIP. The maximum required reporting levels for priority pollutant constituents shall be based on the Minimum Levels (MLs) contained in Appendix 4 of the SIP, determined in accordance with Section 2.4.2 and Section 2.4.3 of the SIP. In accordance with Section 2.4.2 of the SIP, when there is more than one ML value for a given substance, the Central Valley Water Board shall include as RLs, in the permit, all ML values, and their associated analytical methods, listed in Appendix 4 that are below the calculated effluent limitation. The Discharger may select any one of those cited analytical methods for compliance determination. If no ML value is below the effluent limitation, then the Central Valley Water Board shall select as the RL, the lowest ML value, and its associated analytical method, listed in Appendix 4 for inclusion in the permit. Attachment E, Section IX.C provides required maximum reporting levels in accordance with the SIP.
4. **Annual Operations Report.** By 1 February of each year, the Discharger shall submit a written report to the Executive Officer containing the following:
  - a. The names, certificate grades, and general responsibilities of all persons employed at the Facility.
  - b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
  - c. A statement certifying when the flow meter(s) and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration.
  - d. A statement certifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last revised and last reviewed for adequacy.
  - e. The Discharger may also be requested to submit an annual report to the Central Valley Water Board with both tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing.

The report shall discuss the compliance record. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

5. **Annual Pretreatment Reporting Requirements.** The Discharger shall submit annually a report to the Central Valley Water Board, with copies to USEPA Region 9 and the State Water Board, describing the Discharger's pretreatment activities over the previous 12 months (1 January through 31 December). In the event that the Discharger is not in compliance with any conditions or requirements of this Order, including noncompliance with pretreatment audit/compliance inspection requirements, then the Discharger shall also include the reasons for noncompliance and state how and when the Discharger shall comply with such conditions and requirements.

The Discharger may combine annual pretreatment reporting requirements for both this Facility and their Pleasant Grove Wastewater Treatment Plant (NPDES No. CA0084573). If the reports are combined for both plants, then the Discharger shall note so in its transmittal letter accompanying the submission of the annual report.

An annual report shall be submitted by **28 February** and include at least the following items:

- a. A summary of analytical results from representative, flow proportioned, 24-hour composite sampling of the POTW's influent and effluent for those pollutants USEPA has identified under section 307(a) of the CWA which are known or suspected to be discharged by nondomestic users.

Sludge shall be sampled during the same 24-hour period and analyzed for the same pollutants as the influent and effluent sampling and analysis. The sludge analyzed shall be a composite sample of a number of discrete samples taken at equal time intervals during operational hours. Wastewater and sludge sampling and analysis shall be performed at least annually. The Discharger shall also provide any influent, effluent or sludge monitoring data for non-priority pollutants which may be causing or contributing to interference, pass-through or adversely impacting sludge quality. Sampling and analysis shall be performed in accordance with the techniques prescribed in 40 CFR Part 136 and amendments thereto.

- b. A discussion of upset, interference, or pass-through incidents, if any, at the treatment plant, which the Discharger knows or suspects were caused by nondomestic users of the POTW. The discussion shall include the reasons why the incidents occurred, the corrective actions taken and, if known, the name and address of, the nondomestic user(s) responsible. The discussion shall also include a review of the applicable pollutant limitations to determine whether any additional limitations, or changes to existing requirements, may be necessary to prevent pass-through, interference, or noncompliance with sludge disposal requirements.
- c. The cumulative number of nondomestic users that the Discharger has notified regarding Baseline Monitoring Reports and the cumulative number of nondomestic user responses.
- d. An updated list of the Discharger's significant industrial users (SIUs) including their names and addresses, or a list of deletions, additions and SIU name changes keyed to a previously submitted list. The Discharger shall provide a brief explanation for each change. The list shall identify the SIUs subject to federal categorical standards by specifying which set(s) of standards are applicable to each SIU. The list shall indicate which SIUs, or specific pollutants from each industry, are

subject to local limitations. Local limitations that are more stringent than the federal categorical standards shall also be identified.

- e. The Discharger shall characterize the compliance status through the year of record of each SIU by employing the following descriptions:
  - i. complied with baseline monitoring report requirements (where applicable);
  - ii. consistently achieved compliance;
  - iii. inconsistently achieved compliance;
  - iv. significantly violated applicable pretreatment requirements as defined by 40 CFR 403.8(f)(2)(vii);
  - v. complied with schedule to achieve compliance (include the date final compliance is required);
  - vi. did not achieve compliance and not on a compliance schedule; and
  - vii. compliance status unknown.
- f. A report describing the compliance status of each SIU characterized by the descriptions in items iii through vii above shall be submitted for each calendar quarter by the first day of the second month following the end of the quarter. The report shall identify the specific compliance status of each such SIU and shall also identify the compliance status of the POTW with regards to audit/pretreatment compliance inspection requirements. If none of the aforementioned conditions exist, at a minimum, a letter indicating that all industries are in compliance and no violations or changes to the pretreatment program have occurred during the quarter must be submitted. The information required in the fourth quarter report shall be included as part of the annual report due every 28 February. This quarterly reporting requirement shall commence upon issuance of this Order.
- g. A summary of the inspection and sampling activities conducted by the Discharger during the past year to gather information and data regarding the SIUs. The summary shall include:
  - i. The names and addresses of the SIUs subjected to surveillance and an explanation of whether they were inspected, sampled, or both and the frequency of these activities at each user; and
  - ii. The conclusions or results from the inspection or sampling of each industrial user.
- h. The Discharger shall characterize the compliance status of each SIU by providing a list or table which includes the following information:
  - i. Name of SIU;
  - ii. Category, if subject to federal categorical standards;
  - iii. The type of wastewater treatment or control processes in place;
  - iv. The number of samples taken by the POTW during the year;
  - v. The number of samples taken by the SIU during the year;
  - vi. For an SIU subject to discharge requirements for total toxic organics, whether all required certifications were provided;

- vii. A list of the standards violated during the year. Identify whether the violations were for categorical standards or local limits.
- viii. Whether the facility is in significant noncompliance (SNC) as defined at 40 CFR 403.8(f)(2)(viii) at any time during the year; and
- ix. A summary of enforcement or other actions taken during the year to return the SIU to compliance. Describe the type of action (e.g., warning letters or notices of violation, administrative orders, civil actions, and criminal actions), final compliance date, and the amount of fines and penalties collected, if any. Describe any proposed actions for bringing the SIU into compliance;
- x. Restriction of flow to the POTW.
- xi. Disconnection from discharge to the POTW.
- i. A brief description of any programs the POTW implements to reduce pollutants from nondomestic users that are not classified as SIUs;
- j. A brief description of any significant changes in operating the pretreatment program which differ from the previous year including, but not limited to, changes concerning: the program's administrative structure, local limits, monitoring program or monitoring frequencies, legal authority, enforcement policy, funding levels, or staffing levels;
- k. A summary of the annual pretreatment budget, including the cost of pretreatment program functions and equipment purchases; and
- l. A summary of activities to involve and inform the public of the program including a copy of the newspaper notice, if any, required under 40 CFR 403.8(f)(2)(viii).

Duplicate signed copies of these Pretreatment Program reports shall be submitted to the Central Valley Water Board and the:

State Water Resources Control Board  
Division of Water Quality  
1001 I Street or P.O. Box 100  
Sacramento, CA 95812

and the

Regional Administrator  
U.S. Environmental Protection Agency WTR-5  
75 Hawthorne Street  
San Francisco, CA 94105

## ATTACHMENT F – FACT SHEET

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## ATTACHMENT F – FACT SHEET

As described in section I, the Central Valley Water Board incorporates this Fact Sheet as findings of the Central Valley Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

### I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

**Table F-1. Facility Information**

|                                              |                                                            |
|----------------------------------------------|------------------------------------------------------------|
| WDID                                         | 5A310106001                                                |
| Discharger                                   | City of Roseville                                          |
| Name of Facility                             | Dry Creek Wastewater Treatment Plant                       |
| Facility Address                             | 1800 Booth Road                                            |
|                                              | Roseville, CA 95747                                        |
|                                              | Placer County                                              |
| Facility Contact, Title and Phone            | Alfred Lawrence, Chief Operator, (916) 746-1802            |
| Authorized Person to Sign and Submit Reports | Ken Glotzbach, Wastewater Utility Manager, (916) 774-5754  |
| Mailing Address                              | 2005 Hilltop Circle, Roseville, CA 95747                   |
| Billing Address                              | Same as mailing address                                    |
| Type of Facility                             | Publicly Owned Treatment Works (POTW)                      |
| Major or Minor Facility                      | Major                                                      |
| Threat to Water Quality                      | 1                                                          |
| Complexity                                   | A                                                          |
| Pretreatment Program                         | Yes                                                        |
| Recycling Requirements                       | Not Applicable                                             |
| Facility Permitted Flow                      | 18 million gallons per day (MGD), average dry weather flow |
| Facility Design Flow                         | 18 MGD, average dry weather flow                           |
| Watershed                                    | Lower American                                             |
| Receiving Water                              | Dry Creek                                                  |
| Receiving Water Type                         | Inland Surface Water                                       |

- A.** The City of Roseville (hereinafter Discharger) is the owner and operator of the Dry Creek Wastewater Treatment Plant (hereinafter Facility), a POTW.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B.** The Facility discharges wastewater to Dry Creek, a water of the United States, and a tributary to the Sacramento River via the Natomas East Main Drainage Canal, within the Lower American watershed. The Discharger was previously regulated by National Pollutant



Discharge Elimination System (NPDES) Permit No. CA0079502 adopted on 12 June 2013 (Order R5-2008-0077) and amended on 31 May 2013 (Order R5-2008-0077-01), and expired on 1 June 2013. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.

Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. The State Water Board retains the jurisdictional authority to enforce such requirements under Water Code section 1211.

- C. The Discharger filed a report of waste discharge (ROWD) and submitted an application for reissuance of its WDR's and NPDES permit on 30 November 2012. The application was deemed complete on 27 May 2013. A site visit was conducted on 3 June 2013 to observe operations and collect additional data to develop permit limitations and requirements for waste discharge.

## **II. FACILITY DESCRIPTION**

The Discharger provides sewerage service for the southeast portion of the City of Roseville and portions of Placer County and the South Placer Municipal Utility District, and serves a population of approximately 111,000. The Discharger owns and operates portions of the wastewater collection system. Placer County and South Placer Municipal Utility District own and operate the remaining portions of the wastewater collection system. The design average dry weather flow capacity of the Facility is 18 MGD.

### **A. Description of Wastewater and Biosolids Treatment and Controls**

The treatment system at the Facility system consists of a headworks (mechanically cleaned bar screens, influent pump station, and grit chambers), primary clarifiers, and secondary treatment consisting of nitrification and denitrification, aeration, secondary clarification, and tertiary treatment. Tertiary treatment is provided by chemical coagulation with organic polymers, followed by pH adjustment, filtration, ultraviolet light (UV) disinfection, and cascade aeration.

Two on-site emergency storage basins are utilized to store influent during large storm events to prevent overwhelming of the treatment process, to store partially treated wastewater, and to prevent plant upsets by the influent that would be harmful to the treatment process. The influent is stored in the emergency storage basins usually no more than 2 days and all of the wastewater in the basins is re-routed through the plant for treatment. The smaller 2-acre basin is concrete-lined and the larger 16-acre basin possesses a high-density polyethylene (HDPE) liner.

Biosolids treatment consists of a gravity belt thickener, anaerobic digesters, a sludge holding tank, and a belt filter press for dewatering. Biosolids are land applied at the Silva Ranch application site in Sacramento County by the contractor Synagro Technologies, Inc. When land application is not available or inappropriate (e.g., during the winter), biosolids are disposed of at Allied Waste (Forward Landfill).

### **B. Discharge Points and Receiving Waters**

1. The Facility is located in Section 9, T10N, R6E, MDB&M, as shown in Attachment B, a part of this Order.
2. Treated municipal wastewater is discharged at Discharge Point 001 to Dry Creek, a water of the United States and tributary to the Sacramento River via the Natomas East Main Drainage Canal, at a point latitude 38° 43' 53" N and longitude 121° 18' 58" W.

**C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data**

Effluent limitations contained in Order R5-2008-0077-01 for discharges from Discharge Point 001 (Monitoring Location EFF-001) and representative monitoring data from the term of Order R5-2008-0077-01 are as follows:

**Table F-2. Historic Effluent Limitations and Monitoring Data**

| Parameter                                | Units          | Effluent Limitation  |                  |                                    | Monitoring Data<br>(August 2008 – April 2013) |                                  |                         |
|------------------------------------------|----------------|----------------------|------------------|------------------------------------|-----------------------------------------------|----------------------------------|-------------------------|
|                                          |                | Average Monthly      | Average Weekly   | Maximum Daily                      | Highest Average Monthly Discharge             | Highest Average Weekly Discharge | Highest Daily Discharge |
| Average Dry Weather Flow                 | MGD            | --                   | --               | 18                                 | --                                            | --                               | 22 <sup>1</sup>         |
| Biochemical Oxygen Demand (5-day @ 20°C) | mg/L           | 10                   | 15               | 20                                 | 4.4                                           | 6.5                              | 9                       |
|                                          | lbs/day        | 1,500                | 2,250            | 3,000                              | 247                                           | 412                              | 572                     |
|                                          | % removal      | 85                   | --               | --                                 | 98.6 <sup>2</sup>                             | --                               | --                      |
| Total Suspended Solids                   | mg/L           | 10                   | 15               | 20                                 | 3                                             | 4.3                              | 6                       |
|                                          | lbs/day        | 1,500                | 2,250            | 3,000                              | 250                                           | 372                              | 555                     |
|                                          | % removal      | 85                   | --               | --                                 | NR                                            | --                               | --                      |
| pH                                       | standard units | --                   | --               | 6.5 – 8.0                          |                                               | --                               | 6.5 – 7.9               |
| Chlorine, Total Residual                 | mg/L           | 0.011 <sup>3</sup>   | --               | 0.019 <sup>4</sup>                 | --                                            | --                               | ND                      |
| Settleable Solids                        | ml/L           | 0.1                  | --               | 0.2                                | 0.1                                           | --                               | 0.2                     |
| Aluminum, Total Recoverable              | µg/L           | 374/200 <sup>5</sup> | --               | 750                                | 360/121 <sup>6</sup>                          | --                               | 360                     |
| Ammonia Nitrogen, Total (as N)           | mg/L           | 2.0                  | --               | 5.7                                | 0.31                                          | --                               | 0.98                    |
|                                          | lbs/day        | 300                  | --               | 856                                | 11                                            | --                               | 39                      |
| Cadmium, Total Recoverable               | µg/L           | 0.94                 | --               | 2.25                               | 0.1                                           | --                               | 0.1                     |
| Carbon Tetrachloride                     | µg/L           | 0.25                 | --               | 0.50                               | <0.16                                         | --                               | <0.16                   |
| Cyanide, Total (as CN)                   | µg/L           | 4.1                  | --               | 8.9                                | 3.2                                           | --                               | 3.2                     |
| Chlorodibromomethane                     | µg/L           | 0.41                 | --               | 0.82                               | 0.9                                           | --                               | 0.9                     |
| Dichlorobromomethane                     | µg/L           | 0.56                 | --               | 1.12                               | 11                                            | --                               | 11                      |
| Iron, Total Recoverable                  | µg/L           | 300 <sup>5</sup>     | --               | --                                 | 80 <sup>6</sup>                               | --                               | --                      |
| Manganese, Total Recoverable             | µg/L           | 50 <sup>5</sup>      | --               | --                                 | 11 <sup>6</sup>                               | --                               | --                      |
| Mercury, Total Recoverable               | µg/L           | 0.05                 |                  | 0.10                               | 0.006                                         |                                  | 0.006                   |
|                                          | lbs/day        | 0.008                | --               | 0.015                              | --                                            | --                               | --                      |
| Zinc, Total Recoverable                  | µg/L           | 35                   | --               | 71                                 | 51                                            | --                               | 51                      |
| Total Coliform Organisms                 | MPN/100 mL     | --                   | 2.2 <sup>7</sup> | 23 <sup>8</sup> /240 <sup>9</sup>  | --                                            | --                               | 22                      |
| Turbidity                                | NTU            | --                   | 2 <sup>10</sup>  | 5 <sup>11</sup> /10 <sup>9</sup>   | --                                            | --                               | 2                       |
| Acute Toxicity                           | % Survival     | --                   | --               | 70 <sup>12</sup> /90 <sup>13</sup> | --                                            | --                               | 95 <sup>14</sup>        |

| Parameter | Units | Effluent Limitation |                   |                  | Monitoring Data<br>(August 2008 – April 2013) |                                           |                               |
|-----------|-------|---------------------|-------------------|------------------|-----------------------------------------------|-------------------------------------------|-------------------------------|
|           |       | Average<br>Monthly  | Average<br>Weekly | Maximum<br>Daily | Highest<br>Average<br>Monthly<br>Discharge    | Highest<br>Average<br>Weekly<br>Discharge | Highest<br>Daily<br>Discharge |

- 1 Represents the maximum observed average daily flow.
- 2 Represents the minimum observed percent removal.
- 3 Applied as a 4-day average effluent limitation.
- 4 Applied as a 1-hour average effluent limitation.
- 5 Applied as an annual average effluent limitation.
- 6 Represents the maximum observed annual average concentration.
- 7 Applied as a 7-day median effluent limitation.
- 8 Not to be exceeded more than once in any 30-day period.
- 9 Applied as an instantaneous maximum effluent limitation.
- 10 Applied as a daily average effluent limitation.
- 11 Effluent turbidity shall not exceed 5 NTU more than 5% of the time within a 24-hour period.
- 12 Minimum for any one bioassay.
- 13 Median for any three consecutive bioassays.
- 14 Represents the minimum observed percent survival.

#### D. Compliance Summary

1. The Central Valley Water Board issued Administrative Civil Liability (ACL) Complaint Order R5-2010-0510 on 1 March 2010 which proposed to assess a civil liability of \$9,000 against the Discharger for violations of effluent limitations for total coliform organisms during the term of Order R5-2008-0077-01. The Discharger settled the ACL by payment.
2. A Compliance and Enforcement Inspection (CEI) was conducted at the Facility on 19 February 2009. The inspection produced the following adverse findings:
  - a. Standard Provision VI.A.2.n of Order R5-2008-0077-01 states that laboratories performing analyses must be identified on all monitoring reports. The self-monitoring reports (SMRs) submitted by the Discharger did not identify the laboratories performing the analyses.
  - b. Attachment D, Standard Provision V.B.1 of Order R5-2008-0077-01 requires all applications, reports, or information submitted to the Central Valley Water Board to include a certification statement. The *Ultraviolet (UV) Disinfection System Operations Plan for the Dry Creek Wastewater Treatment Plant*, dated 22 January 2009 did not contain the required certification statement.
  - c. Attachment E, Monitoring and Reporting Program Provision X.B.6 of Order R5-2008-0077-01 requires that the letter transmitting the SMRs contain the penalty of perjury statement described in the Standard Provisions. The monthly SMRS did not contain the required penalty of perjury statement.
  - d. Attachment A, Definitions of Order R5-2008-0077-01 includes a definition for 'Daily Discharge', which states "*The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day. For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.*" The influent and effluent

composite samples are taken over the 24-hour period of 6:30 AM to 6:30 AM (i.e., not a calendar day). The Discharger reports the analytical result for the 24-hour period as the result for the calendar day in which the 24-hour period began, not when the period ends, as required.

- e. Attachment E, Monitoring and Reporting Program Provision I.D of Order R5-2008-0077-01 includes a requirement to calibrate all flow measurement devices at least once per year to ensure continued accuracy of the device. The Discharger had not calibrated the low-flow effluent flow measurement device (sonic level sensor) within the past year. As noted in a previous inspection report the Discharger had not calibrated the device since it was installed "several years" ago.
3. A CEI was conducted at the Facility on 8 December 2009. The inspection produced the following adverse finding:
- a. Attachment E, Monitoring and Reporting Program Provision I.A. of Order R5-2008-0077-01 requires all samples and measurements shall be representative of the volume and nature of the monitored discharge. Table E-1 describes Monitoring Location EFF-001 as location(s) representative of the final effluent from the Facility. Table E-3 requires a once/day grab sample for total coliform organisms at Monitoring Location EFF-001. The Discharger collects the once/day grab at the end of the UV disinfection channel; however, only one of the channels is sampled daily prior to the mixing of the UV disinfected effluent from remaining channels, thereby resulting in the possible collection of a sample that is not representative of the discharge. The Facility representative stated that the sampling of the channels is rotated daily and that the results of a study performed by the Discharger documented that sampling in this manner (only one channel daily for reporting daily coliform) is representative of the effluent being discharged from the entire UV disinfection system. The Facility representative also stated that sampling in the sump after all the channels mix could result in a non-representative sample due to regrowth in the sump. Contamination of the samples collected at the end of the UV disinfection system has been a problem.
4. A CEI was conducted at the Facility on 15 December 2010. The inspection produced the following adverse findings:
- a. Attachment E, Monitoring and Reporting Program Section X.B.6 of Order R5-2008-0077-01 requires that *"A letter transmitting the self-monitoring reports shall accompany each report. Such a letter shall include a discussion of requirement violations found during the reporting period, and actions taken or planned for correcting noted violations, such as operation or facility modifications."* The Discharger did not identify that the temperature of the receiving water downstream of Discharge Point 001 was more than 5°F greater than the temperature of the receiving water upstream of the discharge point on four occasions in November 2009 and once in December 2009.
  - b. Receiving Water Limitation V.A.15.b of Order R5-2008-0077-01 requires that the Facility's discharge must not cause the *"natural temperature [of the receiving water] to exceed a daily average temperature of 15.6°C (60°F) and a daily maximum temperature of 16.7°C (62°F) between 1 November through 31 March."* During a review of the November 2009 SMR, the inspectors noted exceedances of the daily average temperature requirements for the receiving water on 2 November 2009 and 13 November 2009, and were 62°F and 61°F, respectively. Because only one sample for temperature was collected on each day, the reported value represents both the daily average and daily maximum temperature.

- c. Receiving Water Limitation V.A.15.a of Order R5-2008-0077-01 requires that the Facility's discharge must not cause the "*natural temperature [of the receiving water] to be increased by more than 5°F.*" In accordance with the Monitoring and Reporting Program for Order R5-2008-0077-01, the Discharger collects temperature readings in the receiving water both upstream and downstream of Discharge Point 001 to Dry Creek at Monitoring Locations RSW-001 and RSW-002, respectively. During a review of the November and December 2009 SMRs, the inspectors noted that the temperature of the receiving water downstream of Discharge Point 001 was more than 5°F greater than the temperature of the receiving water upstream of the discharge point on four occasions in November 2009 and once in December 2009. The Discharger did not identify the above-described receiving water exceedances in the November and December 2009 SMRs.
5. A CEI was conducted at the Facility on 15 July 2011. The inspection produced the following adverse findings:
  - a. Influent and effluent pH are monitored both continuously and with daily grab samples. However, only the continuous monitoring results were being reported.
  - b. Vegetation was growing around the effluent flow measurement weir, which could interfere with the flow measurement.
  - c. Some of the secondary clarifiers contained scum between the launder and the outside wall. The automatic skimmer blade did not appear to reach the outside wall.
6. A CEI was conducted at the Facility on 26 February 2013. No adverse findings were noted during the inspection.

**E. Planned Changes – Not Applicable**

**III. APPLICABLE PLANS, POLICIES, AND REGULATIONS**

The requirements contained in this Order are based on the requirements and authorities described in this section.

**A. Legal Authorities**

This Order serves as WDR's pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters.

**B. California Environmental Quality Act (CEQA)**

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of Chapter 3 of CEQA, (commencing with section 21100) of Division 13 of the Public Resources Code.

**C. State and Federal Laws, Regulations, Policies, and Plans**

1. **Water Quality Control Plans.** Requirements of this Order specifically implement the applicable Water Quality Control Plans.
  - a. **Basin Plan.** The Central Valley Water Board adopted a Water Quality Control Plan, Fourth Edition (Revised October 2011), for the Sacramento and San Joaquin River Basins (hereinafter Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve

those objectives for all waters addressed through the plan. Requirements in this Order implement the Basin Plan.

The Basin Plan at II-2.00 states that the beneficial uses of any specifically identified water body generally apply to its tributary streams. The Basin Plan in Table II-1, Section II, does not specifically identify beneficial uses for Dry Creek, but does identify present and potential uses for the Sacramento River from the Colusa Basin Drain to the I Street Bridge, to which Dry Creek, via Natomas East Main Drainage Canal, is tributary. In addition, the Basin Plan implements State Water Board Resolution 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Thus, beneficial uses applicable to Dry Creek are as follows:

**Table F-3. Basin Plan Beneficial Uses**

| Discharge Point | Receiving Water Name | Beneficial Use(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001             | Dry Creek            | <u>Existing:</u><br>Municipal and domestic supply (MUN); agricultural supply for irrigation (AGR); water contact recreation, including canoeing and rafting (REC-1); non-contact water recreation (REC-2); warm freshwater aquatic habitat (WARM); cold freshwater aquatic habitat (COLD); warm and cold migration of aquatic organisms (MIGR); warm and cold spawning, reproduction, and/or early development (SPWN); wildlife habitat (WILD); and navigation (NAV). |
| 002             | Groundwater          | Municipal and domestic supply (MUN); Industrial service supply (IND); Industrial process supply (PRO); and Agricultural irrigation (AGR);                                                                                                                                                                                                                                                                                                                             |

2. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** U.S. EPA adopted the NTR on 22 December 1992, and later amended it on 4 May 1995 and 9 November 1999. About forty criteria in the NTR applied in California. On 18 May 2000, U.S. EPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on 13 February 2001. These rules contain federal water quality criteria for priority pollutants.
3. **State Implementation Policy.** On 2 March 2000, the State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on 28 April 2000, with respect to the priority pollutant criteria promulgated for California by the U.S. EPA through the NTR and to the priority pollutant objectives established by the Central Valley Water Board in the Basin Plan. The SIP became effective on 18 May 2000, with respect to the priority pollutant criteria promulgated by the U.S. EPA through the CTR. The State Water Board adopted amendments to the SIP on 24 February 2005, that became effective on 13 July 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.
4. **Antidegradation Policy.** Federal regulation 40 C.F.R. section 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State

Water Board Resolution 68-16. Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Central Valley Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. The permitted discharge must be consistent with the antidegradation provision of 40 C.F.R. section 131.12 and State Water Board Resolution 68-16.

5. **Anti-Backsliding Requirements.** Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 C.F.R. section 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed.
6. **Human Right to Water Act.** In compliance with Water Code section 106.3, it is the policy of the State of California that every human being has the right to safe, clean, affordable, and accessible water adequate for human consumption, cooking, and sanitary purposes. This Order promotes that policy by requiring discharges to meet maximum contaminant levels (MCLs) designed to protect human health and ensure that water is safe for domestic use.
7. **Endangered Species Act Requirements.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code, §§ 2050 to 2097) or the Federal Endangered Species Act (16 U.S.C.A. §§ 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the state. The discharger is responsible for meeting all requirements of the applicable Endangered Species Act.
8. **Emergency Planning and Community Right to Know Act.** Section 13263.6(a) of the Water Code, requires that *"the Regional Water Board shall prescribe effluent limitations as part of the waste discharge requirements of a POTW for all substances that the most recent toxic chemical release data reported to the state emergency response commission pursuant to Section 313 of the Emergency Planning and Community Right to Know Act of 1986 (42 U.S.C. Sec. 11023) (EPCRA) indicate as discharged into the POTW, for which the State Water Board or the Regional Water Board has established numeric water quality objectives, and has determined that the discharge is or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to, an excursion above any numeric water quality objective"*.

The most recent toxic chemical data report does not indicate any reportable off-site releases or discharges to the collection system for this Facility. Therefore, a reasonable potential analysis (RPA) based on information from EPCRA cannot be conducted. Based on information from EPCRA, there is no reasonable potential to cause or contribute to an excursion above any numeric water quality objectives included within the Basin Plan or in any State Water Board plan, so no effluent limitations are included in this permit pursuant to Water Code section 13263.6(a).

However, as detailed elsewhere in this Order, available effluent data indicate that there are constituents present in the effluent that have a reasonable potential to cause or contribute to exceedances of water quality standards and require inclusion of effluent limitations based on federal and state laws and regulations.

9. **Storm Water Requirements.** USEPA promulgated federal regulations for storm water on 16 November 1990 in 40 CFR Parts 122, 123, and 124. The NPDES Industrial Storm Water Program regulates storm water discharges from wastewater treatment facilities. Wastewater treatment plants are applicable industries under the storm water program and are obligated to comply with the federal regulations. The Discharger has submitted a Notice of Intent (NOI) and been approved for coverage under the State Water Board's Industrial Storm water General Order. Therefore, this Order does not regulate storm water.

**D. Impaired Water Bodies on CWA 303(d) List**

1. Under section 303(d) of the 1972 CWA, states, territories and authorized tribes are required to develop lists of water quality limited segments. The waters on these lists do not meet water quality standards, even after point sources of pollution have installed the minimum required levels of pollution control technology. On 11 October 2011 USEPA gave final approval to California's 2008-2010 section 303(d) List of Water Quality Limited Segments (WQLSs). The Basin Plan references this list of WQLSs, which are defined as "...those sections of lakes, streams, rivers or other fresh water bodies where water quality does not meet (or is not expected to meet) water quality standards even after the application of appropriate limitations for point sources (40 CFR Part 130, et seq.)." The Basin Plan also states, "Additional treatment beyond minimum federal standards will be imposed on dischargers to [WQLSs]. Dischargers will be assigned or allocated a maximum allowable load of critical pollutants so that water quality objectives can be met in the segment." Dry Creek is not listed as an impaired waterbody on the 2010 303(d) list. Dry Creek is tributary to the Natomas East Main Drainage Canal upstream of Arcade Creek, which is listed as an impaired waterbody for polychlorinated biphenyls (PCBs) on the 2010 303(d) list. The Natomas East Main Drainage Canal downstream of Arcade Creek is listed as an impaired waterbody for diazinon, mercury, and polychlorinated biphenyls (PCBs) on the 2010 303(d) list.
2. **Total Maximum Daily Loads (TMDLs).** USEPA requires the Central Valley Water Board to develop TMDLs for each 303(d) listed pollutant and water body combination. Table F-4, below, identifies the 303(d) listings and the status of each TMDL.

**Table F-4. 303 (d) List for Natomas East Main Drainage Canal**

| Pollutant | Potential Sources                                                    | Proposed TMDL Completion |
|-----------|----------------------------------------------------------------------|--------------------------|
| Diazinon  | Agriculture, urban runoff/storm sewers                               | 2008                     |
| Mercury   | Source Unknown                                                       | 2021                     |
| PCBs      | Urban runoff/storm sewers, agriculture, and industrial point sources | 2020                     |

3. The 303(d) listings and TMDLs have been considered in the development of the Order.

The Basin Plan contains a diazinon and chlorpyrifos waste load allocation (WLA) that applies to the Sacramento and Feather Rivers only. Yet, the Central Valley Water Board may apply the WLA to tributary waters to protect beneficial uses. However, analytical results of the effluent and receiving water monitoring indicated that concentrations of diazinon or chlorpyrifos were not present in either the effluent discharge or in the receiving water upstream of the discharge. Therefore this Order does not contain an effluent limitation based on the WLA.



This Order includes a mass-based effluent limitation for mercury designed to maintain the Facility's current level of treatment performance until a mercury TMDL is approved and implemented.

Effluent and receiving water monitoring indicated that concentrations of PCBs were not present in either the effluent discharge or in the receiving water upstream of the discharge.

#### **E. Other Plans, Policies and Regulations**

1. **Title 27.** The discharge authorized herein and the treatment and storage facilities associated with the discharge of treated municipal wastewater, except for discharges of residual sludge and solid waste, are exempt from the requirements of Title 27, California Code of Regulations (CCR), section 20005 *et seq* (hereafter Title 27). The exemption, pursuant to Title 27 CCR section 20090(a), is based on the following:
  - a. The waste consists primarily of domestic sewage and treated effluent;
  - b. The waste discharge requirements are consistent with water quality objectives; and
  - c. The treatment and storage facilities described herein are associated with a municipal wastewater treatment plant.

#### **IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS**

Effluent limitations and toxic and pretreatment effluent standards established pursuant to sections 301 (Effluent Limitations), 302 (Water Quality Related Effluent Limitations), 304 (Information and Guidelines), and 307 (Toxic and Pretreatment Effluent Standards) of the CWA and amendments thereto are applicable to the discharge.

The CWA mandates the implementation of effluent limitations that are as stringent as necessary to meet water quality standards established pursuant to state or federal law [33 U.S.C., §1311(b)(1)(C); 40 CFR 122.44(d)(1)]. NPDES permits must incorporate discharge limits necessary to ensure that water quality standards are met. This requirement applies to narrative criteria as well as to criteria specifying maximum amounts of particular pollutants. Pursuant to federal regulations, 40 CFR 122.44(d)(1)(i), NPDES permits must contain limits that control all pollutants that *"are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any state water quality standard, including state narrative criteria for water quality."* Federal regulations, 40 CFR 122.44(d)(1)(vi), further provide that *"[w]here a state has not established a water quality criterion for a specific chemical pollutant that is present in an effluent at a concentration that causes, has the reasonable potential to cause, or contributes to an excursion above a narrative criterion within an applicable State water quality standard, the permitting authority must establish effluent limits."*

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 CFR 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 CFR 122.44(d) requires that permits include WQBELs to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water where numeric water quality objectives have not been established. The Basin Plan at page IV-17.00 contains an implementation policy, "Policy for Application of Water Quality Objectives", that specifies that the Central Valley Water Board *"will, on a case-by-case basis, adopt numerical limitations in orders which will implement the narrative objectives."* This Policy complies with 40 CFR 122.44(d)(1). With respect to narrative objectives, the Central Valley

Water Board must establish effluent limitations using one or more of three specified sources, including: (1) USEPA's published water quality criteria, (2) a proposed state criterion (i.e., water quality objective) or an explicit state policy interpreting its narrative water quality criteria (i.e., the Central Valley Water Board's "Policy for Application of Water Quality Objectives")(40 CFR 122.44(d)(1)(vi)(A), (B) or (C)), or (3) an indicator parameter.

The Basin Plan includes numeric site-specific water quality objectives and narrative objectives for toxicity, chemical constituents, discoloration, radionuclides, and tastes and odors. The narrative toxicity objective states: "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*" (Basin Plan at III-8.00) The Basin Plan states that material and relevant information, including numeric criteria, and recommendations from other agencies and scientific literature will be utilized in evaluating compliance with the narrative toxicity objective. The narrative chemical constituents objective states that waters shall not contain chemical constituents in concentrations that adversely affect beneficial uses. At minimum, "...*water designated for use as domestic or municipal supply (MUN) shall not contain concentrations of chemical constituents in excess of the maximum contaminant levels (MCLs)*" in Title 22 of CCR. The Basin Plan further states that, to protect all beneficial uses, the Central Valley Water Board may apply limits more stringent than MCLs. The narrative tastes and odors objective states: "*Water shall not contain taste- or odor-producing substances in concentrations that impart undesirable tastes or odors to domestic or municipal water supplies or to fish flesh or other edible products of aquatic origin, or that cause nuisance, or otherwise adversely affect beneficial uses.*"

#### **A. Discharge Prohibitions**

1. **Prohibition III.A (No discharge or application of waste other than that described in this Order).** This prohibition is based on Water Code section 13260 that requires filing of a ROWD before discharges can occur. The Discharger submitted a ROWD for the discharges described in this Order; therefore, discharges not described in this Order are prohibited.
2. **Prohibition III.B (No bypasses or overflow of untreated wastewater, except under the conditions at CFR Part 122.41(m)(4)).** As stated in section I.G of Attachment D, Standard Provisions, this Order prohibits bypass from any portion of the treatment facility. Federal regulations, 40 CFR 122.41(m), define "bypass" as the intentional diversion of waste streams from any portion of a treatment facility. This section of the federal regulations, 40 CFR 122.41(m)(4), prohibits bypass unless it is unavoidable to prevent loss of life, personal injury, or severe property damage. In considering the Regional Water Board's prohibition of bypasses, the State Water Board adopted a precedential decision, Order No. WQO 2002-0015, which cites the federal regulations, 40 CFR 122.41(m), as allowing bypass only for essential maintenance to assure efficient operation.
3. **Prohibition III.C (No controllable condition shall create a nuisance).** This prohibition is based on Water Code section 13050 that requires water quality objectives established for the prevention of nuisance within a specific area. The Basin Plan prohibits conditions that create a nuisance.
4. **Prohibition III.D (No inclusion of pollutant free wastewater shall cause improper operation of the Facility's systems).** This prohibition is based on CFR Part 122.41 et seq. that requires the proper design and operation of treatment facilities.

## B. Technology-Based Effluent Limitations

### 1. Scope and Authority

Section 301(b) of the CWA and implementing U.S. EPA permit regulations at 40 C.F.R. section 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at 40 C.F.R. part 133.

Regulations promulgated in 40 C.F.R. section 125.3(a)(1) require technology-based effluent limitations for municipal Dischargers to be placed in NPDES permits based on Secondary Treatment Standards or Equivalent to Secondary Treatment Standards.

The Federal Water Pollution Control Act Amendments of 1972 (PL 92-500) established the minimum performance requirements for POTWs [defined in section 304(d)(1)]. Section 301(b)(1)(B) of that Act requires that such treatment works must, as a minimum, meet effluent limitations based on secondary treatment as defined by the U.S. EPA Administrator.

Based on this statutory requirement, U.S. EPA developed secondary treatment regulations, which are specified in 40 C.F.R. part 133. These technology-based regulations apply to all municipal wastewater treatment plants and identify the minimum level of effluent quality attainable by secondary treatment in terms of biochemical oxygen demand (BOD<sub>5</sub>), total suspended solids (TSS), and pH.

### 2. Applicable Technology-Based Effluent Limitations

- a. **BOD<sub>5</sub> and TSS.** Federal regulations, 40 CFR Part 133, establish the minimum weekly and monthly average level of effluent quality attainable by secondary treatment for BOD<sub>5</sub> and TSS. This Order establishes WQBELs that are equal to or more stringent than the secondary technology-based treatment described in 40 CFR Part 133 and are necessary to protect the beneficial uses of the receiving stream. (See section IV.C.3.b of this Attachment for the discussion on WQBELs for pathogens.) In addition, 40 CFR 133.102, in describing the minimum level of effluent quality attainable by secondary treatment, states that the 30-day average percent removal shall not be less than 85 percent. This Order contains a limitation requiring an average of 85 percent removal of BOD<sub>5</sub> and TSS over each calendar month.
- b. **Flow.** The Facility was designed to provide a tertiary level of treatment for up to a design flow of 18 MGD. Therefore, this Order contains an average dry weather discharge flow effluent limit of 18 MGD.
- c. **pH.** The secondary treatment regulations at 40 CFR Part 133 also require that pH be maintained between 6.0 and 9.0 standard units.

#### Summary of Technology-based Effluent Limitations Discharge Point No. 001

Table F-5. Summary of Technology-based Effluent Limitations

| Parameter                | Units | Effluent Limitations |                |               |                       |                       |
|--------------------------|-------|----------------------|----------------|---------------|-----------------------|-----------------------|
|                          |       | Average Monthly      | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Average Dry Weather Flow | MGD   | 18 <sup>1</sup>      | --             | --            | --                    | --                    |

| Parameter                                             | Units                | Effluent Limitations |                |               |                       |                       |
|-------------------------------------------------------|----------------------|----------------------|----------------|---------------|-----------------------|-----------------------|
|                                                       |                      | Average Monthly      | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Conventional Pollutants                               |                      |                      |                |               |                       |                       |
| Biochemical Oxygen Demand (5-day @ 20°C) <sup>2</sup> | mg/L                 | 30                   | 45             | --            | --                    | --                    |
|                                                       | lbs/day <sup>3</sup> | 4,500                | 6,800          | --            | --                    | --                    |
|                                                       | % Removal            | 85                   | --             | --            | --                    | --                    |
| pH <sup>2</sup>                                       | standard units       | --                   | --             | --            | 6.0                   | 9.0                   |
| Total Suspended Solids <sup>2</sup>                   | mg/L                 | 30                   | 45             | --            | --                    | --                    |
|                                                       | lbs/day              | 4,500                | 6,800          | --            | --                    | --                    |

<sup>1</sup> The average dry weather flow shall not exceed 18 MGD.

<sup>2</sup> Note that more stringent WQBELs for BOD<sub>5</sub>, pH, and TSS are applicable and are established as final effluent limitations in this Order (see section IV.C.3.b of this Fact Sheet).

<sup>3</sup> Based on an average dry weather flow of 18 MGD.

## C. Water Quality-Based Effluent Limitations (WQBELs)

### 1. Scope and Authority

CWA Section 301(b) and 40 C.F.R. section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains requirements, expressed as a technology equivalence requirement, more stringent than secondary treatment requirements that are necessary to meet applicable water quality standards. The rationale for these requirements, which consist of tertiary treatment or equivalent requirements or other provisions, is discussed in section IV.C.3.b.iv of the Fact Sheet.

Section 122.44(d)(1)(i) of 40 C.F.R. requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBELs must be established using: (1) U.S. EPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

### 2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. In addition, the Basin Plan implements State Water Board Resolution No. 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply.

The Basin Plan on page II-1.00 states: *“Protection and enhancement of existing and potential beneficial uses are primary goals of water quality planning...”* and with respect to disposal of wastewaters states that *“...disposal of wastewaters is [not] a prohibited use of waters of the State; it is merely a use which cannot be satisfied to the detriment of beneficial uses.”*

The federal CWA section 101(a)(2), states: *“it is the national goal that wherever attainable, an interim goal of water quality which provides for the protection and propagation of fish, shellfish, and wildlife, and for recreation in and on the water be achieved by July 1, 1983.”* Federal Regulations, developed to implement the requirements of the CWA, create a rebuttable presumption that all waters be designated as fishable and swimmable. Federal Regulations, 40 CFR sections 131.2 and 131.10, require that all waters of the State regulated to protect the beneficial uses of public water supply, protection and propagation of fish, shell fish and wildlife, recreation in and on the water, agricultural, industrial and other purposes including navigation. Section 131.3(e), 40 CFR, defines existing beneficial uses as those uses actually attained after 28 November 1975, whether or not they are included in the water quality standards. Federal Regulation, 40 CFR section 131.10 requires that uses be obtained by implementing effluent limitations, requires that all downstream uses be protected and states that in no case shall a state adopt waste transport or waste assimilation as a beneficial use for any waters of the United States.

- a. **Receiving Water and Beneficial Uses.** Refer to III.C.1. above for a complete description of the receiving water and beneficial uses.

Based on letters dated 31 July 1990 and 15 August 1990 from the California Department of Fish and Game (now the Department of Fish and Wildlife or DFW), DFW stated that Dry Creek is used by adult Chinook salmon migrating upstream to spawning areas and by young salmon migrating downstream, in November through March, in normal years.

The Basin Plan, on page IV-24, prohibits the direct discharge of municipal and industrial wastewater into the Sacramento River from the confluence with the Feather River to the Freeport Bridge. The Basin Plan also prohibits the direct discharge of wastewater to the American River. The effluent discharged to Dry Creek flows through western Placer County and northwestern Sacramento County before entering Natomas East Main Drainage Canal and further to the Sacramento River between the Feather River and the Freeport Bridge, or to the American River during some flood events. The discharge to the Sacramento or American Rivers from the Facility is not a direct discharge.

- b. **Effluent and Ambient Background Data.** The reasonable potential analysis (RPA), as described in section IV.C.3 of this Fact Sheet, was based on data from 1 August 2008 through 30 April 2013, which includes effluent and ambient background data submitted in SMRs.

The Discharger converted from chlorine disinfection to UV disinfection in May 2009. Therefore, effluent data collected prior to 1 May 2009 is not representative of the Facility's typical functional capabilities and was, therefore, not considered in the RPA for evaluating chlorine disinfection byproducts (i.e., chlorodibromomethane and dichlorobromomethane) and salinity.

- c. **Assimilative Capacity/Mixing Zone.** Based on the available information, the worst-case dilution is assumed to be zero to provide protection for the receiving water beneficial uses. The impact of assuming zero assimilative capacity within the

receiving water is that discharge limitations are end-of-pipe limits with no allowance for dilution within the receiving water.

- d. **Conversion Factors.** The CTR contains aquatic life criteria for arsenic, cadmium, chromium III, chromium VI, copper, lead, nickel, silver, and zinc which are presented in dissolved concentrations. USEPA recommends conversion factors to translate dissolved concentrations to total concentrations. The default USEPA conversion factors contained in Appendix 3 of the SIP were used to convert the applicable dissolved criteria to total recoverable criteria.
- e. **Hardness-Dependent CTR Metals Criteria.** The *California Toxics Rule* and the *National Toxics Rule* contain water quality criteria for seven metals that vary as a function of hardness. The lower the hardness the lower the water quality criteria. The metals with hardness-dependent criteria include cadmium, copper, chromium III, lead, nickel, silver, and zinc.

This Order has established the criteria for hardness-dependent metals based on the reasonable worst-case ambient hardness as required by the SIP<sup>1</sup>, the CTR<sup>2</sup> and State Water Board Order No. WQO 2008-0008 (City of Davis). The SIP and the CTR require the use of “receiving water” or “actual ambient” hardness, respectively, to determine effluent limitations for these metals. (SIP, § 1.2; 40 CFR § 131.38(c)(4)) The CTR does not define whether the term “ambient,” as applied in the regulations, necessarily requires the consideration of upstream as opposed to downstream hardness conditions. Therefore, where reliable, representative data are available, the hardness value for calculating criteria can be the downstream receiving water hardness, after mixing with the effluent (Order WQO 2008-0008, p. 11). The Central Valley Water Board thus has considerable discretion in determining ambient hardness (*Id.*, p.10).

As discussed below, scientific literature provides a reliable method for calculating protective hardness-dependent CTR criteria, considering all discharge conditions. This methodology produces hardness-dependent CTR criteria based on the reasonable worst-case downstream ambient hardness that ensure these metals do not cause receiving water toxicity under any downstream receiving water condition. Under this methodology, the Central Valley Water Board considers all hardness conditions that could occur in the ambient downstream receiving water after the effluent has mixed with the water body<sup>3</sup>. This ensures that effluent limitations are fully protective of aquatic life in all areas of the receiving water affected by the discharge under all flow conditions, at the fully mixed location, and throughout the water body including at the point of discharge into the water body.

- i. **Conducting the Reasonable Potential Analysis (RPA).** The SIP in Section 1.3 states, “*The RWQCB shall...determine whether a discharge may: (1) cause, (2) have a reasonable potential to cause, or (3) contribute to an excursion above any applicable priority pollutant criterion or objective.*” Section 1.3 provides a step-by-step procedure for conducting the RPA. The procedure

<sup>1</sup> The SIP does not address how to determine the hardness for application to the equations for the protection of aquatic life when using hardness-dependent metals criteria. It simply states, in Section 1.2, that the criteria shall be properly adjusted for hardness using the hardness of the receiving water.

<sup>2</sup> The CTR requires that, for waters with a hardness of 400 mg/L (as CaCO<sub>3</sub>), or less, the actual ambient hardness of the surface water must be used. It further requires that the hardness values used must be consistent with the design discharge conditions for design flows and mixing zones.

<sup>3</sup> All effluent discharges will change the ambient downstream metals concentration and hardness. It is not possible to change the metals concentration without also changing the hardness.

requires the comparison of the maximum effluent concentration (MEC) and maximum ambient background concentration to the applicable criterion that has been properly adjusted for hardness. Unless otherwise noted, for the hardness-dependent CTR metals criteria the following procedures were followed for properly adjusting the criterion for hardness when conducting the RPA.

- (a) The SIP requires WQBELs if the MEC is equal to or exceeds the applicable criterion, adjusted for hardness. For comparing the MEC to the applicable criterion, the “fully mixed” reasonable worst-case downstream ambient hardness was used to adjust the criterion. In this evaluation the portion of the receiving water affected by the discharge is analyzed. For hardness-dependent criteria, the hardness of the effluent has an impact on the determination of the applicable criterion in areas of the receiving water affected by the discharge. Therefore, for comparing the MEC to the applicable criterion, the reasonable worst-case downstream ambient hardness was used to adjust the criterion. For this situation it is necessary to consider the hardness of the effluent in determining the applicable hardness to adjust the criterion. The procedures for determining the applicable criterion after proper adjustment using the reasonable worst-case downstream ambient hardness is outlined in subsection ii, below.
- (b) The SIP requires WQBELs if the receiving water is impaired upstream (outside the influence) of the discharge, i.e., if the maximum ambient background concentration of a pollutant exceeds the applicable criterion, adjusted for hardness<sup>1</sup>. For comparing the maximum ambient background concentration to the applicable criterion, the reasonable worst-case upstream ambient hardness was used to adjust the criteria. This is appropriate, because this area is outside the influence of the discharge. Since the discharge does not impact the upstream hardness, the effect of the effluent hardness was not included in this evaluation.

- ii. **Calculating WQBELs.** The remaining discussion in this section relates to the development of WQBELs when it has been determined that the discharge has reasonable potential to cause or contribute to an exceedance of the CTR hardness-dependent metals criteria in the receiving water.

A 2006 Study<sup>2</sup> developed procedures for calculating the effluent concentration allowance (ECA)<sup>3</sup> for CTR hardness-dependent metals. The 2006 Study demonstrated that it is necessary to evaluate all discharge conditions (e.g., high and low flow conditions) and the hardness and metals concentrations of the effluent and receiving water when determining the appropriate ECA for these hardness-dependent metals. This method is superior to relying on downstream receiving water samples alone because it captures all possible mixed conditions in the receiving water. Both receiving water and effluent hardness vary based on flow and other factors, but the variability of receiving water and effluent hardness is sometimes independent. Using a calculated

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<sup>1</sup> The pollutant must also be detected in the effluent.

<sup>2</sup> Emerick, R.W.; Borroum, Y.; & Pedri, J.E., 2006. California and National Toxics Rule Implementation and Development of Protective Hardness Based Metal Effluent Limitations. WEFTEC, Chicago, Ill.

<sup>3</sup> The ECA is defined in Appendix 1 of the SIP (page Appendix 1-2). The ECA is used to calculate WQBELs in accordance with Section 1.4 of the SIP.

hardness value ensures that the Central Valley Water Board considers all possible mixed downstream values that may result from these two independent variables. Relying on receiving water sampling alone is less likely to capture all possible mixed downstream conditions.

The equation describing the total recoverable regulatory criterion, as established in the CTR<sup>1</sup>, is as follows:

$$\text{CTR Criterion} = \text{WER} \times (e^{m[\ln(H)]+b}) \text{ (Equation 1)}$$

Where:

H = hardness (as CaCO<sub>3</sub>)<sup>2</sup>

WER = water-effect ratio

m, b = metal- and criterion-specific constants

In accordance with the CTR, the default value for the WER is 1. A WER study must be conducted to use a value other than 1. The constants “m” and “b” are specific to both the metal under consideration, and the type of total recoverable criterion (i.e., acute or chronic). The metal-specific values for these constants are provided in the CTR at paragraph (b)(2), Table 1.

The equation for the ECA is defined in Section 1.4, Step 2, of the SIP and is as follows:

$$\text{ECA} = C \quad (\text{when } C \leq B)^3 \text{ (Equation 2)}$$

Where:

C = the priority pollutant criterion/objective, adjusted for hardness (see Equation 1, above)

B = the ambient background concentration

The 2006 Study demonstrated that the relationship between hardness and the calculated criteria is the same for some metals, so the same procedure for calculating the ECA may be used for these metals. The same procedure can be used for chronic cadmium, chromium III, copper, nickel, and zinc. These metals are hereinafter referred to as “Concave Down Metals”. “Concave Down” refers to the shape of the curve represented by the relationship between hardness and the CTR criteria in Equation 1. Another similar procedure can be used for determining the ECA for acute cadmium, lead, and acute silver, which are referred to hereafter as “Concave Up Metals”.

***ECA for Chronic Cadmium, Chromium III, Copper, Nickel, and Zinc*** – For Concave Down Metals (i.e., chronic cadmium, chromium III, copper, nickel, and zinc) the 2006 Study demonstrates that when the effluent is in compliance with the CTR criteria and the upstream receiving water is in compliance with the CTR criteria, any mixture of the effluent and receiving water will always be in compliance with the CTR criteria<sup>4</sup>. The 2006 Study proves that regardless of whether the effluent hardness is lower or greater than the upstream hardness, the reasonable worst-case flow condition is the effluent dominated condition

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<sup>1</sup> 40 CFR § 131.38(b)(2).

<sup>2</sup> For this discussion, all hardness values are in mg/L as CaCO<sub>3</sub>.

<sup>3</sup> The 2006 Study assumes the ambient background metals concentration is equal to the CTR criterion (i.e.  $C \leq B$ )

<sup>4</sup> 2006 Study, p. 5700



(i.e., no receiving water flow)<sup>1</sup>. Consequently, for Concave Down Metals, the CTR criteria have been calculated using the downstream ambient hardness under this condition.

The effluent hardness ranged from 71 mg/L to 123 mg/L, based on 56 samples collected between August 2008 and April 2013. The upstream receiving water hardness varied from 33 mg/L to 101 mg/L, based on 61 samples collected between August 2008 and April 2013. Under the effluent dominated condition, the reasonable worst-case downstream ambient hardness is 71 mg/L. As demonstrated in the example shown in Table F-6, below, using this hardness to calculate the ECA for all Concave Down Metals will result in WQBELs that are protective under all flow conditions, from the effluent dominated condition to high flow condition. This example for zinc assumes the following conservative conditions for the upstream receiving water:

- Upstream receiving water always at the lowest observed upstream receiving water hardness (i.e., 33 mg/L)
- Upstream receiving water zinc concentration always at the CTR criteria (i.e., no assimilative capacity).

Using these reasonable worst-case receiving water conditions, a simple mass balance (as shown in Equation 3, below) accounts for all possible mixtures of effluent and receiving water under all flow conditions.

$$C_{MIX} = C_{RW} \times (1-EF) + C_{Eff} \times (EF) \quad (\text{Equation 3})$$

Where:

$C_{MIX}$  = Mixed concentration (e.g. metals or hardness)  
 $C_{RW}$  = Upstream receiving water concentration  
 $C_{Eff}$  = Effluent concentration  
EF = Effluent Fraction

In this example, for zinc, for any receiving water flow condition (high flow to low flow), the fully-mixed downstream ambient zinc concentration is in compliance with the CTR criteria<sup>2</sup>.

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<sup>1</sup> There are two typographical errors in the 2006 Study in the discussion of Concave Down Metals when the effluent hardness is less than the receiving water hardness. The effluent and receiving water hardness were transposed in the discussion, but the correct hardness values were used in the calculations. The typographical errors were confirmed by the author of the 2006 Study, by email dated 1 April 2011, from Dr. Robert Emerick to Mr. James Marshall, Central Valley Water Board.

<sup>2</sup> This method considers the actual lowest observed upstream hardness and actual lowest observed effluent hardness to determine the reasonable worst-case ambient downstream hardness under all possible receiving water flow conditions. Table F-6 demonstrates that the receiving water is always in compliance with the CTR criteria at the fully-mixed location in the receiving water. It also demonstrates that the receiving water is in compliance with the CTR criteria for all mixtures from the point of discharge to the fully-mixed location. Therefore, a mixing zone is not used for compliance.

**Table F-6. Zinc ECA Evaluation**

| Lowest Observed Effluent Hardness                                                                          |      | 71 mg/L (as CaCO <sub>3</sub> )              |                                     |                             |                            |
|------------------------------------------------------------------------------------------------------------|------|----------------------------------------------|-------------------------------------|-----------------------------|----------------------------|
| Lowest Observed Upstream Receiving Water Hardness                                                          |      | 33 mg/L (as CaCO <sub>3</sub> )              |                                     |                             |                            |
| Highest Assumed Upstream Receiving Water Zinc Concentration                                                |      | 47 µg/L <sup>1</sup>                         |                                     |                             |                            |
| Zinc ECA <sub>chronic</sub> <sup>2</sup>                                                                   |      | 90 µg/L                                      |                                     |                             |                            |
| Effluent Fraction <sup>6</sup>                                                                             |      | Fully Mixed Downstream Ambient Concentration |                                     |                             |                            |
|                                                                                                            |      | Hardness <sup>3</sup><br>(mg/L)              | CTR Criteria <sup>4</sup><br>(µg/L) | Zinc <sup>5</sup><br>(µg/L) | Complies with CTR Criteria |
| High Flow<br><br>Low Flow | 1%   | 33                                           | 47                                  | 47                          | Yes                        |
|                                                                                                            | 5%   | 35                                           | 49                                  | 49                          | Yes                        |
|                                                                                                            | 15%  | 39                                           | 54                                  | 53                          | Yes                        |
|                                                                                                            | 25%  | 43                                           | 58                                  | 58                          | Yes                        |
|                                                                                                            | 50%  | 52                                           | 69                                  | 68                          | Yes                        |
|                                                                                                            | 75%  | 62                                           | 79                                  | 79                          | Yes                        |
|                                                                                                            | 100% | 71                                           | 90                                  | 90                          | Yes                        |

<sup>1</sup> Highest assumed upstream receiving water zinc concentration calculated using Equation 1 for chronic criterion at a hardness of 33 mg/L.

<sup>2</sup> ECA calculated using Equation 1 for chronic criterion at a hardness of 71 mg/L.

<sup>3</sup> Fully mixed downstream ambient hardness is the mixture of the receiving water and effluent hardness at the applicable effluent fraction using Equation 3.

<sup>4</sup> Fully mixed downstream ambient criteria are the chronic criteria calculated using Equation 1 at the mixed hardness.

<sup>5</sup> Fully mixed downstream ambient zinc concentration is the mixture of the receiving water and effluent zinc concentrations at the applicable effluent fraction using Equation 3.

<sup>6</sup> The effluent fraction ranges from 1% at the high receiving water flow condition, to 100% at the lowest receiving water flow condition (i.e., effluent dominated).

**ECA for Acute Cadmium, Lead, and Acute Silver** – For Concave Up Metals (i.e., acute cadmium, lead, and acute silver), the relationship between hardness and the metals criteria is different than for Concave Down Metals. The 2006 Study demonstrates that for Concave Up Metals, the effluent and upstream receiving water can be in compliance with the CTR criteria, but the resulting mixture may contain metals concentrations that exceed the CTR criteria and could cause toxicity. For these metals, the 2006 Study provides a mathematical approach to calculate the ECA that is protective of aquatic life, in all areas of the receiving water affected by the discharge, under all discharge and receiving water flow conditions (see Equation 4, below).

The ECA, as calculated using Equation 4, is based on the reasonable worst-case upstream receiving water hardness, the lowest observed effluent hardness, and assuming no receiving water assimilative capacity for metals (i.e., ambient background metals concentrations are at their respective CTR criterion). Equation 4 is not used in place of the CTR equation (Equation 1). Rather, Equation 4, which is derived using the CTR equation, is used as a direct approach for calculating the ECA. This replaces an iterative approach for calculating the ECA. The CTR equation has been used to evaluate the receiving water downstream of the discharge at all discharge and flow conditions to ensure the ECA is protective (e.g., see Table F-7).

$$ECA = \frac{m(H_e - H_{rw})}{H_{rw}} \left( e^{\frac{m \{ \ln(H_{rw}) \} + b}{\theta}} \right) + e^{m \{ \ln(H_{rw}) \} + b} \quad (\text{Equation 4})$$

Where:

- m, b= criterion specific constants (from CTR)  
 H<sub>e</sub> = lowest observed effluent hardness  
 H<sub>rw</sub> = reasonable worst-case upstream receiving water hardness

An example similar to the Concave Down Metals is shown for lead, a Concave Up Metal, in Table F-7, below. As previously mentioned, the lowest effluent hardness is 71 mg/L, while the upstream receiving water hardness ranged from 33 mg/L to 101 mg/L. In this case, the reasonable worst-case upstream receiving water hardness to use in Equation 4 to calculate the ECA is 33 mg/L.

Using the procedures discussed above to calculate the ECA for all Concave Up Metals will result in WQBELs that are protective under all potential effluent/receiving water flow conditions (high flow to low flow) and under all known hardness conditions, as demonstrated in Table F-7, for lead.

**Table F-7. Lead ECA Evaluation**

|                                                                          |      |                                                                  |                                            |                                    |                                       |
|--------------------------------------------------------------------------|------|------------------------------------------------------------------|--------------------------------------------|------------------------------------|---------------------------------------|
| <b>Lowest Observed Effluent Hardness</b>                                 |      | <b>71 mg/L</b>                                                   |                                            |                                    |                                       |
| <b>Reasonable Worst-case Upstream Receiving Water Hardness</b>           |      | <b>33 mg/L</b>                                                   |                                            |                                    |                                       |
| <b>Reasonable Worst-case Upstream Receiving Water Lead Concentration</b> |      | <b>0.78 µg/L<sup>1</sup></b>                                     |                                            |                                    |                                       |
| <b>Lead ECA<sub>chronic</sub><sup>2</sup></b>                            |      | <b>1.9 µg/L</b>                                                  |                                            |                                    |                                       |
| <b>Effluent Fraction<sup>6</sup></b>                                     |      | <b>Fully Mixed Downstream Ambient Concentration</b>              |                                            |                                    |                                       |
|                                                                          |      | <b>Hardness<sup>3</sup><br/>(mg/L)<br/>(as CaCO<sub>3</sub>)</b> | <b>CTR Criteria<sup>4</sup><br/>(µg/L)</b> | <b>Lead<sup>5</sup><br/>(µg/L)</b> | <b>Complies with<br/>CTR Criteria</b> |
| <div> <div>High Flow</div> <div>↓</div> <div>Low Flow</div> </div>       | 1%   | 33                                                               | 0.79                                       | 0.79                               | <b>Yes</b>                            |
|                                                                          | 5%   | 35                                                               | 0.83                                       | 0.83                               | <b>Yes</b>                            |
|                                                                          | 15%  | 39                                                               | 0.95                                       | 0.95                               | <b>Yes</b>                            |
|                                                                          | 25%  | 43                                                               | 1.1                                        | 1.1                                | <b>Yes</b>                            |
|                                                                          | 50%  | 52                                                               | 1.4                                        | 1.3                                | <b>Yes</b>                            |
|                                                                          | 75%  | 62                                                               | 1.7                                        | 1.6                                | <b>Yes</b>                            |
|                                                                          | 100% | 71                                                               | 2.1                                        | 1.9                                | <b>Yes</b>                            |

<sup>1</sup> Reasonable worst-case upstream receiving water lead concentration calculated using Equation 1 for chronic criterion at a hardness of 33 mg/L.

<sup>2</sup> ECA calculated using Equation 4 for chronic criteria.

<sup>3</sup> Fully mixed downstream ambient hardness is the mixture of the receiving water and effluent hardness at the applicable effluent fraction.

<sup>4</sup> Fully mixed downstream ambient criteria are the chronic criteria calculated using Equation 1 at the mixed hardness.

<sup>5</sup> Fully mixed downstream ambient lead concentration is the mixture of the receiving water and effluent lead concentrations at the applicable effluent fraction.

<sup>6</sup> The effluent fraction ranges from 1% at the high receiving water flow condition, to 100% at the lowest receiving water flow condition (i.e., effluent dominated).

Based on the procedures discussed above, Table F-8 lists all the CTR hardness-dependent metals and the associated ECA used in this Order.

**Table F-8. Summary of ECA Evaluations for CTR Hardness-dependent Metals**

| CTR Metals   | ECA (µg/L, total recoverable) <sup>1</sup> |         |
|--------------|--------------------------------------------|---------|
|              | acute                                      | chronic |
| Copper       | 10                                         | 7.0     |
| Chromium III | 1,300                                      | 160     |
| Cadmium      | 3.0                                        | 1.9     |
| Lead         | 49                                         | 1.9     |
| Nickel       | 350                                        | 39      |
| Silver       | 1.8                                        | --      |
| Zinc         | 90                                         | 90      |

<sup>1</sup> Metal criteria established as two significant figures in accordance with the CTR.

### 3. Determining the Need for WQBELs

- a. **Constituents with No Reasonable Potential.** WQBELs are not included in this Order for constituents that do not demonstrate reasonable potential (i.e., constituents were not detected in the effluent or receiving water); however, monitoring for those pollutants is established in this Order as required by the SIP. If the results of effluent monitoring demonstrate reasonable potential, this Order may be reopened and modified by adding an appropriate effluent limitation.

Most constituents with no reasonable potential are not discussed in this Order. However, the following constituents were found to have no reasonable potential after assessment of the data:

#### i. Aluminum

Aluminum is the third most abundant element in the earth's crust and is ubiquitous in both soils and aquatic sediments. When mobilized in surface waters, aluminum has been shown to be toxic to various fish species. However, the potential for aluminum toxicity in surface waters is directly related to the chemical form of aluminum present, and the chemical form is highly dependent on water quality characteristics that ultimately determine the mechanism of aluminum toxicity. Surface water characteristics, including pH, temperature, colloidal material, fluoride and sulfate concentrations, and total organic carbon, all influence aluminum speciation and its subsequent bioavailability to aquatic life. Calcium [hardness] concentrations in surface water may also reduce aluminum toxicity by competing with monomeric aluminum ( $Al^{3+}$ ) binding to negatively charged fish gills.

- (a) **WQO.** The Code of Federal Regulations promulgated criteria for priority toxic pollutants for California's surface waters as part of section 131.38 Establishment of Numeric Criteria for Priority Toxic Pollutants for the State of California (California Toxics Rule or CTR), including metals criteria. However, aluminum criteria were not promulgated as part of the CTR. Absent numeric aquatic life criteria for aluminum, WQBEL's in the Central Valley Region's NPDES permits are based on the Basin Plans' narrative toxicity objective. The Basin Plans' Policy for Application of Water Quality Objectives requires the Central Valley Water Board to consider, "*on a case-by-case basis, direct evidence of beneficial use impacts, all material and relevant information submitted by the discharger and other interested parties, and relevant numerical criteria and guidelines developed and/or published by other agencies and organizations. In considering such criteria, the Board evaluates whether the specific numerical criteria which*

*are available through these sources and through other information supplied to the Board, are relevant and appropriate to the situation at hand and, therefore, should be used in determining compliance with the narrative objective."* Relevant information includes, but is not limited to (1) USEPA Ambient Water Quality Criteria (NAWQC) and subsequent Correction, (2) site-specific conditions of Dry Creek, the receiving water, and (3) site-specific aluminum studies conducted by dischargers within the Central Valley Region. (Basin Plan, p. IV.-17.00; see also, 40 CFR 122.44(d)(vi).)

**USEPA NAWQC.** USEPA recommended the NAWQC aluminum acute criterion at 750 µg/L based on test waters with a pH of 6.5 to 9.0. USEPA also recommended the NAWQC aluminum chronic criterion at 87 µg/L based upon the following two toxicity tests. All test waters contained hardness at 12 mg/L as CaCO<sub>3</sub>.

- (1) Acute toxicity tests at various aluminum doses were conducted in various acidic waters (pH 6.0 – 6.5) on 159- and 160-day old striped bass. The 159-day old striped bass showed no mortality in waters with pH at 6.5 and aluminum doses at 390 µg/L, and the 160-day old striped bass showed 58% mortality at a dose of 174.4 µg/L in same pH waters. However, the 160-day old striped bass showed 98% mortality at aluminum dose of 87.2 µg/L in waters with pH at 6.0, which is USEPA's basis for the 87 µg/L chronic criterion. The varied results draw into question this study and the applicability of the NAWQC chronic criterion of 87 µg/L.
- (2) Chronic toxicity effects on 60-day old brook trout were evaluated in circumneutral pH waters (6.5-6.9 pH) in five cells at various aluminum doses (4, 57, 88, 169, and 350 µg/L). Chronic evaluation started upon hatching of eyed eggs of brook trout, and their weight and length were measure after 45 days and 60 days. The 60-day old brook trout showed 24% weight loss at 169 µg/L of aluminum and 4% weight loss at 88 µg/L of aluminum, which is the basis for USEPA's chronic criteria. Though this test study shows chronic toxic effects 4% reduction in weight after exposure for 60-days, the chronic criterion is based on 4-day exposure; so again, the applicability of the NAWQC chronic criterion of 87 µg/L is questionable.

**Site-specific Conditions.** Effluent and receiving water monitoring data indicate that the pH and hardness values are not similar to the low pH and hardness conditions under which the chronic criterion for aluminum was developed, as shown in the table below, and therefore, the Central Valley Water Board does not expect aluminum to be as reactive in Dry Creek as in the previously described toxicity tests. The pH of Dry Creek, the receiving water, ranged from 7.0 to 9.0 with an average of 7.58 based on 247 monitoring results obtained between August 2008 and April 2013. These water conditions are circumneutral pH where aluminum is predominately in the form of Al(OH)<sub>3</sub> and non-toxic to aquatic life. The hardness of Dry Creek ranged from 33 mg/L to 101 mg/L based on 61 samples, which is above the conditions, and thus less toxic, than the tests used to develop the chronic criterion. Dry Creek supports aquatic species such as Chinook salmon and steelhead (rainbow trout). Brook trout and striped bass have not been surveyed in Dry Creek

(<http://bios.dfg.ca.gov/>), and brook trout is not expected to be present since they reside in higher elevation lakes and streams. However, striped bass, although non-native to California, are present within the Delta and thus could potentially migrate upstream to Dry Creek. Nevertheless, as described in detail in the following section (i.e. Local Environmental Conditions and Studies), site-specific aluminum toxicity tests show that aluminum does not have toxic effects to aquatic species in receiving waters with similar water quality characteristics.

| Parameter                               | Units          | Test Conditions for Applicability of Chronic Criterion | Effluent  | Receiving Water |
|-----------------------------------------|----------------|--------------------------------------------------------|-----------|-----------------|
| pH                                      | standard units | 6.0 – 6.5                                              | 6.5 – 7.9 | 7.0 – 9.0       |
| Hardness, Total (as CaCO <sub>3</sub> ) | mg/L           | 12                                                     | 71 – 123  | 33 – 101        |
| Aluminum, Total Recoverable             | µg/L           | 87.2 - 390                                             | 10 – 360  | 103 – 285       |

**Local Environmental Conditions and Studies.** Twenty-one site-specific aluminum toxicity tests have been conducted within the Central Valley Region. The pH and hardness of Dry Creek are similar, as shown in the table below, and thus the results of these site-specific aluminum toxicity tests are relevant and appropriate for Dry Creek. As shown in the following table, all EC<sub>50</sub> toxicity study result values are at concentrations of aluminum above 5,000 µg/L. Thus, the toxic effects of aluminum in surface waters within the Central Valley Region, including Dry Creek, is less toxic (or less reactive) to aquatic species than demonstrated in the toxicity tests that USEPA used for the basis of establishing the chronic criterion of 87 µg/L. This new information, and review of the toxicity tests USEPA used to establish the chronic criterion, indicates that 87 µg/L is overly stringent and not applicable to Dry Creek.

| Discharger (City) | Species                   | Test Waters            | Hardness Value       | Total Aluminum EC <sub>50</sub> Value | pH        | WER   |
|-------------------|---------------------------|------------------------|----------------------|---------------------------------------|-----------|-------|
| Auburn            | <i>Ceriodaphnia dubia</i> | Effluent               | 99                   | >5270                                 | 7.44      | >19.3 |
|                   | “ “                       | Surface Water          | 16                   | >5160                                 | 7.44      | >12.4 |
| Manteca           | “ “                       | Surface Water/Effluent | 124                  | >8800                                 | 9.14      | N/C   |
|                   | “ “                       | Effluent               | 117                  | >8700                                 | 7.21      | >27.8 |
|                   | “ “                       | Surface Water          | 57                   | 7823                                  | 7.58      | 25.0  |
|                   | “ “                       | Effluent               | 139                  | >9500                                 | 7.97      | >21.2 |
|                   | “ “                       | Surface Water          | 104                  | >11000                                | 8.28      | >24.5 |
|                   | “ “                       | Effluent               | 128                  | >9700                                 | 7.78      | >25.0 |
|                   | “ “                       | Surface Water          | 85                   | >9450                                 | 7.85      | >25.7 |
|                   | “ “                       | Effluent               | 106                  | >11900                                | 7.66      | >15.3 |
|                   | “ “                       | Surface Water          | 146                  | >10650                                | 7.81      | >13.7 |
| Modesto           | “ “                       | Surface Water/Effluent | 120/156              | 31604                                 | 8.96      | 211   |
| Yuba City         | “ “                       | Surface Water/Effluent | 114/164 <sup>1</sup> | >8000                                 | 7.60/7.46 | >53.5 |
| Placer County     | “ “                       | Effluent               | 150                  | >5000                                 | 7.4 – 8.7 | >13.7 |
| Manteca           | <i>Daphnia magna</i>      | Surface Water/Effluent | 124                  | >8350                                 | 9.14      | N/C   |
| Modesto           | “ “                       | Surface Water/Effluent | 120/156              | >11900                                | 8.96      | >79.6 |
| Yuba City         | “ “                       | Surface Water/Effluent | 114/164 <sup>1</sup> | >8000                                 | 7.60/7.46 | >53.5 |

| Discharger<br>(City) | Species                                       | Test Waters            | Hardness<br>Value    | Total<br>Aluminum<br>EC <sub>50</sub> Value | pH        | WER   |
|----------------------|-----------------------------------------------|------------------------|----------------------|---------------------------------------------|-----------|-------|
| Manteca              | <i>Oncorhynchus mykiss</i><br>(rainbow trout) | Surface Water/Effluent | 124                  | >8600                                       | 9.14      | N/C   |
| Auburn               | " "                                           | Surface Water          | 16                   | >16500                                      | 7.44      | N/C   |
| Modesto              | " "                                           | Surface Water/Effluent | 120/156              | >34250                                      | 8.96      | >229  |
| Yuba City            | " "                                           | Surface Water/Effluent | 114/164 <sup>1</sup> | >8000                                       | 7.60/7.46 | >53.5 |

<sup>1</sup> Hardness values may be biased high because the EDTA titrimetric method is subject to interferences that measure as hardness (barium, cadmium, lead, manganese, strontium, and zinc will be measured as hardness) producing hardness numbers that are likely to be greater than the calculation of hardness based upon the ICP analysis of calcium and magnesium. Upstream receiving water hardness ranged from 30 to 50.9 mg/L as CaCO<sub>3</sub> between January 2008 and August 2011. Furthermore, the upstream receiving water hardness was 37 mg/L as CaCO<sub>3</sub> on 4 October 2005, 7 days prior to the Feasibility Assessment (first phase of a Water Effects Ratio study) sample collection date of 11 October 2005. It is likely that matrix interferences from other metals were responsible for the unexpected hardness values reported by Pacific EcoRisk.

The Discharger has not conducted a toxicity test for aluminum; however, the City of Auburn conducted two toxicity tests in Auburn Ravine, shown highlighted in the previous table. The City of Auburn is located at an elevation of approximately 1,400 feet above sea level, and is surrounded by forest. As shown, the test water quality characteristics of Auburn Ravine are similar to Dry Creek, with the pH at 7.4 and hardness at 16 mg/L as CaCO<sub>3</sub> in comparison to the mean pH at 7.58 and the minimum hardness at 33 mg/L (mean hardness at 65 mg/L) as CaCO<sub>3</sub>, respectively. Thus, based on these two similar primary water quality characteristics (pH and hardness) that drive aluminum speciation, the aluminum toxicity within Auburn Ravine is expected to be similar in Dry Creek. Therefore, the Auburn Ravine aluminum toxicity test study is relevant and appropriate in this case for use in determining the specific numerical criteria to be used in determining compliance with the Basin Plan's narrative toxicity objective. The Auburn Ravine aluminum toxicity study resulted in a site-specific aluminum objective at 1,079 µg/L. Thus, these results support the conclusion that the 87 µg/L chronic criterion is overly stringent for Dry Creek.

Order R5-2008-0077-01 contained effluent limitations of aluminum based on the State of California Department of Public Health (DPH) Secondary MCLs, established to assist public drinking water systems in managing their drinking water for aesthetic conditions such as taste, color, and odor. The Secondary MCL for aluminum is 200 µg/L. U.S. EPA has also adopted a NAWQC acute criterion of 750 µg/L for the protection of aquatic life.

Order R5-2008-0077-01 required the Discharger to complete and submit a site-specific study for aluminum. The Discharger's 30 June 2010 *Aluminum Site-Specific Study for City of Roseville Pleasant Grove and Dry Creek WWTPs* summarized receiving water pH and hardness conditions and compared application of the chronic criterion in several Central Valley Permits. Based on the study results, the Discharger concluded that the NAWQC chronic criterion of 87 µg/L was overly stringent for Dry Creek, and that the Secondary MCL and NAWQC acute criterion are adequately protective of the beneficial uses in Dry Creek.

- (b) **RPA Results.** For priority pollutants, the SIP dictates the procedures for conducting the RPA. Aluminum is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used its judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent. The most stringent objective is the Secondary MCL, which is derived from human welfare considerations (e.g., taste, odor, laundry staining), not for toxicity. Secondary MCLs are drinking water standards contained in Title 22 of the California Code of Regulations. Title 22 requires compliance with these standards on an annual average basis, when sampling at least quarterly. To be consistent with how compliance with the standards is determined, the RPA was conducted based on the calendar annual average effluent aluminum concentrations.

The maximum annual average effluent concentration for aluminum was 121 µg/L based on 56 samples collected between August 2008 and April 2013. Thus, annual average effluent concentrations for aluminum are less than the concentrations in the receiving water and below the Secondary MCL. Therefore, the Central Valley Water Board finds the discharge does not have reasonable potential to cause or contribute to an exceedance in the receiving water and the Facility is adequately controlling the discharge of aluminum. Since the discharge does not demonstrate reasonable potential, the effluent limitations for aluminum have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

ii. **Cadmium**

- (a) **WQO.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for cadmium. These criteria for cadmium are presented in dissolved concentrations, as 1-hour acute criteria and 4-day chronic criteria. USEPA recommends conversion factors to translate dissolved concentrations to total concentrations. Default USEPA translators were used for the receiving water and effluent. Order R5-2008-0077-01 included effluent limitations for cadmium based on the CTR criteria.
- (b) **RPA Results.** Section IV.C.2.e of this Fact Sheet includes procedures for conducting the RPA for hardness-dependent CTR metals, such as cadmium. The CTR includes hardness-dependent criteria for cadmium for the receiving water. The RPA was conducted using the upstream receiving water hardness to calculate the criteria for comparison to the maximum ambient background concentration, and likewise using the reasonable worst-case downstream hardness to compare the MEC. The table below shows the specific criteria used for the RPA.

**Table F-9. Cadmium RPA**

|                 | CTR Chronic Criterion<br>(Total Recoverable) | Maximum Concentration<br>(Total Recoverable) | Reasonable Potential?<br>(Y/N) |
|-----------------|----------------------------------------------|----------------------------------------------|--------------------------------|
| Receiving Water | 1.0 µg/L <sup>1</sup>                        | 0.26 µg/L                                    | No <sup>2</sup>                |



|          | CTR Chronic<br>Criterion<br>(Total Recoverable) | Maximum<br>Concentration<br>(Total Recoverable) | Reasonable<br>Potential?<br>(Y/N) |
|----------|-------------------------------------------------|-------------------------------------------------|-----------------------------------|
| Effluent | 1.9 µg/L <sup>3</sup>                           | 0.1 µg/L                                        | No <sup>4</sup>                   |

<sup>1</sup> Based on lowest observed upstream hardness of 33 mg/L (as CaCO<sub>3</sub>)

<sup>3</sup> Per Section 1.3, step 6 of the SIP.

<sup>2</sup> Based on reasonable worst-case downstream hardness of 71 mg/L (as CaCO<sub>3</sub>)

<sup>4</sup> Per Section 1.3, step 4 of the SIP.

Based on the available data, cadmium in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR chronic criterion, and the WQBELs for cadmium have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

### iii. Carbon Tetrachloride

- (a) **WQO.** The CTR includes a criterion of 0.25 µg/L for carbon tetrachloride for the protection of human health for waters from which both water and organisms are consumed. Order R5-2008-0077-01 included effluent limitations for carbon tetrachloride based on the CTR criterion.
- (b) **RPA Results.** Carbon tetrachloride was not detected in the effluent based on 56 samples collected between August 2008 and April 2013 (minimum MDL 0.16 µg/L, minimum RL 0.5 µg/L). Carbon tetrachloride was not detected in the upstream receiving water based on four samples collected between August 2008 and April 2013 (minimum MDL 0.16 µg/L, minimum RL 0.5 µg/L). Therefore, carbon tetrachloride in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR criterion, and the effluent limitations for carbon tetrachloride have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

### iv. Chlorine Residual

- (a) **WQO.** USEPA developed NAWQC for protection of freshwater aquatic life for chlorine residual. The recommended 4-day average (chronic) and 1-hour average (acute) criteria for chlorine residual are 0.011 mg/L and 0.019 mg/L, respectively. These criteria are protective of the Basin Plan's narrative toxicity objective. Order R5-2008-0077-01 included effluent limitations for chlorine residual based on the NAWQC criteria.
- (b) **RPA Results.** The Discharger converted from chlorine disinfection to UV disinfection in May 2009 and no longer uses chlorine-containing compounds within the treatment process. Therefore, the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of the Basin Plan's narrative toxicity objective and the effluent limitations for chlorine residual have not been retained. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

v. **Chlorodibromomethane**

- (a) **WQO.** The CTR includes criteria of 0.41 µg/L for chlorodibromomethane for the protection of human health for waters where both water and organisms are consumed. Order R5-2008-0077-01 included effluent limitations for chlorodibromomethane based on the CTR criterion.
- (b) **RPA Results.** Chlorodibromomethane was detected once in the effluent based on 48 samples collected between May 2009 and April 2013 (minimum MDL 0.17 µg/L, minimum RL 0.5 µg/L). The single detection occurred on 2 July 2010 at a concentration of 0.90 µg/L. In the cover letter for the July 2010 SMR, the Discharger noted that this sample was collected on the same day in which maintenance on the UV equipment occurred involving the use of a chlorinated detergent. The chlorinated detergent was inadvertently used by a new maintenance staff person who was not fully trained on the UV maintenance procedures. The Discharger subsequently implemented procedures to ensure that chlorine-containing compounds are not used in the treatment process. SIP Section 1.2 states that the Central Valley Water Board has the discretion to consider if any data are inappropriate or insufficient for use in implementing the SIP. Therefore, in accordance with Section 1.2 of the SIP, the Central Valley Water Board has determined that the 2 July 2010 sample is inappropriate and insufficient to be used in the RPA. Chlorodibromomethane was not detected in the effluent in the remaining 47 samples. Chlorodibromomethane was not detected in the upstream receiving water based on four samples collected between August 2008 and April 2013 (minimum MDL 0.17 µg/L, minimum RL 0.5 µg/L). Therefore, chlorodibromomethane in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR human health criterion, and the WQBELs for the chlorodibromomethane have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

vi. **Copper**

- (a) **WQO.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for copper. These criteria for copper are presented in dissolved concentrations, as 1-hour acute criteria and 4-day chronic criteria. USEPA recommends conversion factors to translate dissolved concentrations to total concentrations. Default USEPA translators were used for the receiving water and effluent.
- (b) **RPA Results.** Section IV.C.2.e of this Fact Sheet includes procedures for conducting the RPA for hardness-dependent CTR metals, such as copper. When conducting the RPA for CTR metals with hardness-dependent criteria, the maximum ambient background concentration is compared with the CTR criteria calculated using the upstream receiving water hardness. The maximum observed upstream total recoverable copper concentration was 3.9 µg/L on 9 March 2011. The observed hardness on that day was 90 mg/L, which correlates with a chronic criterion of 8.5 µg/L (as total recoverable). Therefore, the upstream receiving water does not exceed the applicable criteria for copper. As shown in the table below, an evaluation of the known situations where metals and hardness were

measured indicates that there are no instances where the upstream receiving water exceeded the criteria for copper.

**Table F-10. Copper Receiving Water CTR Criteria Comparison**

| Sample Date       | RW Hardness (mg/L) | Copper                       |                  |
|-------------------|--------------------|------------------------------|------------------|
|                   |                    | CTR Chronic Criterion (µg/L) | RW Copper (µg/L) |
| 16 September 2010 | 33                 | 3.6                          | 1.9              |
| 3 November 2010   | 64                 | 6.4                          | 2.1              |
| 9 March 2011      | 90                 | 8.5                          | 3.9              |
| 9 June 2011       | 72                 | 7.0                          | 2.6              |

Likewise, the reasonable worst-case downstream hardness was used to compare to the MEC. The table below shows the specific criteria used for the RPA.

**Table F-11. Copper RPA**

|                 | CTR Chronic Criterion (Total Recoverable) | Maximum Concentration (Total Recoverable) | Reasonable Potential? (Y/N) |
|-----------------|-------------------------------------------|-------------------------------------------|-----------------------------|
| Receiving Water | 8.5 µg/L <sup>1</sup>                     | 3.9 µg/L                                  | No <sup>2</sup>             |
| Effluent        | 7.0 µg/L <sup>3</sup>                     | 4.2 µg/L                                  | No <sup>4</sup>             |

<sup>1</sup> Based on upstream hardness of 90 mg/L (as CaCO<sub>3</sub>) which was collected on the same day (9 March 2011) as the maximum observed upstream receiving water copper concentration.

<sup>2</sup> Per Section 1.3, step 6 of the SIP.

<sup>3</sup> Based on reasonable worst-case effluent hardness of 71 mg/L (as CaCO<sub>3</sub>).

<sup>4</sup> Per Section 1.3, step 4 of the SIP.

Based on the available data, copper in the discharge does not have a reasonable potential to cause or contribute to an in-stream excursion above the CTR criteria for the protection of freshwater aquatic life criterion.

#### vii. Cyanide

- (a) **WQO.** The CTR includes maximum 1-hour average and 4-day average criteria of 5.2 µg/L and 22 µg/L, respectively, for cyanide, for the protection of freshwater aquatic life. Order R5-2008-0077-01 included effluent limitations for cyanide based on the CTR freshwater aquatic life criteria. Order R5-2008-0077-01 included effluent limitations for cyanide based on the CTR criteria.
- (b) **RPA Results.** The MEC for cyanide was 3.2 µg/L based on 56 samples collected between August 2008 and April 2013 (minimum MDL 0.6 µg/L, minimum RL 3 µg/L). The maximum observed upstream receiving water concentration for cyanide was 1.3 µg/L based on four samples collected between August 2008 and April 2013 (minimum MDL 0.6 µg/L, minimum RL 3 µg/L). Therefore, cyanide in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR criteria, and the effluent limitations for cyanide have not

been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

viii. **Dichlorobromomethane**

- (a) **WQO.** The CTR includes criteria of 0.56 µg/L for the protection of human health for waters where both water and organisms are consumed. Order R5-2008-0077-01 included effluent limitations for dichlorobromomethane based on the CTR criterion.
- (b) **RPA Results.** Dichlorobromomethane was detected once in the effluent based on 48 samples collected between May 2009 and April 2013 (minimum MDL 0.16 µg/L, minimum RL 0.5 µg/L). The single detection occurred on 2 July 2010 at a concentration of 11 µg/L. In the cover letter for the July 2010 SMR, the Discharger noted that this sample was collected on the same day in which maintenance on the UV equipment occurred involving the use of a chlorinated detergent. The chlorinated detergent was inadvertently used by a new maintenance staff person who was not fully trained on the UV maintenance procedures. The Discharger subsequently implemented procedures to ensure that chlorine-containing compounds are not used in the treatment process. SIP section 1.2 states that the Central Valley Water Board has the discretion to consider if any data are inappropriate or insufficient for use in implementing the SIP. Therefore, in accordance with Section 1.2 of the SIP, the Central Valley Water Board has determined that the 2 July 2010 sample is inappropriate and insufficient to be used in the RPA. Dichlorobromomethane was not detected in the effluent in the remaining 47 samples. Dichlorobromomethane was not detected in the upstream receiving water based on four samples collected between August 2008 and April 2013 (minimum MDL 0.16 µg/L, minimum RL 0.5 µg/L). Therefore, dichlorobromomethane in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR human health criterion, and the WQBELs for the dichlorobromomethane have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

ix. **Iron**

- (a) **WQO.** DPH has developed a Secondary MCL for total recoverable iron of 300 µg/L. Order R5-2008-0077-01 has established an effluent limitation for iron based on the Secondary MCL.
- (b) **RPA Results.** For priority pollutants, the SIP dictates the procedures for conducting the RPA. Iron is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used its judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent. The most stringent objective is the Secondary MCL, which is derived from human welfare considerations (e.g., taste, odor, laundry staining), not for toxicity. Secondary MCLs are drinking water standards contained in Title 22 of the California Code of Regulations. Title 22 requires compliance with these standards on an annual average basis, when sampling at least quarterly.

To be consistent with how compliance with the standards is determined, the RPA was conducted based on the calendar annual average effluent iron concentrations.

The maximum annual average effluent concentration for iron was 80 µg/L based on 56 samples collected between August 2008 and April 2013. Although the receiving water contains iron exceeding the Secondary MCL, the effluent iron is consistently less than the concentrations in the receiving water and below the Secondary MCL. Therefore, the Central Valley Water Board finds the discharge does not have reasonable potential to cause or contribute to an exceedance in the receiving water and the Facility is adequately controlling the discharge of iron. Since the discharge does not demonstrate reasonable potential, the effluent limitation for iron has not been retained in this Order. Removal of this effluent limitation is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

x. **Manganese, Total Recoverable**

- (a) **WQO.** USEPA has developed a Secondary MCL for total recoverable manganese of 50 µg/L. Order R5-2008-0077-01 established an effluent limitation for manganese based on the Secondary MCL.
- (b) **RPA Results.** For priority pollutants, the SIP dictates the procedures for conducting the RPA. Manganese is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used its judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent. The most stringent objective is the Secondary MCL, which is derived from human welfare considerations (e.g., taste, odor, laundry staining), not for toxicity. Secondary MCLs are drinking water standards contained in Title 22 of the California Code of Regulations. Title 22 requires compliance with these standards on an annual average basis, when sampling at least quarterly. To be consistent with how compliance with the standards is determined, the RPA was conducted based on the calendar annual average effluent manganese concentrations.

The maximum observed annual average effluent concentration for manganese was 11 µg/L based on 56 samples collected between August 2008 and April 2013. Although the receiving water contains manganese exceeding the Secondary MCL, the effluent manganese is consistently less than the concentrations in the receiving water and below the Secondary MCL. Therefore, the Central Valley Water Board finds the discharge does not have reasonable potential to cause or contribute to an exceedance in the receiving water and the Facility is adequately controlling the discharge of manganese. Since the discharge does not demonstrate reasonable potential, the effluent limitation for manganese has not been retained in this Order. Removal of this effluent limitation is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

xi. **Salinity**

- (a) **WQO.** The Basin Plan contains a chemical constituent objective that incorporates state MCLs, contains a narrative objective, and contains numeric water quality objectives for certain specified water bodies for electrical conductivity, total dissolved solids, sulfate, and chloride. The USEPA Ambient Water Quality Criteria for Chloride recommends acute and chronic criteria for the protection of aquatic life. There are no USEPA water quality criteria for the protection of aquatic life for electrical conductivity, total dissolved solids, and sulfate. Additionally, there are no USEPA numeric water quality criteria for the protection of agricultural, livestock, and industrial uses. Numeric values for the protection of these uses are typically based on site specific conditions and evaluations to determine the appropriate constituent threshold necessary to interpret the narrative chemical constituent Basin Plan objective. The Central Valley Water Board must determine the applicable numeric limit to implement the narrative objective for the protection of agricultural supply. The Central Valley Water Board is currently implementing the CV-SALTS initiative to develop a Basin Plan Amendment that will establish a salt and nitrate Management Plan for the Central Valley. Through this effort the Basin Plan will be amended to define how the narrative water quality objective is to be interpreted for the protection of agricultural use. All studies conducted through this Order to establish an agricultural limit to implement the narrative objective will be reviewed by and consistent with the efforts currently underway by CV-SALTS.

**Table F-12. Salinity Water Quality Criteria/Objectives**

| Parameter       | Agricultural WQ Objective <sup>1</sup> | Secondary MCL <sup>2</sup> | USEPA NAWQC           | Effluent             |         |
|-----------------|----------------------------------------|----------------------------|-----------------------|----------------------|---------|
|                 |                                        |                            |                       | Average <sup>3</sup> | Maximum |
| EC (µmhos/cm)   | Varies                                 | 900, 1600, 2200            | N/A                   | 452                  | 710     |
| TDS (mg/L)      | Varies                                 | 500, 1000, 1500            | N/A                   | 295                  | 410     |
| Sulfate (mg/L)  | Varies                                 | 250, 500, 600              | N/A                   | 23                   | 25      |
| Chloride (mg/L) | Varies                                 | 250, 500, 600              | 860 1-hr<br>230 4-day | 40                   | 42      |

<sup>1</sup> Narrative chemical constituent objective of the Basin Plan. Procedures for establishing the applicable numeric limitation to implement the narrative objective can be found in the Policy for Application of Water Quality, Chapter IV, Section 8 of the Basin Plan. However, the Basin Plan does not require improvement over naturally occurring background concentrations. In cases where the natural background concentration of a particular constituent exceeds an applicable water quality objective, the natural background concentration will be considered to comply with the objective.

<sup>2</sup> The Secondary MCLs are stated as a recommended level, upper level, and a short-term maximum level.

<sup>3</sup> Maximum calendar annual average.

- (1) **Chloride.** The Secondary MCL for chloride is 250 mg/L, as a recommended level, 500 mg/L as an upper level, and 600 mg/L as a short-term maximum.

- (2) **Electrical Conductivity.** The Secondary MCL for EC is 900  $\mu\text{mhos/cm}$  as a recommended level, 1600  $\mu\text{mhos/cm}$  as an upper level, and 2200  $\mu\text{mhos/cm}$  as a short-term maximum.
- (3) **Sulfate.** The Secondary MCL for sulfate is 250 mg/L as a recommended level, 500 mg/L as an upper level, and 600 mg/L as a short-term maximum.
- (4) **Total Dissolved Solids.** The Secondary MCL for TDS is 500 mg/L as a recommended level, 1000 mg/L as an upper level, and 1500 mg/L as a short-term maximum.

(b) **RPA Results**

- (1) **Chloride.** Chloride concentrations in the effluent ranged from 37 mg/L to 42 mg/L, with an average of 40 mg/L. These levels do not exceed the Secondary MCL. Background concentrations in Dry Creek ranged from 6 mg/L to 16 mg/L, with an average of 11.5 mg/L, for four samples collected by the Discharger from August 2008 through April 2013.
- (2) **Electrical Conductivity.** A review of the Discharger's monitoring reports shows an average effluent EC of 452  $\mu\text{mhos/cm}$ , with a range from 330  $\mu\text{mhos/cm}$  to 710  $\mu\text{mhos/cm}$ . The background receiving water EC ranged from 89  $\mu\text{mhos/cm}$  to 270  $\mu\text{mhos/cm}$  with an average of 174  $\mu\text{mhos/cm}$ . These levels do not exceed the Secondary MCL.
- (3) **Sulfate.** Sulfate concentrations in the effluent ranged from 20 mg/L to 25 mg/L, with an average of 23 mg/L. These levels do not exceed the Secondary MCL. Background concentrations in Dry Creek ranged from 4.7 mg/L to 13 mg/L, with an average of 9.3 mg/L.
- (4) **Total Dissolved Solids.** The average TDS effluent concentration was 295 mg/L with concentrations ranging from 250 mg/L to 410 mg/L. These levels do not exceed the Secondary MCL. Background concentrations in Dry Creek ranged from 97 mg/L to 170 mg/L, with an average of 127 mg/L.

Order R5-2008-0077-01 required the Discharger to complete a site-specific study of appropriate EC levels to protect the AGR and MUN beneficial uses, a salinity reduction goal of the water supply plus 500  $\mu\text{mhos/cm}$ , and a salinity evaluation and minimization plan. The Discharger's November 2010 *Dry Creek Wastewater Treatment Plant Site-Specific Salinity Study* concluded that the Secondary MCL of 900  $\mu\text{mhos/cm}$  is protective of both the AGR and MUN beneficial uses. In the 9 March 2009 *Salinity Evaluation and Minimization Plan*, the Discharger identified salinity contributions from the water supply and residential, commercial, and industrial users, and indicated that salinity may increase when water from their Aquifer Storage and Recovery wells is used, but that the discharge would still be in compliance with the Salinity Goal. In the November 2012 *Salinity Reduction Progress Report for Dry Creek Wastewater Treatment Plant*, the Discharger indicated that the conversion from chlorine disinfection to UV disinfection resulted in decreases in effluent EC. Based on monitoring data collected after conversion to UV disinfection, effluent EC ranged from 330  $\mu\text{mhos/cm}$  to 710  $\mu\text{mhos/cm}$ ,

with a maximum annual average of 453 µmhos/cm (2010). Based on the relatively low reported salinity since the conversion to UV disinfection, the discharge does not have reasonable potential to cause or contribute to an in-stream excursion of water quality objectives for salinity. In order to ensure that the Discharger will continue to control the discharge of salinity, this Order includes a requirement to continue to implement the salinity evaluation and minimization plan.

xii. **Settleable Solids**

- (a) **WQO.** For inland surface waters, the Basin Plan states that “[w]ater shall not contain substances in concentrations that result in the deposition of material that causes nuisance or adversely affects beneficial uses.” Order R5-2008-0077-01 established an average monthly effluent limitation (AMEL) of 0.1 ml/L and maximum daily effluent limitation (MDEL) of 0.2 ml/L for settleable solids to implement the narrative settleable solids objective.
- (b) **RPA Results.** The maximum observed effluent settleable solids concentration was 0.1 ml/L based on 181 samples collected between August 2008 and April 2013. Therefore, the discharge from the Facility does not have reasonable potential to cause or contribute to an excursion above the Basin Plan’s narrative objective for settleable solids and the effluent limitations for settleable solids have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

xiii. **Zinc**

- (a) **WQO.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for zinc. These criteria for zinc are presented in dissolved concentrations, as 1-hour acute criteria and 4-day chronic criteria. USEPA recommends conversion factors to translate dissolved concentrations to total concentrations. Default USEPA translators were used for the receiving water and effluent. Order R5-2008-0077 included effluent limitations for zinc based on the CTR criteria.
- (b) **RPA Results.** Section IV.C.2 of this Fact Sheet includes procedures for conducting the RPA for hardness-dependent CTR metals, such as zinc. The CTR includes hardness-dependent criteria for zinc for the receiving water. The RPA was conducted using the upstream receiving water hardness to calculate the criteria for comparison to the maximum ambient background concentration, and likewise using the reasonable worst-case downstream hardness to compare the MEC. The table below shows the specific criteria used for the RPA.

**Table F-13. Zinc RPA**

|                 | CTR Chronic Criterion<br>(Total Recoverable) | Maximum Concentration<br>(Total Recoverable) | Reasonable Potential?<br>(Y/N) |
|-----------------|----------------------------------------------|----------------------------------------------|--------------------------------|
| Receiving Water | 47 µg/L <sup>1</sup>                         | 6.1 µg/L                                     | No <sup>2</sup>                |
| Effluent        | 90 µg/L <sup>3</sup>                         | 51 µg/L                                      | No <sup>4</sup>                |



|  | CTR Chronic<br>Criterion<br>(Total Recoverable) | Maximum<br>Concentration<br>(Total Recoverable) | Reasonable<br>Potential?<br>(Y/N) |
|--|-------------------------------------------------|-------------------------------------------------|-----------------------------------|
|--|-------------------------------------------------|-------------------------------------------------|-----------------------------------|

<sup>1</sup> Based on lowest observed upstream hardness of 33 mg/L (as CaCO<sub>3</sub>)

<sup>3</sup> Per Section 1.3, step 6 of the SIP.

<sup>2</sup> Based on reasonable worst-case downstream hardness of 71 mg/L (as CaCO<sub>3</sub>)

<sup>4</sup> Per Section 1.3, step 4 of the SIP.

Based on the available data, zinc in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR chronic criterion, and the WQBELs for zinc have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

- b. **Constituents with Reasonable Potential.** The Central Valley Water Board finds that the discharge has a reasonable potential to cause or contribute to an in-stream excursion above a water quality standard for ammonia, BOD<sub>5</sub>, mercury, nitrate plus nitrite, pH, total coliform organisms, and TSS. WQBELs for these constituents are included in this Order. A summary of the RPA is provided in Attachment G, and a detailed discussion of the RPA for each constituent is provided below.

i. **Ammonia**

- (a) **WQO.** The NAWQC for the protection of freshwater aquatic life for total ammonia, recommends acute (1-hour average; criteria maximum concentration or CMC) standards based on pH and chronic (30-day average; criteria continuous concentration or CCC) standards based on pH and temperature. USEPA also recommends that no 4-day average concentration should exceed 2.5 times the 30-day CCC. USEPA found that as pH increased, both the acute and chronic toxicity of ammonia increased. Salmonids were more sensitive to acute toxicity effects than other species. However, while the acute toxicity of ammonia was not influenced by temperature, it was found that invertebrates and young fish experienced increasing chronic toxicity effects with increasing temperature. Because Dry Creek has a beneficial use of cold freshwater habitat and the presence of salmonids and early fish life stages in Dry Creek is well-documented, the recommended criteria for waters where salmonids and early life stages are present, were used.

The maximum permitted effluent pH is 8.5, as the Basin Plan objective for pH in the receiving stream is the range of 6.5 to 8.5. In order to protect against the worst-case short-term exposure of an organism, a pH value of 8.5 was used to derive the acute criterion. The resulting acute criterion is 2.14 mg/L.

A chronic criterion was calculated for each day when paired temperature and pH were measured using downstream receiving water data for temperature and pH. Rolling 30-day average criteria were calculated from downstream data using the criteria calculated for each day and the minimum observed 30-day average criterion was established as the applicable 30-day average chronic criterion, or 30-day CCC. The resulting 30-day CCC is 1.58 mg/L (as N). The 4-day average concentration is derived in accordance with the USEPA criterion as 2.5 times the 30-day

CCC. Based on the 30-day CCC of 1.58 mg/L (as N), the 4-day average concentration that should not be exceeded is 3.95 mg/L (as N).

- (b) **RPA Results.** The Facility is a POTW that treats domestic wastewater. Untreated domestic wastewater contains ammonia in concentrations that, without treatment, would be harmful to fish and would violate the Basin Plan narrative toxicity objective if discharged to the receiving water. Reasonable potential therefore exists and effluent limitations are required.

Federal regulations at 40 C.F.R. §122.44(d)(1)(i) requires that, *"Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality."* For priority pollutants, the SIP dictates the procedures for conducting the RPA. Ammonia is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, *"State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBELs are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBELs for pathogens in all permits for POTWs discharging to contact recreational waters)."* USEPA's TSD also recommends that factors other than effluent data should be considered in the RPA, *"When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric or narrative water quality criterion for individual toxicants or for toxicity, the regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data."* With regard to POTWs, USEPA recommends that, *"POTWs should also be characterized for the possibility of chlorine and ammonia problems."* (TSD, p. 50)

Nitrification is a biological process that converts ammonia to nitrite and nitrite to nitrate. Denitrification is a process that converts nitrate to nitrite or nitric oxide and then to nitrous oxide or nitrogen gas, which is then released to the atmosphere. The Discharger currently uses nitrification to remove ammonia from the waste stream. Inadequate or incomplete nitrification may result in the discharge of ammonia to the receiving stream. Ammonia is known to cause toxicity to aquatic organisms in surface waters. Discharges of ammonia in concentrations that produce detrimental physiological responses to human, plant, animal, or aquatic life would violate the Basin Plan narrative toxicity objective. Although the Discharger nitrifies the discharge, inadequate or incomplete nitrification

creates the potential for ammonia to be discharged and provides the basis for the discharge to have a reasonable potential to cause or contribute to an in-stream excursion above the NAWQC. Therefore, the Central Valley Water Board finds the discharge has reasonable potential for ammonia and WQBELs are required.

- (c) **WQBELs.** The Central Valley Water Board calculates WQBELs in accordance with SIP procedures for non-CTR constituents, and ammonia is a non-CTR constituent. The SIP procedure assumes a 4-day averaging period for calculating the long-term average discharge condition (LTA). However, USEPA recommends modifying the procedure for calculating permit limits for ammonia using a 30-day averaging period for the calculation of the LTA corresponding to the 30-day CCC. Therefore, while the LTAs corresponding to the acute and 4-day chronic criteria were calculated according to SIP procedures, the LTA corresponding to the 30-day CCC was calculated assuming a 30-day averaging period. The lowest LTA representing the acute, 4-day CCC, and 30-day CCC is then selected for deriving the AMEL and the maximum daily effluent limitation (MDEL). The remainder of the WQBEL calculation for ammonia was performed according to the SIP procedures. This Order contains a final AMEL and MDEL for ammonia of 0.8 mg/L and 2.2 mg/L, respectively, based on NAWQC.
- (d) **Plant Performance and Attainability.** The Facility is designed to provide complete nitrification of the discharge. Therefore, the Central Valley Water Board concludes that immediate compliance with these effluent limitations is feasible.

ii. **Mercury**

- (a) **WQO.** The current NAWQC for protection of freshwater aquatic life, continuous concentration, for mercury is 0.77 µg/L (30-day average, chronic criteria). The CTR contains a human health criterion (based on a threshold dose level causing neurological effects in infants) of 0.050 µg/L for waters from which both water and aquatic organisms are consumed. Both values are controversial and subject to change. In 40 CFR Part 131, USEPA acknowledges that the human health criteria may not be protective of some aquatic or endangered species and that “...*more stringent mercury limits may be determined and implemented through use of the State’s narrative criterion.*” In the CTR, USEPA reserved the mercury criteria for freshwater and aquatic life and may adopt new criteria at a later date.
- (b) **RPA Results.** The MEC for mercury was 0.006 µg/L based on 56 samples collected between August 2008 and April 2013 (minimum MDL 0.0002 µg/L, minimum RL 0.0005 µg/L). The maximum observed upstream receiving water mercury concentration was 0.011 µg/L based on four samples collected between August 2008 and April 2013 (minimum MDL 0.0002 µg/L, minimum RL 0.0005 µg/L). Therefore, mercury in the discharge does not demonstrate reasonable potential to cause or contribute to an in-stream excursion above the CTR human health criteria, and the WQBELs for mercury have not been retained in this Order. Removal of these effluent limitations is in accordance with federal anti-backsliding regulations (see section IV.D.3 of the Fact Sheet).

Although the discharge does not exceed the CTR human health criteria, mercury bioaccumulates in fish tissue and, therefore, the discharge of mercury to the receiving water may contribute to exceedances of the narrative toxicity objective and impact beneficial uses. The Natomas East Main Drainage Canal has been listed as an impaired water body pursuant to CWA section 303(d) because of mercury and the discharge must not cause or contribute to increased mercury levels. In addition, a TMDL is under development for the Natomas East Main Drainage Canal downstream of the Facility which may include a waste load allocation applicable to the Facility.

- (c) **WQBELs.** This Order contains a performance-based mass effluent limitation of 5.5 lbs/year for the effluent discharged to the receiving water. This limitation is based on maintaining the mercury loading at the current level until a TMDL can be established and USEPA develops mercury standards that are protective of human health. The mass limitation was derived using the MDEL contained in Order R5-2008-0077-01 and the design average dry weather rate of the Facility (18 MGD):

$$0.00010 \text{ (mg/L)} * 18 \text{ MGD} * 8.34 \text{ (conversion factor)} * 365 \text{ days} = \text{lbs/year}$$

If USEPA develops new water quality standards for mercury, this permit may be reopened and the effluent limitations adjusted.

- (d) **Plant Performance and Attainability.** Analysis of the effluent data shows the Facility can meet these WQBELs.

iii. **Nitrate and Nitrite**

- (a) **WQO.** DPH has adopted Primary MCLs for the protection of human health for nitrite and nitrate that are equal to 1 mg/L and 10 mg/L (measured as nitrogen), respectively. DPH has also adopted a Primary MCL of 10 mg/L for the sum of nitrate and nitrite, measured as nitrogen.

USEPA has developed a primary MCL and an MCL goal of 1 mg/L for nitrite (as nitrogen). For nitrate, USEPA has developed Drinking Water Standards (10 mg/L as Primary MCL) and NAWQC for protection of human health (10 mg/L for non-cancer health effects). Recent toxicity studies<sup>1</sup> have indicated a possibility that nitrate is toxic to aquatic organisms.

- (b) **RPA Results.** The Facility is a POTW that treats domestic wastewater. Untreated domestic wastewater contains ammonia in concentrations that, if untreated, will be harmful to fish and will violate the Basin Plan's narrative toxicity objective. This Order, therefore, requires removal of ammonia (i.e., nitrification). Nitrification is a biological process that converts ammonia to nitrate and nitrite, and will result in effluent nitrate concentrations above the primary MCL for nitrate plus nitrite. Nitrate concentrations in a drinking water supply above the Primary MCL threatens the health of human fetuses and newborn babies by reducing

<sup>1</sup> Camargo, Julio A., Alvaro Alonso, Annabella Salamanca, "Nitrate toxicity to aquatic animals: a review with new data for freshwater invertebrates". Accepted 25 October 2004-Chemosphere 58 (2005) 1255-1267; Canadian Council of Ministers of the Environment, National Guidelines and Standards Office, "Canadian Water Quality Guidelines for the Protection of Aquatic Life". 2012; Minnesota Pollution Control Agency, "Developing Surface Water Nitrate Standards and Strategies for Reducing Nitrogen Loading". December 2010.

the oxygen-carrying capacity of the blood (methemoglobinemia). Reasonable potential for nitrate and nitrite therefore exists and WQBELs are required.

Federal regulations at 40 C.F.R. §122.44(d)(1)(i) requires that, *“Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality.”* For priority pollutants, the SIP dictates the procedures for conducting the RPA. Nitrate and nitrite are not priority pollutants. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, *“State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBELs are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBELs for pathogens in all permits for POTWs discharging to contact recreational waters).”* USEPA's TSD also recommends that factors other than effluent data should be considered in the RPA, *“When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric or narrative water quality criterion for individual toxicants or for toxicity, the regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data.”* With regard to POTWS, USEPA recommends that, *“POTWs should also be characterized for the possibility of chlorine and ammonia problems.”* (TSD, p. 50)

The concentration of nitrogen in raw domestic wastewater is sufficiently high that the resultant treated wastewater has a reasonable potential to exceed or threaten to exceed the Primary MCL for nitrate plus nitrite unless the wastewater is treated for nitrogen removal, and therefore an effluent limit for nitrate plus nitrite is required. Denitrification is a process that converts nitrate to nitrite or nitric oxide and then to nitrous oxide or nitrogen gas, which is then released to the atmosphere. The Discharger currently uses nitrification/denitrification to remove ammonia, nitrite, and nitrate from the waste stream. Inadequate or incomplete denitrification may result in the discharge of nitrate and/or nitrite to the receiving stream. Discharges of nitrate plus nitrite in concentrations that exceed the Primary MCL would violate the Basin Plan narrative chemical constituents objective. Although the Discharger denitrifies the discharge, inadequate or incomplete denitrification creates the potential for nitrate and nitrite to be discharged and provides the basis for the discharge to have a

reasonable potential to cause or contribute to an in-stream excursion above the Primary MCL. Therefore, the Central Valley Water Board finds the discharge has reasonable potential for nitrate plus nitrite and WQBELs are required.

- (c) **WQBELs.** This Order contains a final AMEL for nitrate plus nitrite of 10 mg/L (total as N), based on the Primary MCL. This effluent limitation is included in this Order to assure the treatment process adequately nitrifies and denitrifies the waste stream to protect the beneficial use of municipal and domestic supply.
- (d) **Plant Performance and Attainability.** Analysis of the effluent data shows that the MEC of 15 mg/L for nitrate is greater than applicable WQBEL. Based on the sample results for the effluent, the limitation appears to put the Discharger in immediate non-compliance. New or modified control measures may be necessary in order to comply with the effluent limitation, and the new or modified control measures cannot be designed, installed and put into operation within 30 calendar days. Furthermore, the effluent limitation for nitrate plus nitrite is a new regulatory requirement within this permit, which becomes applicable to the waste discharge with the adoption of this Order, which was adopted after 1 July 2000. Therefore, a compliance time schedule for compliance with the nitrate plus nitrite effluent limitation is established in Time Schedule Order (TSO) No. R5-2014-0050 in accordance with Water Code section 13300, that requires preparation and implementation of a pollution prevention plan in compliance with Water Code section 13263.3.

iv. **Pathogens**

- (a) **WQO.** DPH has developed reclamation criteria, CCR, Division 4, Chapter 3 (Title 22), for the reuse of wastewater. Title 22 requires that for spray irrigation of food crops, parks, playgrounds, schoolyards, and other areas of similar public access, wastewater be adequately disinfected, oxidized, coagulated, clarified, and filtered, and that the effluent total coliform levels not exceed 2.2 MPN/100 mL as a 7-day median; 23 MPN/100 mL, not to be exceeded more than once in a 30-day period; and 240 MPN/100 mL, at any time.

Title 22 also requires that recycled water used as a source of water supply for non-restricted recreational impoundments be disinfected tertiary recycled water that has been subjected to conventional treatment. A non-restricted recreational impoundment is defined as “...an impoundment of recycled water, in which no limitations are imposed on body-contact water recreational activities.” Title 22 is not directly applicable to surface waters; however, the Central Valley Water Board finds that it is appropriate to apply an equivalent level of treatment to that required by the DPH’s reclamation criteria because the receiving water is used for irrigation of agricultural land and for contact recreation purposes. The stringent disinfection criteria of Title 22 are appropriate since the undiluted effluent may be used for the irrigation of food crops and/or for body-contact water recreation. Coliform organisms are intended as an indicator of the effectiveness of the entire treatment train and the effectiveness of removing other pathogens.

- (b) **RPA Results.** Raw domestic wastewater inherently contains human pathogens that threaten human health and life, and constitute a threatened pollution and nuisance under CWC Section 13050 if discharged untreated to the receiving water. Reasonable potential for pathogens therefore exists and WQBELs are required.

Federal regulations at 40 C.F.R. §122.44(d)(1)(i) requires that, *"Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality."* For priority pollutants, the SIP dictates the procedures for conducting the RPA. Pathogens are not priority pollutants. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, *"State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBELs are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBELs for pathogens in all permits for POTWs discharging to contact recreational waters)." USEPA's TSD also recommends that factors other than effluent data should be considered in the RPA, "When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric or narrative water quality criterion for individual toxicants or for toxicity, the regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data."* (TSD, p. 50)

The beneficial uses of Dry Creek include municipal and domestic supply, water contact recreation, and agricultural irrigation supply, and there is, at times, less than 20:1 dilution. To protect these beneficial uses, the Central Valley Water Board finds that the wastewater must be disinfected and adequately treated to prevent disease. Although the Discharger provides disinfection, inadequate or incomplete disinfection creates the potential for pathogens to be discharged. Therefore, the Central Valley Water Board finds the discharge has reasonable potential for pathogens and WQBELs are required.

- (c) **WQBELs.** In accordance with the requirements of Title 22, this Order includes effluent limitations for total coliform organisms of 2.2 MPN/100 mL as a 7-day median; 23 MPN/100 mL, not to be exceeded more than once in a 30-day period; and 240 MPN/100 mL as an instantaneous maximum.

The tertiary treatment process, or equivalent, is capable of reliably treating wastewater to a turbidity level of 2 nephelometric turbidity units (NTU) as a daily average. Failure of the filtration system such that virus removal is impaired would normally result in increased particles in the effluent, which result in higher effluent turbidity. Turbidity has a major advantage for monitoring filter performance. Coliform testing, by comparison, is not conducted continuously and requires several hours, to days, to identify high coliform concentrations. Therefore, to ensure compliance with the DPH recommended Title 22 disinfection criteria, weekly average specifications are impracticable for turbidity. This Order includes operational specifications for turbidity of 2 NTU as a daily average; 5 NTU, not to be exceeded more than 5 percent of the time within a 24-hour period; and 10 NTU as an instantaneous maximum.

This Order contains effluent limitations for BOD<sub>5</sub>, total coliform organisms, and TSS and requires a tertiary level of treatment, or equivalent, necessary to protect the beneficial uses of the receiving water. The Central Valley Water Board has previously considered the factors in Water Code section 13241 in establishing these requirements.

Final WQBELs for BOD<sub>5</sub> and TSS are based on the technical capability of the tertiary process, which is necessary to protect the beneficial uses of the receiving water. BOD<sub>5</sub> is a measure of the amount of oxygen used in the biochemical oxidation of organic matter. The tertiary treatment standards for BOD<sub>5</sub> and TSS are indicators of the effectiveness of the tertiary treatment process. The principal design parameter for wastewater treatment plants is the daily BOD<sub>5</sub> and TSS loading rates and the corresponding removal rate of the system. The application of tertiary treatment processes results in the ability to achieve lower levels for BOD<sub>5</sub> and TSS than the secondary standards currently prescribed. Therefore, this Order requires AMELs for BOD<sub>5</sub> and TSS of 10 mg/L, which is technically based on the capability of a tertiary system. In addition to the average weekly effluent limitations (AWELs) and AMELs, MDELs for BOD<sub>5</sub> and TSS are included in the Order to ensure that the treatment works are not organically overloaded and operate in accordance with design capabilities.

- (d) **Plant Performance and Attainability.** The Facility possesses a media filtration and UV disinfection system which was designed to achieve Title 22 criteria. The Central Valley Water Board concludes, therefore, that immediate compliance with these effluent limitations is feasible.

v. **pH**

- (a) **WQO.** The Basin Plan includes a water quality objective for surface waters (except for Goose Lake) that the “...*pH shall not be depressed below 6.5 nor raised above 8.5.*”
- (b) **RPA Results.** Raw domestic wastewater inherently has variable pH. Additionally, some wastewater treatment processes can increase or decrease wastewater pH which if not properly controlled, would violate the Basin Plan’s numeric objective for pH in the receiving water. Therefore, reasonable potential exists for pH and WQBELs are required.



Federal regulations at 40 C.F.R. §122.44(d)(1)(i) requires that, *"Limitations must control all pollutants or pollutant parameters (either conventional, nonconventional, or toxic pollutants) which the Director determines are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality."* For priority pollutants, the SIP dictates the procedures for conducting the RPA. pH is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA for this non-priority pollutant constituent.

USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, *"State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBELs are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBELs for pathogens in all permits for POTWs discharging to contact recreational waters)."* USEPA's TSD also recommends that factors other than effluent data should be considered in the RPA, *"When determining whether or not a discharge causes, has the reasonable potential to cause, or contributes to an excursion of a numeric or narrative water quality criterion for individual toxicants or for toxicity, the regulatory authority can use a variety of factors and information where facility-specific effluent monitoring data are unavailable. These factors also should be considered with available effluent monitoring data."* (TSD, p. 50)

The Facility is a POTW that treats domestic wastewater. Based on 1,186 samples taken from January 2010 to March 2013, the maximum pH reported was 7.9 and the minimum was 6.5. The Facility did not exceed either the instantaneous maximum or minimum effluent limitation. Although the Discharger has proper pH controls in place, the pH for the Facility's influent varies due to the nature of municipal sewage, which provides the basis for the discharge to have a reasonable potential to cause or contribute to an in-stream excursion above the Basin Plan's numeric objective for pH in the receiving water. Therefore, WQBELs for pH are required in this Order.

- (c) **WQBELs.** Order R5-2008-0077-01 contained minimum and maximum effluent limitations of 6.5 and 8.0 at Discharge Point 001. The maximum effluent limitation of 8.0 is more stringent than required by the Basin Plan pH objectives and was based on the treatment capabilities of the Facility. The Discharger has requested that the maximum effluent limitation be revised from 8.0 to 8.5 to provide operational flexibility for the operation of the cooling towers, which are necessary to achieve compliance with the receiving water limitations for temperature. Since the effluent pH has not exceeded 8.5 during the term of Order R5-2008-0077-01 and a pH of 8.5 is equivalent to the applicable water quality objective, this Order includes a revised maximum effluent limitation of 8.5.

- (d) **Plant Performance and Attainability.** Analysis of the effluent data shows that the range of observed pH performance levels is within the bounds of the applicable WQBELs. The Central Valley Water Board concludes, therefore, that immediate compliance with these effluent limitations is feasible.

#### 4. WQBEL Calculations

- a. This Order includes WQBELs for ammonia, BOD<sub>5</sub>, mercury, nitrate plus nitrite, pH, total coliform organisms, and TSS. The general methodology for calculating WQBELs based on the different criteria/objectives is described in subsections IV.C.4.b through e, below. See Attachment H for the WQBEL calculations.
- b. **Effluent Concentration Allowance.** For each water quality criterion/objective, the ECA is calculated using the following steady-state mass balance equation from Section 1.4 of the SIP:

$$\begin{aligned} ECA &= C + D(C - B) && \text{where } C > B, \text{ and} \\ ECA &= C && \text{where } C \leq B \end{aligned}$$

where:

ECA = effluent concentration allowance  
 D = dilution credit  
 C = the priority pollutant criterion/objective  
 B = the ambient background concentration.

According to the SIP, the ambient background concentration (B) in the equation above shall be the observed maximum with the exception that an ECA calculated from a priority pollutant criterion/objective that is intended to protect human health from carcinogenic effects shall use the arithmetic mean concentration of the ambient background samples. For ECAs based on MCLs, which implement the Basin Plan's chemical constituents objective and are applied as annual averages, an arithmetic mean is also used for B due to the long-term basis of the criteria.

- c. **Basin Plan Objectives and MCLs.** For WQBELs based on site-specific numeric Basin Plan objectives or MCLs, the effluent limitations are applied directly as the ECA as either an MDEL, AMEL, or average annual effluent limitations, depending on the averaging period of the objective.
- d. **Aquatic Toxicity Criteria.** WQBELs based on acute and chronic aquatic toxicity criteria are calculated in accordance with Section 1.4 of the SIP. The ECAs are converted to equivalent long-term averages (i.e. LTA<sub>acute</sub> and LTA<sub>chronic</sub>) using statistical multipliers and the lowest LTA is used to calculate the AMEL and MDEL using additional statistical multipliers.
- e. **Human Health Criteria.** WQBELs based on human health criteria, are also calculated in accordance with Section 1.4 of the SIP. The ECAs are set equal to the AMEL and a statistical multiplier was used to calculate the MDEL.

$$\begin{aligned} AMEL &= mult_{AMEL} \left[ \min \left( \overbrace{M_A ECA_{acute}, M_C ECA_{chronic}}^{LTA_{acute}} \right) \right] \\ MDEL &= mult_{MDEL} \left[ \min \left( \overbrace{M_A ECA_{acute}, M_C ECA_{chronic}}^{LTA_{chronic}} \right) \right] \end{aligned}$$

$$MDEL_{HH} = \frac{mult_{MDEL} \cdot AMEL_{HH}}{mult_{AMEL}}$$

where:

$mult_{AMEL}$  = statistical multiplier converting minimum LTA to AMEL

$mult_{MDEL}$  = statistical multiplier converting minimum LTA to MDEL

$M_A$  = statistical multiplier converting acute ECA to  $LTA_{acute}$

$M_C$  = statistical multiplier converting chronic ECA to  $LTA_{chronic}$

### Summary of Water Quality-Based Effluent Limitations Discharge Point 001

Table F-14. Summary of Water Quality-Based Effluent Limitations

| Parameter                                | Units                | Effluent Limitations |                  |                 |                       |                       |
|------------------------------------------|----------------------|----------------------|------------------|-----------------|-----------------------|-----------------------|
|                                          |                      | Average Monthly      | Average Weekly   | Maximum Daily   | Instantaneous Minimum | Instantaneous Maximum |
| Conventional Pollutants                  |                      |                      |                  |                 |                       |                       |
| Biochemical Oxygen Demand (5-day @ 20°C) | mg/L                 | 10                   | 15               | 20              | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 1,500                | 2,250            | 3,000           | --                    | --                    |
| pH                                       | standard units       | --                   | --               | --              | 6.5                   | 8.5                   |
| Total Suspended Solids                   | mg/L                 | 10                   | 15               | 20              | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 1,500                | 2,250            | 3,000           | --                    | --                    |
| Priority Pollutants                      |                      |                      |                  |                 |                       |                       |
| Mercury, Total Recoverable               | lbs/year             | 5.5 <sup>2</sup>     | --               | --              | --                    | --                    |
| Non-Conventional Pollutants              |                      |                      |                  |                 |                       |                       |
| Ammonia Nitrogen, Total (as N)           | mg/L                 | 0.8                  | --               | 2.2             | --                    | --                    |
|                                          | lbs/day <sup>1</sup> | 120                  | --               | 330             | --                    | --                    |
| Nitrate plus Nitrite (as N)              | mg/L                 | 10                   | --               | --              | --                    | --                    |
| Total Coliform Organisms                 | MPN/100 mL           | --                   | 2.2 <sup>3</sup> | 23 <sup>4</sup> | --                    | 240                   |

<sup>1</sup> Based on an average dry weather flow of 18 MGD.

<sup>2</sup> The total annual mass discharge of total mercury shall not exceed 5.5 lbs/year.

<sup>3</sup> Applied as a 7-day median effluent limitation.

<sup>4</sup> Not to be exceeded more than once in any 30-day period.

## 5. Whole Effluent Toxicity (WET)

For compliance with the Basin Plan's narrative toxicity objective, this Order requires the Discharger to conduct whole effluent toxicity testing for acute and chronic toxicity, as specified in the Monitoring and Reporting Program (Attachment E section V.). This Order also contains effluent limitations for acute toxicity and requires the Discharger to implement best management practices to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity.

- a. **Acute Aquatic Toxicity.** The Basin Plan contains a narrative toxicity objective that states, *"All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life."* (Basin Plan at page III-8.0) The Basin Plan also states that, *"...effluent limits based upon acute biotoxicity tests of effluents will be prescribed where appropriate..."*

For priority pollutants, the SIP dictates the procedures for conducting the RPA. Acute toxicity is not a priority pollutant. Therefore, the Central Valley Water Board is not restricted to one particular RPA method. Acute whole effluent toxicity is not a priority pollutant. Therefore, due to the site-specific conditions of the discharge, the Central Valley Water Board has used professional judgment in determining the appropriate method for conducting the RPA. USEPA's September 2010 NPDES Permit Writer's Manual, page 6-30, states, *"State implementation procedures might allow, or even require, a permit writer to determine reasonable potential through a qualitative assessment process without using available facility-specific effluent monitoring data or when such data are not available...A permitting authority might also determine that WQBELs are required for specific pollutants for all facilities that exhibit certain operational or discharge characteristics (e.g., WQBELs for pathogens in all permits for POTWs discharging to contact recreational waters)."* Although the discharge has been consistently in compliance with the acute effluent limitations, the Facility is a POTW that treats domestic wastewater containing ammonia and other acutely toxic pollutants. Acute toxicity effluent limits are required to ensure compliance with the Basin Plan's narrative toxicity objective.

USEPA Region 9 provided guidance for the development of acute toxicity effluent limitations in the absence of numeric water quality objectives for toxicity in its document titled "Guidance for NPDES Permit Issuance", dated February 1994. In section B.2. "Toxicity Requirements" (pgs. 14-15) it states that, *"In the absence of specific numeric water quality objectives for acute and chronic toxicity, the narrative criterion 'no toxics in toxic amounts' applies. Achievement of the narrative criterion, as applied herein, means that ambient waters shall not demonstrate for acute toxicity: 1) less than 90% survival, 50% of the time, based on the monthly median, or 2) less than 70% survival, 10% of the time, based on any monthly median. For chronic toxicity, ambient waters shall not demonstrate a test result of greater than 1 TUc."* Accordingly, effluent limitations for acute toxicity have been included in this Order as follows:

**Acute Toxicity.** Survival of aquatic organisms in 96-hour bioassays of undiluted waste shall be no less than:

|                                                 |     |
|-------------------------------------------------|-----|
| Minimum for any one bioassay -----              | 70% |
| Median for any three consecutive bioassays----- | 90% |

- b. **Chronic Aquatic Toxicity.** The Basin Plan contains a narrative toxicity objective that states, *"All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life."* (Basin Plan at page III-8.00) As shown in Table F-15 below, the effluent exhibited chronic toxicity for *Ceriodaphnia dubia* reproduction in the second quarter of 2010 and third quarter of 2011. In both instances, the Discharger conducted four accelerated monitoring tests. No toxicity was detected in the accelerated monitoring samples and the source of the transient toxicity observed is unknown. No toxicity was observed in the remaining 16 3-species toxicity tests. Based on chronic WET testing performed by the Discharger from the fourth quarter

2008 through the first quarter of 2013, the discharge does not have reasonable potential to cause or contribute to an in-stream excursion above of the Basin Plan's narrative toxicity objective.

**Table F-15. Whole Effluent Chronic Toxicity Testing Results**

| Date                         | Fathead Minnow<br><i>Pimephales promelas</i> |                 | Water Flea<br><i>Ceriodaphnia dubia</i> |                       | Green Algae<br><i>Selenastrum capricornutum</i> |
|------------------------------|----------------------------------------------|-----------------|-----------------------------------------|-----------------------|-------------------------------------------------|
|                              | Survival<br>(TUC)                            | Growth<br>(TUC) | Survival<br>(TUC)                       | Reproduction<br>(TUC) | Growth<br>(TUC)                                 |
| 4th Quarter 2008             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1st Quarter 2009             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 2nd Quarter 2009             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 3rd Quarter 2009             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 4th Quarter 2009             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1st Quarter 2010             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 2nd Quarter 2010             | 1                                            | 1               | 1                                       | 4                     | 1                                               |
| 3rd Quarter 2010             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 4th Quarter 2010             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1st Quarter 2011             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 2nd Quarter 2011             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 3 <sup>rd</sup> Quarter 2011 | 1                                            | 1               | 1                                       | 8                     | 1                                               |
| 4th Quarter 2011             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1st Quarter 2012             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 2 <sup>nd</sup> Quarter 2012 | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 3 <sup>rd</sup> Quarter 2012 | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 4th Quarter 2012             | 1                                            | 1               | 1                                       | 1                     | 1                                               |
| 1st Quarter 2013             | 1                                            | 1               | 1                                       | 1                     | 1                                               |

The Monitoring and Reporting Program of this Order requires quarterly chronic WET monitoring for demonstration of compliance with the narrative toxicity objective. In addition to WET monitoring, the Special Provision in section VI.C.2.a of the Order includes a numeric toxicity monitoring trigger, requirements for accelerated monitoring, and requirements for Toxicity Reduction Evaluation (TRE) initiation if toxicity is demonstrated.

Numeric chronic WET effluent limitations have not been included in this Order. The SIP contains implementation gaps regarding the appropriate form and implementation of chronic toxicity limits. This has resulted in the petitioning of a NPDES permit in the Los Angeles Region<sup>1</sup> that contained numeric chronic toxicity effluent limitations. To address the petition, the State Water Board adopted WQO 2003-012 directing its staff to revise the toxicity control provisions in the SIP. The State Water Board states the following in WQO 2003-012, "*In reviewing this petition and receiving comments from numerous interested persons on the propriety of including numeric effluent limitations for chronic toxicity in NPDES permits for publicly-owned treatment works that discharge to inland waters, we have determined that this issue should be considered in a regulatory setting, in order to*

<sup>1</sup> In the Matter of the Review of Own Motion of Waste Discharge Requirements Order Nos. R4-2002-0121 [NPDES No. CA0054011] and R4-2002-0123 [NPDES NO. CA0055119] and Time Schedule Order Nos. R4-2002-0122 and R4-2002-0124 for Los Coyotes and Long Beach Wastewater Reclamation Plants Issued by the California Regional Water Quality Control Board, Los Angeles Region SWRCB/OCC FILES A-1496 AND 1496(a)

*allow for full public discussion and deliberation. We intend to modify the SIP to specifically address the issue. We anticipate that review will occur within the next year. We therefore decline to make a determination here regarding the propriety of the final numeric effluent limitations for chronic toxicity contained in these permits."*

The process to revise the SIP is currently underway. Proposed changes include clarifying the appropriate form of effluent toxicity limits in NPDES permits and general expansion and standardization of toxicity control implementation related to the NPDES permitting process. Since the toxicity control provisions in the SIP are under revision it is infeasible to develop numeric effluent limitations for chronic toxicity. Therefore, this Order requires that the Discharger meet best management practices for compliance with the Basin Plan's narrative toxicity objective, as allowed under 40 CFR 122.44(k).

To ensure compliance with the Basin Plan's narrative toxicity objective, the Discharger is required to conduct chronic WET testing, as specified in the Monitoring and Reporting Program (Attachment E section V.). Furthermore, the Special Provision contained at VI.C.2.a. of this Order requires the Discharger to investigate the causes of, and identify and implement corrective actions to reduce or eliminate effluent toxicity. If the discharge demonstrates toxicity exceeding the numeric toxicity monitoring trigger, the Discharger is required to initiate a TRE in accordance with an approved TRE workplan. The numeric toxicity monitoring trigger is not an effluent limitation; it is the toxicity threshold at which the Discharger is required to perform accelerated chronic toxicity monitoring, as well as, the threshold to initiate a TRE if effluent toxicity has been demonstrated.

#### **D. Final Effluent Limitation Considerations**

##### **1. Mass-based Effluent Limitations**

40 CFR 122.45(f)(1) requires effluent limitations be expressed in terms of mass, with some exceptions, and 40 CFR 122.45(f)(2) allows pollutants that are limited in terms of mass to additionally be limited in terms of other units of measurement. This Order includes effluent limitations expressed in terms of mass and concentration. In addition, pursuant to the exceptions to mass limitations provided in 40 CFR 122.45(f)(1), some effluent limitations are not expressed in terms of mass, such as pH and temperature, and when the applicable standards are expressed in terms of concentration (e.g., CTR criteria and MCLs) and mass limitations are not necessary to protect the beneficial uses of the receiving water.

Mass-based effluent limitations have been established in this Order for ammonia, BOD<sub>5</sub>, and TSS because they are oxygen demanding substances. Except for the pollutants listed above, mass-based effluent limitations are not included in this Order for pollutant parameters for which effluent limitations are based on water quality objectives and criteria that are concentration-based.

Mass-based effluent limitations were calculated based upon the design flow (average dry weather flow) permitted in section IV.A.1.e of this Order.

##### **2. Averaging Periods for Effluent Limitations**

40 CFR 122.45 (d) requires AWELs and AMELs for POTWs unless impracticable. However, for toxic pollutants and pollutant parameters in water quality permitting, USEPA recommends the use of and MDEL in lieu of AWELs for two reasons. *"First, the basis for the 7-day average for POTWs derives from the secondary treatment requirements. This basis is not related to the need for assuring achievement of water quality standards. Second, a 7-day average, which could comprise up to seven or more*

*daily samples, could average out peak toxic concentrations and therefore the discharge's potential for causing acute toxic effects would be missed.*" (TSD, pg. 96) This Order uses MDELs in lieu of AWELs for ammonia as recommended by the TSD for the achievement of water quality standards and for the protection of the beneficial uses of the receiving stream. Furthermore, for BOD<sub>5</sub>, pH, and TSS, AWELs have been replaced or supplemented with effluent limitations utilizing shorter averaging periods. The rationale for using shorter averaging periods for these constituents is discussed in section IV.C.3 of this Fact Sheet.

### 3. Satisfaction of Anti-Backsliding Requirements

The Clean Water Act specifies that a revised permit may not include effluent limitations that are less stringent than the previous permit unless a less stringent limitation is justified based on exceptions to the anti-backsliding provisions contained in Clean Water Act sections 402(o) or 303(d)(4), or, where applicable, 40 CFR 122.44(l).

The effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order, with the exception of effluent limitations for aluminum, cadmium, carbon tetrachloride, chlorine residual, chlorodibromomethane, cyanide, dichlorobromomethane, iron, manganese, mercury, settleable solids, turbidity, and zinc. The effluent limitations for these pollutants are less stringent than those in Order No. R5-2008-0077-01. This relaxation of effluent limitations is consistent with the anti-backsliding requirements of the CWA and federal regulations

- a. **CWA section 402(o)(1) and 303(d)(4).** CWA section 402(o)(1) prohibits the establishment of less stringent WQBELs *"except in compliance with Section 303(d)(4)."* CWA section 303(d)(4) has two parts: paragraph (A) which applies to nonattainment waters and paragraph (B) which applies to attainment waters.
  - i. For waters where standards are not attained, CWA section 304(d)(4)(A) specifies that any effluent limit based on a TMDL or other WLA may be revised only if the cumulative effect of all such revised effluent limits based on such TMDLs or WLAs will assure the attainment of such water quality standards.
  - ii. For attainment waters, CWA section 303(d)(4)(B) specifies that a limitation based on a water quality standard may be relaxed where the action is consistent with the antidegradation policy.

Dry Creek is considered an attainment water for aluminum, cadmium, carbon tetrachloride, chlorine residual, chlorodibromomethane, cyanide, dichlorobromomethane, iron, manganese, mercury, settleable solids, and zinc because the receiving water is not listed as impaired on the 303(d) list for these constituents<sup>1</sup>. As discussed in section IV.D.4, the removal or relaxation of WQBELs for these pollutants from Order R5-2008-0077-01 meets the exception in CWA section 303(d)(4)(B).

- b. **CWA section 402(o)(2).** CWA section 402(o)(2) provides several exceptions to the anti-backsliding regulations. CWA 402(o)(2)(B)(i) allows a renewed, reissued, or modified permit to contain a less stringent effluent limitation for a pollutant if information is available which was not available at the time of permit issuance (other than revised regulations, guidance, or test methods) and which would have

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<sup>1</sup> "The exceptions in Section 303(d)(4) address both waters in attainment with water quality standards and those not in attainment, i.e. waters on the section 303(d) impaired waters list." State Water Board Order WQ 2008-0006, Berry Petroleum Company, Poso Creek/McVan Facility.

justified the application of a less stringent effluent limitation at the time of permit issuance.

As described further in section IV.C.3.a of this Fact Sheet, updated information that was not available at the time Order R5-2008-0077-01 was issued indicates that aluminum, cadmium, carbon tetrachloride, chlorine residual, chlorodibromomethane, cyanide, dichlorobromomethane, iron, manganese, mercury, settleable solids, and zinc do not exhibit reasonable potential to cause or contribute to an exceedance of water quality objectives in the receiving water. The updated information that supports the removal of effluent limitations for these constituents includes the following:

- i. **Aluminum, Total Recoverable.** As discussed in Section IV.C.3 of the Fact Sheet, based on the site-specific study completed by the Discharger and new aluminum toxicity studies developed by other dischargers with similar receiving water conditions, the Central Valley Water Board finds that based on site-specific conditions of Dry Creek, the chronic criterion (87 µg/L) recommended in U.S. EPA's NAWQC for aluminum is not applicable. Effluent monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the Secondary MCL or the NAWQC acute criterion.
- ii. **Cadmium, Total Recoverable.** Effluent and receiving water monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the CTR aquatic life criteria.
- iii. **Carbon Tetrachloride.** Effluent and receiving water monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the CTR human health criteria.
- iv. **Chlorine Residual.** Use of chlorine for disinfection and maintenance processes has been discontinued at the Facility since May 2009. Therefore, the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the Basin Plan's narrative toxicity objective.
- v. **Chlorodibromomethane.** Effluent data collected between May 2009 and April 2013 and receiving water monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the CTR human health criteria.
- vi. **Cyanide.** Effluent and receiving water monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the CTR aquatic life criteria.
- vii. **Dichlorobromomethane.** Effluent data collected between May 2009 and April 2013 and receiving water monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the CTR human health criteria.



- viii. **Iron.** Effluent data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the Secondary MCL.
- ix. **Manganese.** Effluent data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the Secondary MCL.
- x. **Mercury.** Effluent data and receiving water monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the CTR human health criteria.
- xi. **Settleable Solids.** Effluent data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the Basin Plan narrative objective.
- xii. **Zinc.** Effluent data and receiving water monitoring data collected between August 2008 and April 2013 indicates that the discharge does not demonstrate reasonable potential to cause or contribute to an exceedance of the CTR aquatic life criteria.

Thus, removal of the effluent limitations for aluminum, cadmium, carbon tetrachloride, chlorine residual, chlorodibromomethane, cyanide, dichlorobromomethane, iron, manganese, settleable solids, and zinc and relaxation of the effluent limitations for mercury from Order R5-2008-0077-01 is in accordance with CWA section 402(o)(2)(B)(i), which allows for the removal or relaxation of effluent limitations based on information that was not available at the time of permit issuance.

- c. **Turbidity.** Order R5-2008-0077-01 contained effluent limitations for turbidity. The prior limitations were solely an operational check to ensure the treatment system was functioning properly and could meet the limits for solids and coliform. The prior effluent limitations were not intended to regulate turbidity in the receiving water. Rather, turbidity is an operational parameter to determine proper system functioning and not a WQBEL.

This Order contains operational turbidity specifications to be met in lieu of effluent limitations. The revised Order does not include effluent limitations for turbidity. However, the performance-based specification in this Order is an equivalent limit that is not less stringent, and therefore does not constitute backsliding.

The revised operational specifications for turbidity are the same as the effluent limitations in Order R5-2008-0077-01. These revisions are consistent with State regulations implementing recycled water requirements. The revision in the turbidity limitation is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution 68-16 because this Order imposes equivalent or more stringent requirements than Order R5-2008-0077-01 and therefore does not allow degradation.

#### 4. Antidegradation Policies

- a. **Surface Water.** This Order does not allow for an increase in flow or mass of pollutants to the receiving water. Therefore, a complete antidegradation analysis is not necessary. The Order requires compliance with applicable federal technology-based standards and with WQBELs where the discharge could have the reasonable

potential to cause or contribute to an exceedance of water quality standards. The permitted discharge is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution No. 68-16. Compliance with these requirements will result in the use of best practicable treatment or control of the discharge. The impact on existing water quality will be insignificant.

This Order removes existing effluent limitations for constituents in which updated monitoring data demonstrates that the effluent does not cause or contribute to an exceedance of the applicable water quality criteria or objectives in the receiving water (i.e., aluminum, cadmium, carbon tetrachloride, chlorine residual, chlorodibromomethane, cyanide, dichlorobromomethane, iron, manganese, settleable solids, and zinc) and relaxes effluent limitations for mercury. The Central Valley Water Board finds that the removal of the effluent limitations for aluminum, cadmium, carbon tetrachloride, chlorine residual, chlorodibromomethane, cyanide, dichlorobromomethane, iron, manganese, settleable solids, and zinc and the relaxation of effluent limitations for mercury does not result in an allowed increase in pollutants or any additional degradation of the receiving water. Thus, the removal of effluent limitations is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution No. 68-16.

- b. **Groundwater.** The Discharger installed an impermeable HPDE liner in the larger (16 acres) storage basin during the last permit term. The smaller storage basin (2 acres) is concrete lined, which also makes it impermeable. With the lining of both storage basins the potential for future groundwater contamination has been minimized, especially because both basins are not used for long term storage.

## 5. Stringency of Requirements for Individual Pollutants

This Order contains both technology-based effluent limitations and WQBELs for individual pollutants. The technology-based effluent limitations consist of restrictions on flow, BOD<sub>5</sub>, and TSS. Restrictions on these parameters are discussed in section IV.B.2 of the Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum, federal technology-based requirements that are necessary to meet water quality standards.

WQBELs have been derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant WQBELs were derived from the CTR, the CTR is the applicable standard pursuant to 40 C.F.R. section 131.38. The procedures for calculating the individual WQBELs for priority pollutants are based on the CTR implemented by the SIP, which was approved by U.S. EPA on May 18, 2000. All beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by U.S. EPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to U.S. EPA prior to May 30, 2000, but not approved by U.S. EPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 C.F.R. section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

This Order contains pollutant restrictions that are more stringent than applicable federal requirements and standards. Specifically, this Order includes effluent limitations for total coliform organisms that are more stringent than applicable federal standards, but that are nonetheless necessary to meet numeric objectives or protect beneficial uses. The

rationale for including these limitations is explained in section IV.C.3 of this Fact Sheet. In addition, the Central Valley Water Board has considered the factors in Water Code section 13241 in section IV.C.3 of this Fact Sheet.

**Summary of Final Effluent Limitations  
 Discharge Point 001**

**Table F-16. Summary of Final Effluent Limitations**

| Parameter                                | Units                | Effluent Limitations             |                  |                 |                       |                       | Basis <sup>1</sup> |
|------------------------------------------|----------------------|----------------------------------|------------------|-----------------|-----------------------|-----------------------|--------------------|
|                                          |                      | Average Monthly                  | Average Weekly   | Maximum Daily   | Instantaneous Minimum | Instantaneous Maximum |                    |
| Flow                                     | MGD                  | --                               | --               | 18              | --                    | --                    | DC                 |
| <b>Conventional Pollutants</b>           |                      |                                  |                  |                 |                       |                       |                    |
| Biochemical Oxygen Demand (5-day @ 20°C) | mg/L                 | 10                               | 15               | 20              | --                    | --                    | TTC                |
|                                          | lbs/day <sup>2</sup> | 1,500                            | 2,250            | 3,000           | --                    | --                    |                    |
|                                          | % Removal            | 85                               | --               | --              | --                    | --                    | CFR                |
| pH                                       | standard units       | --                               | --               | --              | 6.5                   | 8.5                   | BP                 |
| Total Suspended Solids                   | mg/L                 | 10                               | 15               | 20              | --                    | --                    | TTC                |
|                                          | lbs/day <sup>2</sup> | 1,500                            | 2,250            | 3,000           | --                    | --                    |                    |
|                                          | % Removal            | 85                               | --               | --              | --                    | --                    | CFR                |
| <b>Priority Pollutants</b>               |                      |                                  |                  |                 |                       |                       |                    |
| Mercury, Total Recoverable               | lbs/year             | 5.5 <sup>3</sup>                 | --               | --              | --                    | --                    | Pending TMDL       |
| <b>Non-Conventional Pollutants</b>       |                      |                                  |                  |                 |                       |                       |                    |
| Ammonia Nitrogen, Total (as N)           | mg/L                 | 0.8                              | --               | 2.2             | --                    | --                    | NAWQC              |
|                                          | lbs/day <sup>2</sup> | 120                              | --               | 330             | --                    | --                    |                    |
| Nitrate plus Nitrite (as N)              | mg/L                 | 10                               | --               | --              | --                    | --                    | MCL                |
| Total Coliform Organisms                 | MPN/100 mL           | --                               | 2.2 <sup>4</sup> | 23 <sup>5</sup> | --                    | 240                   | Title 22           |
| Acute Toxicity                           | % Survival           | 70 <sup>6</sup> /90 <sup>7</sup> | --               | --              | --                    | --                    | BP                 |

| Parameter | Units | Effluent Limitations |                |               |                       |                       | Basis <sup>1</sup> |
|-----------|-------|----------------------|----------------|---------------|-----------------------|-----------------------|--------------------|
|           |       | Average Monthly      | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |                    |

- <sup>1</sup> DC – Based on the design capacity of the Facility.  
 TTC – Based on tertiary treatment capability. These effluent limitations reflect the capability of a properly operated tertiary treatment plant.  
 CFR – Based on secondary treatment standards contained in 40 CFR Part 133.  
 BP – Based on water quality objectives contained in the Basin Plan.  
 Pending TMDL - Based on maintaining the mercury loading at the current level until a TMDL can be established.  
 NAWQC – Based on USEPA's National Ambient Water Quality Criteria for the protection of freshwater aquatic life.  
 MCL – Based on the Primary Maximum Contaminant Level.  
 Title 22 – Based on CA Department of Public Health Reclamation Criteria, CCR, Division 4, Chapter 3 (Title 22).
- <sup>2</sup> Based on an average dry weather flow of 18 MGD.
- <sup>3</sup> The total annual mass discharge of total mercury shall not exceed 5.5 lbs/year.
- <sup>4</sup> Applied as a 7-day median effluent limitation.
- <sup>5</sup> Not to be exceeded more than once in any 30-day period.
- <sup>6</sup> 70% minimum of any one bioassay.
- <sup>7</sup> 90% median for any three consecutive bioassays.

**E. Interim Effluent Limitations – Not Applicable**

**F. Land Discharge Specifications – Not Applicable**

**G. Recycling Specifications – Not Applicable**

**V. RATIONALE FOR RECEIVING WATER LIMITATIONS**

Basin Plan water quality objectives to protect the beneficial uses of surface water and groundwater include numeric objectives and narrative objectives, including objectives for chemical constituents, toxicity, and tastes and odors. The toxicity objective requires that surface water and groundwater be maintained free of toxic substances in concentrations that produce detrimental physiological responses in humans, plants, animals, or aquatic life. The chemical constituent objective requires that surface water and groundwater shall not contain chemical constituents in concentrations that adversely affect any beneficial use or that exceed the MCLs in Title 22, CCR. The tastes and odors objective states that surface water and groundwater shall not contain taste- or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses. The Basin Plan requires the application of the most stringent objective necessary to ensure that surface water and groundwater do not contain chemical constituents, toxic substances, radionuclides, or taste and odor producing substances in concentrations that adversely affect domestic drinking water supply, agricultural supply, or any other beneficial use.

**A. Surface Water**

1. CWA section 303(a-c), requires states to adopt water quality standards, including criteria where they are necessary to protect beneficial uses. The Central Valley Water Board adopted water quality criteria as water quality objectives in the Basin Plan. The Basin Plan states that “[t]he numerical and narrative water quality objectives define the least stringent standards that the Regional Water Board will apply to regional waters in order to protect the beneficial uses.” The Basin Plan includes numeric and narrative water quality objectives for various beneficial uses and water bodies. This Order contains receiving surface water limitations based on the Basin Plan numerical and narrative water quality objectives for bacteria, biostimulatory substances, color, chemical constituents, dissolved oxygen, floating material, oil and grease, pH, pesticides,

radioactivity, suspended sediment, settleable substances, suspended material, tastes and odors, temperature, toxicity, and turbidity.

- a. **pH.** Order R5-2008-0077-01 established a receiving water limitation for pH specifying that discharges from the Facility shall not cause the ambient pH to change by more than 0.5 units based on the water quality objective for pH in the Basin Plan, and allowed an annual averaging period for calculating pH change. The Central Valley Water Board adopted Resolution R5-2007-0136 on 25 October 2007, amending the Basin Plan to delete the portion of the pH water quality objective that limits the change in pH to 0.5 units and the allowance of averaging periods for pH. The Basin Plan amendment has been approved by the State Water Board, the Office of Administrative Law, and USEPA. Consistent with the revised water quality objective in the Basin Plan, this Order does not require a receiving water limitation for pH change.

In Finding No. 14 of Resolution R5-2007-0136 the Central Valley Water Board found that the change in the pH receiving water objective is consistent with the State Water Board Resolution No. 68-16, in that the changes to water quality objectives (i) consider maximum benefit to the people of the State, (ii) will not unreasonably affect present and anticipated beneficial use of waters, and (iii) will not result in water quality less than that prescribed in policies, and is consistent with the federal Antidegradation Policy (40 CFR 131.12).

Ammonia is the only constituent in the discharge regulated by this Order directly related to pH. The fixed ammonia effluent limitations in this Order are based on reasonable worse-case conditions. Although ammonia criteria are based on pH, and the pH receiving water limitations are more lenient in this Order than in the previous permit, the fixed ammonia limits are more stringent limits, and are developed to protect under worse-case pH conditions. Therefore the relaxation of the pH receiving water limitation will protect aquatic life and other beneficial uses and will not unreasonably affect present and anticipated beneficial uses nor result in water quality less than described in applicable policies. The relaxation of the receiving water limitation is not expected to cause other impacts on water quality. The Central Valley Water Board finds that the relaxation of the pH receiving water limitation (i) is to the maximum benefit to the people of the State, (ii) will not unreasonably affect present and anticipated beneficial use of waters, and (iii) will not result in water quality less than that prescribed in policies, and is consistent with the federal Antidegradation Policy (40 CFR 131.12).

The revised receiving water limitation for pH, which is based on the amendment to the Basin Plan's pH water quality objective, reflects current scientifically supported pH requirements for the protection of aquatic life and other beneficial uses. The revised receiving water limitation for pH is more consistent with the current USEPA recommended criteria and is fully protective of aquatic life and the other beneficial uses listed in the Basin Plan. Changes in pH when pH is maintained within the range of 6.5 to 8.5 are neither beneficial nor adverse and, therefore, are not considered to be degradation in water quality. Attempting to restrict pH changes to 0.5 pH units would incur substantial costs without demonstrable benefits to beneficial uses. Thus, any changes in pH that would occur under the revised pH limitation would not only be protective of beneficial uses, but also would be consistent with maximum benefit to people of the State. Therefore the proposed amendment will not violate antidegradation policies.

- b. **Turbidity.** Order R5-2008-0077-01 established a receiving water limitation for turbidity specifying that discharges from the Facility shall not cause the turbidity to increase more than 1 NTU where natural turbidity is between 0 and 5 NTU based on the water quality objective for turbidity in the Basin Plan. The Central Valley Water Board adopted Resolution R5-2007-0136 on 25 October 2007, amending the Basin Plan to limit turbidity to 2 NTU when the natural turbidity is less than 1 NTU. The Basin Plan amendment has been approved by the State Water Board, the Office of Administrative Law, and USEPA. Consistent with the revised water quality objective in the Basin Plan, this Order limits turbidity to 2 NTU when the natural turbidity is less than 1 NTU.

In Finding No. 14 of Resolution R5-2007-0136 the Central Valley Water Board found that the change in the turbidity receiving water objective is consistent with the State Water Board Resolution No. 68-16, in that the changes to water quality objectives (i) consider maximum benefit to the people of the State, (ii) will not unreasonably affect present and anticipated beneficial use of waters, and (iii) will not result in water quality less than that prescribed in policies, and is consistent with the federal Antidegradation Policy (40 CFR 131.12).

This Order includes operational specifications that require the Discharger to operate the treatment system to insure that turbidity shall not exceed 2 NTU as a daily average, and 5 NTU more than 5 percent of the time within a 24 hour period, and 10 NTU, at any time. Because this Order limits the average daily discharge of turbidity to 2 NTU, the Order will be protective of the receiving water under all natural background conditions as defined in the Basin Plan's revised water quality objective for turbidity. The relaxation of the turbidity receiving water limitation will protect aquatic life and other beneficial uses and will not unreasonably affect present and anticipated beneficial uses nor result in water quality less than described in applicable policies. The relaxation of the receiving water limitation is not expected to cause other impacts on water quality. The Central Valley Water Board finds that the relaxation of the turbidity receiving water limitation is to the maximum benefit to the people of the State, (ii) will not unreasonably affect present and anticipated beneficial use of waters, and (iii) will not result in water quality less than that prescribed in policies, and is consistent with the federal Antidegradation Policy (40 CFR 131.12).

The revised receiving water limitation for turbidity, which is based on the amendment to the Basin Plan's turbidity water quality objective, reflects current scientifically supported turbidity requirements for the protection of aquatic life and other beneficial uses and, therefore, will be fully protective of aquatic life and the other beneficial uses listed in the Basin Plan. Changes in turbidity allowed by the revised receiving water limitation, when ambient turbidity is below 1 NTU, would not adversely affect beneficial uses and would maintain water quality at a level higher than necessary to protect beneficial uses. Restricting low-level turbidity changes further may require costly upgrades, which would not provide any additional protection of beneficial uses. Thus, any changes in turbidity that would occur under the amended turbidity receiving water limitation would not only be protective of beneficial uses, but also would be consistent with maximum benefit to people of the State. Therefore, the relaxed receiving water limitations for turbidity will not violate antidegradation policies.

- c. **Temperature.** Dry Creek has the beneficial uses of COLD and WARM. The Basin Plan includes the objective that "[a]t not time or place shall the temperature of COLD or WARM intrastate waters be increased more than 5°F above natural

*receiving water temperature.*” This Order includes receiving water limitations based on this objective.

As described in section IV.C.2.a of this Fact Sheet, based on letters of 31 July 1990 and 15 August 1990 from DFW, DFW stated that Dry Creek is used by adult Chinook salmon migrating upstream to spawning areas and by young salmon migrating downstream in November through March, in normal years. Therefore, consistent with Order R5-2008-0077-01, this Order includes a daily average temperature receiving water limitation of 15.6°C (60°F) and a maximum daily temperature receiving water limitation of 16.7°C (62°F) between 1 November and 31 March.

## **B. Groundwater**

1. The beneficial uses of the underlying groundwater are municipal and domestic supply, industrial service supply, industrial process supply, and agricultural supply.
2. Basin Plan water quality objectives include narrative objectives for chemical constituents, tastes and odors, and toxicity of groundwater. The toxicity objective requires that groundwater be maintained free of toxic substances in concentrations that produce detrimental physiological responses in humans, plants, animals, or aquatic life. The chemical constituent objective states groundwater shall not contain chemical constituents in concentrations that adversely affect any beneficial use. The tastes and odors objective prohibits taste- or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses. The Basin Plan also establishes numerical water quality objectives for chemical constituents and radioactivity in groundwaters designated as municipal supply. These include, at a minimum, compliance with MCLs in Title 22 of the CCR. The bacteria objective prohibits coliform organisms at or above 2.2 MPN/100 mL. The Basin Plan requires the application of the most stringent objective necessary to ensure that waters do not contain chemical constituents, toxic substances, radionuclides, taste- or odor-producing substances, or bacteria in concentrations that adversely affect municipal or domestic supply, agricultural supply, industrial supply or some other beneficial use.
3. The previous Order R5-2008-0077-01 contained groundwater limits that preventing an increase in groundwater constituent concentrations above background concentrations. Since the adoption of the Order R5-2008-0077-01 the Discharger lined the large storage pond (16 acres) with an impermeable HPDE liner, making both emergency storage ponds lined with impermeable liners. Minimal to no percolation will occur from both lined ponds and use of the storage ponds for short periods of time (a few days at time) should prevent any degradation of groundwater below the Facility from occurring.

## **VI. RATIONALE FOR PROVISIONS**

### **A. Standard Provisions**

Standard Provisions, which apply to all NPDES permits in accordance with 40 C.F.R. section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 C.F.R. section 122.42, are provided in Attachment D. The Discharger must comply with all standard provisions and with those additional conditions that are applicable under section 122.42.

Sections 122.41(a)(1) and (b) through (n) of 40 C.F.R. establish conditions that apply to all state-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) of 40 C.F.R. allows the state to omit or

modify conditions to impose more stringent requirements. In accordance with 40 C.F.R. section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 C.F.R. sections 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387(e).

## **B. Special Provisions**

### **1. Reopener Provisions**

- a. **Mercury.** This provision allows the Central Valley Water Board to reopen this Order in the event mercury is found to be causing toxicity based on acute or chronic toxicity test results, or if a TMDL program is adopted. In addition, this Order may be reopened if the Central Valley Water Board determines that a mercury offset program is feasible for dischargers subject to NPDES permits.
- b. **Whole Effluent Toxicity.** This Order requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity through a TRE. This Order may be reopened to include a numeric chronic toxicity limitation, a new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE. Additionally, if a numeric chronic toxicity water quality objective is adopted by the State Water Board, this Order may be reopened to include a numeric chronic toxicity limitation based on that objective.
- c. **Water Effects Ratio (WER) and Metal Translators.** A default WER of 1.0 has been used in this Order for calculating criteria for applicable inorganic constituents. In addition, default dissolved-to-total metal translators have been used to convert water quality objectives from dissolved to total recoverable. If the Discharger performs studies to determine site-specific WERs and/or site-specific dissolved-to-total metal translators, this Order may be reopened to modify the effluent limitations for the applicable inorganic constituents.

### **2. Special Studies and Additional Monitoring Requirements**

- a. **Chronic Whole Effluent Toxicity Requirements.** The Basin Plan contains a narrative toxicity objective that states, "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*" (Basin Plan at page III-8.00.) Based on whole effluent chronic toxicity testing performed by the Discharger from December 2008 through March 2013, the discharge does not have reasonable potential to cause or contribute to an in-stream excursion above of the Basin Plan's narrative toxicity objective.

The Monitoring and Reporting Program of this Order requires chronic WET monitoring for demonstration of compliance with the narrative toxicity objective. The provision also includes a numeric toxicity monitoring trigger, requirements for accelerated monitoring, and requirements for TRE initiation if toxicity is demonstrated. As required by Order R5-2008-0077-01, the Discharger submitted the *Dry Creek Wastewater Treatment Plant Initial Investigative TRE Work Plan* on 22 August 2008 to ensure the Discharger has a plan to immediately move forward with the initial tiers of a TRE, in the event effluent toxicity is encountered. Consistent with Order R5-2008-0077-01, this Order requires submission of a full TRE Workplan within 60 days of notification of test results exhibiting toxicity during accelerated monitoring.



**Monitoring Trigger.** A numeric toxicity monitoring trigger of  $> 1$  TUc (where TUc = 100/NOEC) is applied in the provision, because this Order does not allow any dilution for the chronic condition. Therefore, a TRE is triggered when the effluent exhibits toxicity at 100% effluent.

**Accelerated Monitoring.** The provision requires accelerated WET testing when a regular WET test result exceeds the monitoring trigger. The purpose of accelerated monitoring is to determine, in an expedient manner, whether there is toxicity before requiring the implementation of a TRE. Due to possible seasonality of the toxicity, the accelerated monitoring should be performed in a timely manner, preferably taking no more than 2 to 3 months to complete.

The provision requires accelerated monitoring consisting of four chronic toxicity tests in a six-week period (i.e., one test every two weeks) using the species that exhibited toxicity. Guidance regarding accelerated monitoring and TRE initiation is provided in the *Technical Support Document for Water Quality-based Toxics Control*, EPA/505/2-90-001, March 1991 (TSD). The TSD at page 118 states, “EPA recommends if toxicity is repeatedly or periodically present at levels above effluent limits more than 20 percent of the time, a TRE should be required.” Therefore, four accelerated monitoring tests are required in this provision. If no toxicity is demonstrated in the four accelerated tests, then it demonstrates that toxicity is not present at levels above the monitoring trigger more than 20 percent of the time (only 1 of 5 tests are toxic, including the initial test). However, notwithstanding the accelerated monitoring results, if there is adequate evidence of effluent toxicity (i.e. toxicity present exceeding the monitoring trigger more than 20 percent of the time), the Executive Officer may require that the Discharger initiate a TRE.

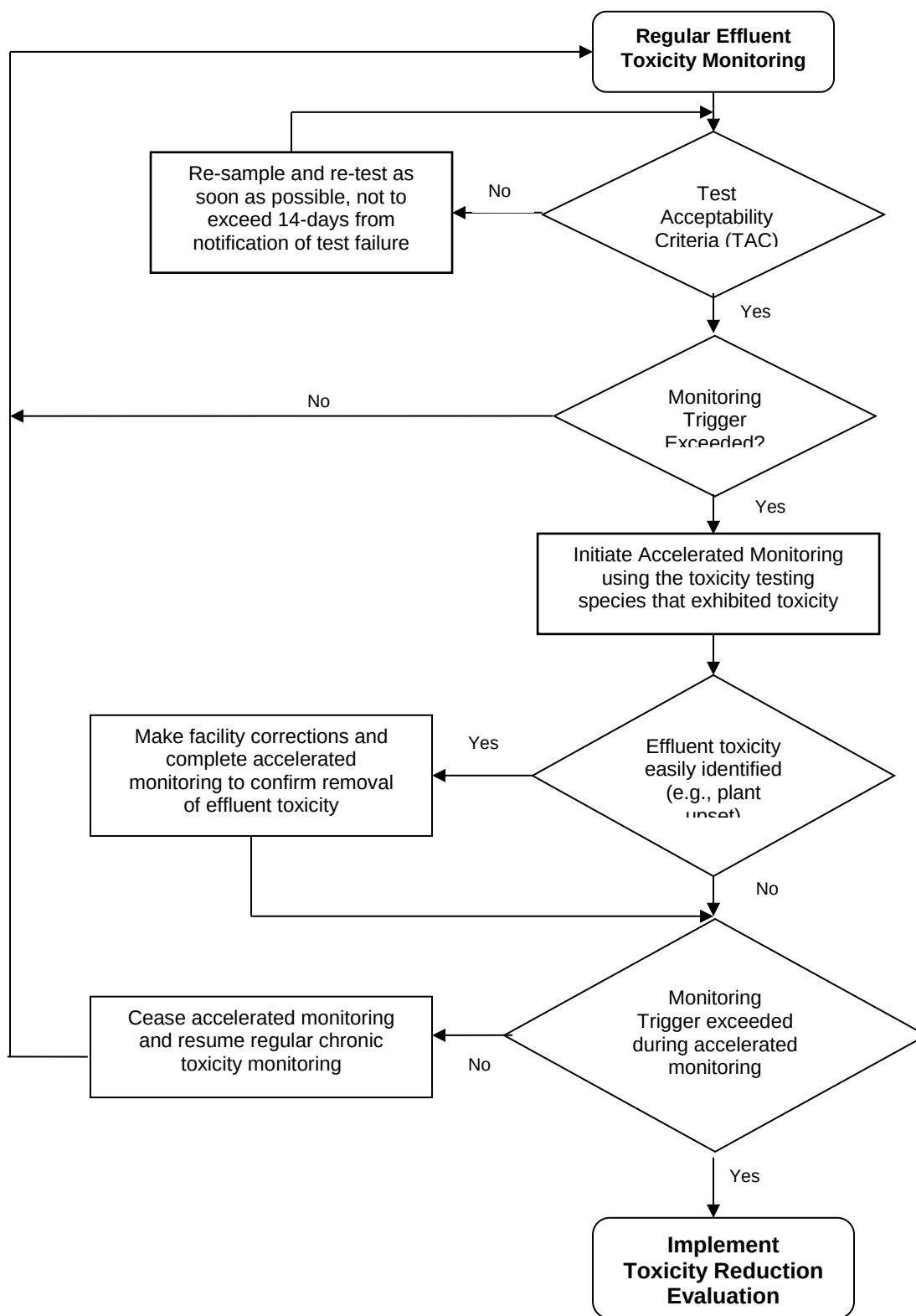
See the WET Accelerated Monitoring Flow Chart (Figure F-1), below, for further clarification of the accelerated monitoring requirements and for the decision points for determining the need for TRE initiation.

**TRE Guidance.** The Discharger is required to prepare a TRE Workplan in accordance with USEPA guidance. Numerous guidance documents are available, as identified below:

- *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants*, EPA/833-B-99/002, August 1999.
- *Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations (TREs)*, EPA/600/2-88/070, April 1989.
- *Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures, Second Edition*, EPA 600/6-91/003, February 1991.
- *Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents, Phase I*, EPA/600/6-91/005F, May 1992.
- *Methods for Aquatic Toxicity Identification Evaluations: Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity, Second Edition*, EPA/600/R-92/080, September 1993.

- *Methods for Aquatic Toxicity Identification Evaluations: Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity, Second Edition, EPA 600/R-92/081, September 1993.*
- *Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms, Fifth Edition, EPA-821-R-02-012, October 2002.*
- *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition, EPA-821-R-02-013, October 2002.*
- *Technical Support Document for Water Quality-based Toxics Control, EPA/505/2-90-001, March 1991.*

**Figure F-1**  
**WET Accelerated Monitoring Flow Chart**



### 3. Best Management Practices and Pollution Prevention

- a. **Salinity Evaluation and Minimization Plan.** The Discharger submitted a *DCWWTP and PG WWTP Salinity Evaluation and Minimization Plan* on 9 March 2009, which identified salinity contributions from the water supply and residential, commercial, and industrial users, and indicated that salinity may increase when water from their Aquifer Storage and Recovery wells is used, but that the discharge would still be in compliance with the Salinity Goal contained in Order R5-2008-0077-01. This Order requires the Discharger to continue to implement its Salinity Evaluation and Minimization Plan to ensure adequate measures are developed and implemented by the Discharger to reduce the discharge of salinity to Dry Creek.
- b. **Pollution Prevention Plan (PPP).** As required by Time Schedule Order (TSO) R5-2008-0078, the Discharger submitted a PPP for mercury on 11 June 2009. As discussed in section IV.C.3.b of this Fact Sheet, a TMDL is under development for the Natomas East Main Drainage Canal downstream of the Facility which may include a waste load allocation applicable to the Facility. Therefore, this Order requires the Discharger to continue to implement their PPP for mercury per Water Code section 13263.3(d)(1)(D). Progress reports shall be submitted annually in accordance with the Monitoring and Reporting Program (Attachment E, section X.D.1.). The progress reports shall discuss the effectiveness of the PPP in the reduction of mercury in the discharge, include a summary of mercury monitoring results, and discuss updates to the PPP.

### 4. Construction, Operation, and Maintenance Specifications

- a. **Filtration System Operating Specifications.** Turbidity is included as an operational specification as an indicator of the effectiveness of the filtration system for providing adequate disinfection. The tertiary treatment process utilized at this Facility is capable of reliably meeting a turbidity limitation of 2 NTU as a daily average. Failure of the treatment system such that virus removal is impaired would normally result in increased particles in the effluent, which result in higher effluent turbidity and could impact UV dosage. Turbidity has a major advantage for monitoring filter performance, allowing immediate detection of filter failure and rapid corrective action. The operational specification requires that turbidity prior to disinfection shall not exceed 2 NTU as a daily average; 5 NTU, more than 5 percent of the time within a 24-hour period, and an instantaneous maximum of 10 NTU.
- b. **Ultraviolet Light (UV) Disinfection System Operating Specifications.** This Order requires that wastewater shall be oxidized, coagulated, filtered, and adequately disinfected pursuant to DPH reclamation criteria, CCR, Title 22, division 4, chapter 3, (Title 22), or equivalent. To ensure that the UV disinfection system is operated to achieve the required pathogen removal, this Order includes effluent limits for total coliform organisms, filtration system operating specifications, and UV disinfection system operating specifications. Compliance with total coliform effluent limits alone does not ensure that pathogens in the municipal wastewater have been deactivated by the UV disinfection system. Compliance with the effluent limits and the filtration system and UV disinfection operating specifications demonstrates compliance with the equivalency to Title 22 disinfection requirement.

The National Water Research Institute (NWRI) and American Water Works Association Research Foundation (AWWRF) *Ultraviolet Disinfection Guidelines for Drinking Water and Water Reuse* first published in December 2000 and revised as a Third Edition dated August 2012 (NWRI guidelines) includes UV operating

specifications for compliance with Title 22. For water recycling in accordance with Title 22, the UV system shall be an approved system included in the Treatment Technology Report for Recycled Water, December 2009 (or a later version, as applicable) published by the DPH. The UV system shall also conform to all requirements and operating specifications of the NWRI guidelines. A Memorandum dated 1 November 2004 issued by DPH to Regional Water Board executive offices recommended that provisions be included in permits for water recycling treatment plants employing UV disinfection requiring Dischargers to establish fixed cleaning frequency of lamp sleeves, as well as, include provisions that specify minimum delivered UV dose that must be maintained (per the NWRI Guidelines). The NWRI Guidelines recommend a minimum hourly average UV dose of 100 mJ/cm<sup>2</sup> and a minimum hourly average UV transmittance of 55%.

The Discharger submitted the UV Field Commissioning Test Engineering Report dated October 2008 (*City of Roseville, California, Dry Creek WWTP UV Field Commissioning Tests, Summary Report – Trojan UV3000plus System; Final October 2008; Carollo*) to DPH and the Central Valley Water Board. In a letter dated 25 February 2009 to the Central Valley Water Board, based on review of the UV Field Commissioning Test Engineering Report, DPH recommended that the Board include in the City's water reclamation permit that the "UV system must be operated to deliver a minimum UV dose of 100 mJ/cm<sup>2</sup> at all times", and also recommended several operational and maintenance requirements. [The Discharger is permitted to use recycled water under Master Reclamation Permit, Order No. 97-147.] This Order includes ~~an~~ UV operating specifications for a minimum hourly average UV dose of 100 mJ/cm<sup>2</sup> and a minimum hourly average UV transmittance of 55%, based on the NWRI Guidelines and DPH recommendations. DPH's February 2009 letter also recommends the following operational parameter:

*"Flow should be maintained at or below 34.06 MGD....unless the UVT is at or above 66 percent. This will maintain one redundant bank of UV lamps out of service. If during short-term, unexpected conditions the flow is greater than this and the UVT is less than 71 percent, the redundant bank would need to be utilized....to maintain the required 100-mJ/cm<sup>2</sup> dose, otherwise, the flow must be diverted."*

The UV system at this facility is set to automatically divert any flows that do not receive the 100 mJ/cm<sup>2</sup> dose. In addition, the permitted average dry weather flow is 18 MGD, peak daily flow has never been greater than 22 MGD (daily average, based on data from 2010-2013). The facility also has 45 million gallons of storage for diversions during peak flows (if the 100 mJ/cm<sup>2</sup> dose is not achieved). Because the flows will rarely exceed a peak of 34.06 MGD and only for short time periods (much less than a day), there is 45 million gallons of emergency storage, and the system is automatically set to divert, the recommended redundancy may be achieved at this site without requiring the 34.06 MGD (UV) flow limit. The recommended operational parameter will be considered further when the Central Valley Water Board renews the Discharger's Master Reclamation Permit.

- c. **Pond Operating Requirements.** This operating and maintenance specifications for the storage basins are necessary to protect the beneficial uses of the groundwater. The specifications included in this Order are retained from Order R5-2008-0077-01. In addition, reporting requirements related to use of the basins are required to monitor their use and the potential impact on groundwater.

## 5. Special Provisions for Municipal Facilities (POTWs Only)

### a. Pretreatment Requirements

- i. The federal CWA section 307(b), and federal regulations, 40 CFR Part 403, require POTWs to develop an acceptable industrial pretreatment program. A pretreatment program is required to prevent the introduction of pollutants, which will interfere with treatment plant operations or sludge disposal, and prevent pass through of pollutants that exceed water quality objectives, standards or permit limitations. Pretreatment requirements are imposed pursuant to 40 CFR Part 403.
- ii. The Discharger shall implement and enforce its approved pretreatment program and is an enforceable condition of this Order. If the Discharger fails to perform the pretreatment functions, the Central Valley Water Board, the State Water Board or USEPA may take enforcement actions against the Discharger as authorized by the CWA.
- iii. The Discharger operates a pretreatment program that addresses discharges to both the Facility and the Discharger's Pleasant Grove Wastewater Treatment Plant. Although the pretreatment program requirements, including the requirements to submit annual reports on program progress, are included in the permits for both plants, the Discharger will be allowed to submit one report that addresses the Discharger's activities that affect both plants.

- b. The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (General Order) on 2 May 2006. The Monitoring and Reporting Requirements for the General Order were amended by Water Quality Order WQ 2008-0002-EXEC on 20 February 2008. The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. The Discharger is enrolled under State Water Board Order 2006-0003-DWQ.

## 6. Other Special Provisions

- a. **Title 22, or Equivalent, Disinfection Requirements.** Consistent with Order R5-2008-0077-01, this Order requires wastewater to be oxidized, coagulated, filtered, and adequately disinfected pursuant to DPH reclamation criteria, CCR, Title 22, division 4, chapter 3 (Title 22), or equivalent.
- b. By **1 October 2014**, the Discharger shall submit a *Groundwater Monitoring Well Destruction Workplan* for the wells that are not used at the Facility. The workplan shall describe the proposed abandonment procedures, which shall comply with *California Well Standards Bulletin 74-90* (June 1991); *State of California Bulletin 94-81* (December 1981); and any more stringent standards adopted by the state or county pursuant to Water Code section 13801. At a minimum, the workplan shall specifically address the monitoring wells listed in Table 1 of the Monitoring and Reporting Program, Attachment E. For each well that will not be destroyed, provide an explanation of the need to maintain the well for monitoring purposes.

- c. By **1 October 2015**, a *Well Destruction Report* shall be submitted to the Central Valley Water Board that describes in detail the methods used to abandon each well and includes copies of the well abandonment permits issued by the City of Roseville Environmental Utilities Department.

#### **7. Compliance Schedules – Not Applicable**

### **VII. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS**

Section 122.48 of 40 C.F.R. requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 authorize the Central Valley Water Board to require technical and monitoring reports. The Monitoring and Reporting Program (MRP), Attachment E, establishes monitoring and reporting requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

#### **A. Influent Monitoring**

1. Influent monitoring is required to collect data on the characteristics of the wastewater and to assess compliance with effluent limitations (e.g., BOD<sub>5</sub> and TSS reduction requirements). The monitoring frequency for flow (continuous) has been retained from Order R5-2008-0077-01.
2. Order R5-2008-0077-01 required daily monitoring for BOD<sub>5</sub> and TSS. The monitoring frequencies for BOD<sub>5</sub> and TSS have been reduced to three times per week. The Central Valley Water Board finds that this frequency is adequate to assess compliance with effluent limitations and monitor the performance of the Facility.
3. Order R5-2008-0077-01 required monitoring for electrical conductivity, pH, and hardness. Monitoring requirements for these constituents have not been retained from Order R5-2008-0077-01 as monitoring is not necessary to determine compliance with permit requirements.

#### **B. Effluent Monitoring**

1. Pursuant to the requirements of 40 CFR 122.44(i)(2) effluent monitoring is required for all constituents with effluent limitations. Effluent monitoring is necessary to assess compliance with effluent limitations, assess the effectiveness of the treatment process, and to assess the impacts of the discharge on the receiving stream and groundwater.
2. Effluent monitoring frequencies and sample types for flow (continuous), ammonia (weekly), hardness (monthly), mercury (monthly), nitrate (monthly), pH (continuous), temperature (daily), and total dissolved solids (monthly) have been retained from Order R5-2008-0077-01 to determine compliance with effluent limitations for these parameters.
3. Monitoring data collected during the term of Order R5-2008-0077-01 indicates that the Facility has maintained compliance with the applicable effluent limitations for BOD<sub>5</sub> and TSS. Therefore, the effluent monitoring frequencies for BOD<sub>5</sub> and TSS have been decreased from daily to three times per week. The Central Valley Water Board finds that this frequency is sufficient to determine compliance with the applicable effluent limitations and monitor the performance of the Facility.
4. Monitoring data collected during the term of Order R5-2008-0077-01 indicates that the Facility does not have reasonable potential for salinity. However, monitoring for electrical conductivity and total dissolved solids, indicator parameters for salinity, is necessary to characterize the discharge and ensure that the Discharger continues to implement adequate measures to reduce the discharge of salinity to Dry Creek. Therefore, the effluent monitoring frequency for electrical conductivity has been decreased from daily to

weekly. The Central Valley Water Board finds that this frequency is sufficient to monitor the performance of the Facility.

5. Monitoring data collected over the term of Order R5-2008-0077-01 for 1,1-dichloroethylene, aluminum, bis (2-ethylhexyl) phthalate, cadmium, carbon tetrachloride, chlorine residual, chlorodibromomethane, cyanide, dichlorobromomethane, iron, manganese, persistent chlorinated hydrocarbon pesticides, settleable solids, and zinc did not demonstrate reasonable potential to exceed water quality objectives/criteria. Thus, specific monitoring requirements for these parameters have not been retained from Order R5-2008-0077-01.
6. This Order establishes a new effluent limitation for nitrate plus nitrite. Therefore, this Order establishes monthly monitoring for nitrate and nitrite to determine compliance with the applicable effluent limitation.
7. Monitoring data collected during the term of Order R5-2008-0077-01 indicates that the Facility has maintained compliance with the applicable effluent limitations for total coliform organisms. Therefore, the effluent monitoring frequency for total coliform organisms has been decreased from daily to five times per week. The Central Valley Water Board finds that this frequency is sufficient to determine compliance with the applicable effluent limitations and monitor the performance of the Facility. This Order moves the point of compliance from Monitoring Location EFF-001 to an internal compliance point following the UV disinfection system. Therefore, monitoring for total coliform organisms is required at Monitoring Location UVS-002.
8. This Order includes operational specifications for turbidity. Although turbidity monitoring is necessary to monitor filter performance, the Discharger does not have a monitoring location for the combined effluent from the filters prior to the UV disinfection system. Rather, the Discharger monitors turbidity at a point downstream of the UV disinfection system and, if the turbidity specifications are exceeded, the Discharger diverts the wastewater to the emergency storage basins to avoid discharges of effluent that have not been adequately disinfected. Since the UV disinfection process has no effect on the turbidity of the wastewater and the Discharger has a system in place to automatically divert wastewater to the emergency storage basins if turbidity specifications are exceeded, monitoring for turbidity at a location downstream of the UV disinfection process and upstream of the point of diversion to the emergency storage basins is sufficient to determine compliance with the turbidity specifications and to ensure that the filtration system is operating properly to provide adequate disinfection. Therefore, this Order moves the point of compliance from the final effluent after disinfection to an internal compliance point following the filtration system and UV disinfection system but prior to the point of diversion to the effluent storage basins. Therefore, monitoring for turbidity is required at Monitoring Location FIL-001.
9. In accordance with Section 1.3 of the SIP, periodic monitoring is required for priority pollutants for which criteria or objectives apply and for which no effluent limitations have been established. Consistent with Order R5-2008-0077-01, this Order requires effluent monitoring for priority pollutants quarterly during the third year of the permit term. See Attachment E, Section IX.C for more detailed requirements related to performing priority pollutant monitoring.
10. California Water Code section 13176, subdivision (a), states: *"The analysis of any material required by [Water Code sections 13000-16104] shall be performed by a laboratory that has accreditation or certification pursuant to Article 3 (commencing with Section 100825) of Chapter 4 of Part 1 of Division 101 of the Health and Safety Code."*



DPH certifies laboratories through its Environmental Laboratory Accreditation Program (ELAP).

California Water Code section 13176, subdivision (a), states: "*The analysis of any material required by [Water Code sections 13000-16104] shall be performed by a laboratory that has accreditation or certification pursuant to Article 3 (commencing with Section 100825) of Chapter 4 of Part 1 of Division 101 of the Health and Safety Code.*" DPH certifies laboratories through its Environmental Laboratory Accreditation Program (ELAP). The Facility has an ELAP-certified laboratory on-site.

### **C. Whole Effluent Toxicity Testing Requirements**

1. **Acute Toxicity.** Based on acute toxicity testing conducted during the term of Order R5-2008-0077-01, the discharge has been in compliance with the effluent limitations for acute toxicity. Therefore, this Order reduces the frequency for 96-hour bioassay testing from quarterly to semiannually to demonstrate compliance with the effluent limitation for acute toxicity.
2. **Chronic Toxicity.** Consistent with Order R5-2008-0077-01, quarterly chronic whole effluent toxicity testing is required in order to demonstrate compliance with the Basin Plan's narrative toxicity objective.

### **D. Receiving Water Monitoring**

#### **1. Surface Water**

- a. Receiving water monitoring is necessary to assess compliance with receiving water limitations and to assess the impacts of the discharge on the receiving stream.
- b. The receiving water monitoring frequency and sample type for dissolved oxygen, hardness, pH, electrical conductivity, and temperature have been retained from Order R5-2008-0077-01.
- c. In accordance with Section 1.3 of the SIP, periodic monitoring for priority pollutants for which criteria or objectives apply and for which no effluent limitations have been established. This Order requires monitoring for priority pollutants and other pollutants of concern at Monitoring Location RSW-001 quarterly during the third year of the permit term, concurrent with effluent monitoring, in order to collect data to conduct an RPA for the next permit renewal. See Attachment E, Section IX.C for more detailed requirements related to performing priority pollutant monitoring.

#### **2. Groundwater**

- a. Groundwater monitoring has been discontinued because both emergency storage basins are now lined with impermeable liners and are used infrequently for short periods of time (a few days at a time).

### **E. Other Monitoring Requirements**

#### **1. Biosolids Monitoring**

Biosolids monitoring is required to ensure compliance with the biosolids disposal requirements contained in the Special Provision contained in section VI.C.6.b. of this Order. Biosolids disposal requirements are imposed pursuant to 40 CFR Part 503 to protect public health and prevent groundwater degradation.

#### **2. Water Supply Monitoring**

Order R5-2008-0077-01 contains water supply monitoring requirements which have not been retained in this Order. Due to the relatively low effluent salinity concentrations, the

Central Valley Water Board finds that water supply monitoring is no longer necessary to evaluate salinity sources in the wastewater.

**3. UV Disinfection System Monitoring**

UV system monitoring and reporting are required to ensure that the UV system is operated to adequately inactivate pathogens in the wastewater. UV disinfection system monitoring is imposed to achieve equivalency to requirements established by DPH, and the NWRI/AWWARF's "*Ultraviolet Disinfection Guidelines for Drinking Water and Water Reuse*."

**4. Land Discharge Monitoring**

Emergency storage basin monitoring has been retained from Order R5-2008-0077-01 to ensure proper operation of the storage basins.

**VIII. PUBLIC PARTICIPATION**

The Central Valley Water Board has considered the issuance of WDR's that will serve as an NPDES permit for the Dry Creek Wastewater Treatment Plant. As a step in the WDR adoption process, the Central Valley Water Board staff has developed tentative WDR's and has encouraged public participation in the WDR adoption process.

**A. Notification of Interested Parties**

The Central Valley Water Board notified the Discharger and interested agencies and persons of its intent to prescribe WDR's for the discharge and provided an opportunity to submit written comments and recommendations. Notification was provided through the following, Notice of Public Hearings were posted at the Discharger's Facility, City Hall and in the Roseville Press Tribune on 24 January 2014.

The public had access to the agenda and any changes in dates and locations through the Central Valley Water Board's website at:  
<http://www.waterboards.ca.gov/centralvalley/>

**B. Written Comments**

Interested persons were invited to submit written comments concerning tentative WDR's as provided through the notification process. Comments were due either in person or by mail to the Executive Office at the Central Valley Water Board at the address on the cover page of this Order.

To be fully responded to by staff and considered by the Central Valley Water Board, the written comments were due at the Central Valley Water Board office by 5:00 p.m. on 17 February 2014.

**C. Public Hearing**

The **Central Valley Water Board** held a public hearing on the tentative WDR's during its regular Board meeting on the following date and time and at the following location:

Date: 27/28 March 2014  
Time: 9:00 a.m.  
Location: Regional Water Quality Control Board, Central Valley Region  
11020 Sun Center Dr., Suite #200  
Rancho Cordova, CA 95670

Interested persons were invited to attend. At the public hearing, the Central Valley Water Board heard testimony pertinent to the discharge, WDR's, and permit. For accuracy of the record, important testimony was requested in writing.

**D. Reconsideration of Waste Discharge Requirements**

Any aggrieved person may petition the State Water Board to review the decision of the Central Valley Water Board regarding the final WDR's. The petition must be received by the State Water Board at the following address within 30 calendar days of the Central Valley Water Board's action:

State Water Resources Control Board  
Office of Chief Counsel  
P.O. Box 100, 1001 I Street  
Sacramento, CA 95812-0100

For instructions on how to file a petition for review, see  
[http://www.waterboards.ca.gov/public\\_notices/petitions/water\\_quality/wqpetition\\_instr.shtml](http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtml)

**E. Information and Copying**

The Report of Waste Discharge, other supporting documents, and comments received are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Central Valley Water Board by calling (916) 464-3291.

**F. Register of Interested Persons**

Any person interested in being placed on the mailing list for information regarding the WDR's and NPDES permit should contact the Central Valley Water Board, reference this facility, and provide a name, address, and phone number.

**G. Additional Information**

Requests for additional information or questions regarding this order should be directed to Joshua Palmer at (916) 464-4674.

**ATTACHMENT G – SUMMARY OF REASONABLE POTENTIAL ANALYSIS FOR CONSTITUENTS OF CONCERN**

| Constituent                    | Units    | MEC              | B                | C                                  | CMC                                | CCC                                | Water & Org | Org. Only | Basin Plan | MCL             | Reasonable Potential |
|--------------------------------|----------|------------------|------------------|------------------------------------|------------------------------------|------------------------------------|-------------|-----------|------------|-----------------|----------------------|
| Aluminum, Total Recoverable    | µg/L     | 121 <sup>1</sup> | 237 <sup>1</sup> | 200                                | 750 <sup>2</sup>                   | --                                 | --          | --        | --         | 200             | No                   |
| Ammonia Nitrogen, Total (as N) | mg/L     | 0.98             | 0.07             | 1.58                               | 2.14 <sup>2</sup>                  | 1.58 <sup>3</sup>                  | --          | --        | --         | --              | Yes                  |
| Cadmium, Total Recoverable     | µg/L     | 0.1              | 0.26             | 1.9 <sup>4</sup> /1.0 <sup>5</sup> | 3.0 <sup>4</sup> /1.3 <sup>5</sup> | 1.9 <sup>4</sup> /1.0 <sup>5</sup> | --          | --        | --         | 5               | No                   |
| Carbon Tetrachloride           | µg/L     | <0.16            | <0.16            | 0.25                               | --                                 | --                                 | 0.25        | 4.4       | --         | 0.5             | No                   |
| Chloride                       | mg/L     | 42               | 16               | 230                                | 860 <sup>2</sup>                   | 230 <sup>6</sup>                   | --          | --        | --         | 250             | No                   |
| Chlorodibromomethane           | µg/L     | <0.17            | <0.17            | 0.41                               | --                                 | --                                 | 0.41        | 34        | --         | 80 <sup>7</sup> | No                   |
| Copper, Total Recoverable      | µg/L     | 4.2              | 3.9              | 7.0 <sup>4</sup> /8.5 <sup>5</sup> | 10 <sup>4</sup> /4.9 <sup>5</sup>  | 7.0 <sup>4</sup> /8.5 <sup>5</sup> | 1,300       | --        | --         | 1,000           | No                   |
| Cyanide, Total (as CN)         | µg/L     | 3.2              | 1.3              | 5.2                                | 22                                 | 5.2                                | 700         | 220,000   | --         | 150             | No                   |
| Dichlorobromomethane           | µg/L     | <0.16            | <0.16            | 0.56                               | --                                 | --                                 | 0.56        | 46        | --         | 80 <sup>7</sup> | No                   |
| Electrical Conductivity @ 25°C | µmhos/cm | 453 <sup>1</sup> | 182 <sup>1</sup> | 900                                | --                                 | --                                 | --          | --        | --         | 900             | No                   |
| Iron, Total Recoverable        | µg/L     | 80 <sup>1</sup>  | 815 <sup>1</sup> | 300                                | --                                 | 1,000                              | --          | --        | --         | 300             | No                   |
| Manganese, Total Recoverable   | µg/L     | 11 <sup>1</sup>  | 75 <sup>1</sup>  | 50                                 | --                                 | --                                 | --          | 100       | --         | 50              | No                   |
| Mercury, Total Recoverable     | µg/L     | 0.006            | 0.011            | 0.050                              | --                                 | --                                 | 0.050       | 0.051     | --         | 2               | Yes <sup>8</sup>     |
| Nitrate Nitrogen, Total (as N) | mg/L     | 15               | 0.542            | 10                                 | --                                 | --                                 | --          | --        | --         | 10              | Yes                  |
| Sulfate                        | mg/L     | 25 <sup>1</sup>  | 13 <sup>1</sup>  | 250                                | --                                 | --                                 | --          | --        | --         | 250             | No                   |
| Total Dissolved Solids         | mg/L     | 410 <sup>1</sup> | 170 <sup>1</sup> | 500                                | --                                 | --                                 | --          | --        | --         | 500             | No                   |
| Zinc, Total Recoverable        | µg/L     | 51               | 6.1              | 90 <sup>4</sup> /90 <sup>5</sup>   | 90 <sup>4</sup> /90 <sup>5</sup>   | 90 <sup>4</sup> /90 <sup>5</sup>   | 7,400       | 26,000    | --         | 5,000           | No                   |

| Constituent | Units | MEC | B | C | CMC | CCC | Water & Org | Org. Only | Basin Plan | MCL | Reasonable Potential |
|-------------|-------|-----|---|---|-----|-----|-------------|-----------|------------|-----|----------------------|
|-------------|-------|-----|---|---|-----|-----|-------------|-----------|------------|-----|----------------------|

General Note: All inorganic concentrations are given as a total recoverable.

MEC = Maximum Effluent Concentration

B = Maximum Receiving Water Concentration or lowest detection level, if non-detect

C = Criterion used for Reasonable Potential Analysis

CMC = Criterion Maximum Concentration (CTR or NTR)

CCC = Criterion Continuous Concentration (CTR or NTR)

Water & Org = Human Health Criterion for Consumption of Water & Organisms (CTR or NTR)

Org. Only = Human Health Criterion for Consumption of Organisms Only (CTR or NTR)

Basin Plan = Numeric Site-specific Basin Plan Water Quality Objective

MCL = Drinking Water Standards Maximum Contaminant Level

NA = Not Available

ND = Non-detect

Footnotes:

- (1) Represents the maximum observed annual average concentration for comparison with the Secondary MCL.
- (2) USEPA National Recommended Ambient Water Quality Criteria, Freshwater Aquatic Life Protection, 1-hour average.
- (3) USEPA National Recommended Ambient Water Quality Criteria, Freshwater Aquatic Life Protection, 30-day average.
- (4) Criterion to be compared to the MEC.
- (5) Criterion to be compared to the maximum upstream upstream receiving water concentration.
- (6) USEPA National Recommended Ambient Water Quality Criteria, Freshwater Aquatic Life Protection, 4-day average.
- (7) Represents the Primary MCL for Total Trihalomethanes which include bromoform, chlorodibromomethane, chloroform, and dichlorobromomethane.
- (8) See section IV.C.3.b of the Fact Sheet (Attachment F) for a discussion of the RPA results.

### ATTACHMENT H – CALCULATION OF WQBELS

| Parameter                      | Units | Most Stringent Criteria |      |      | HH Calculations <sup>1</sup>           |                                    |                    | Aquatic Life Calculations <sup>1</sup> |                      |                                   |                        |            |                               |                    |                               |                    | Final Effluent Limitations |             |
|--------------------------------|-------|-------------------------|------|------|----------------------------------------|------------------------------------|--------------------|----------------------------------------|----------------------|-----------------------------------|------------------------|------------|-------------------------------|--------------------|-------------------------------|--------------------|----------------------------|-------------|
|                                |       | HH                      | CMC  | CCC  | ECA <sub>HH</sub> = AMEL <sub>HH</sub> | AMEL/MDEL Multiplier <sub>HH</sub> | MDEL <sub>HH</sub> | ECA Multiplier <sub>acute</sub>        | LTA <sub>acute</sub> | ECA Multiplier <sub>chronic</sub> | LTA <sub>chronic</sub> | Lowest LTA | AMEL Multiplier <sub>95</sub> | AMEL <sub>AL</sub> | MDEL Multiplier <sub>99</sub> | MDEL <sub>AL</sub> | Lowest AMEL                | Lowest MDEL |
| Ammonia Nitrogen, Total (as N) | mg/L  | --                      | 2.14 | 1.58 | --                                     | --                                 | --                 | 0.14                                   | 0.30                 | 0.53                              | 0.84                   | 0.30       | 2.5                           | 0.8                | 7.36                          | 2.2                | 0.8                        | 2.2         |

<sup>1</sup> As described in section IV.C.2.c of the Fact Sheet (Attachment F), calculation of effluent limitations for the protection of human health and aquatic life are determined without the allowance of dilution credits.